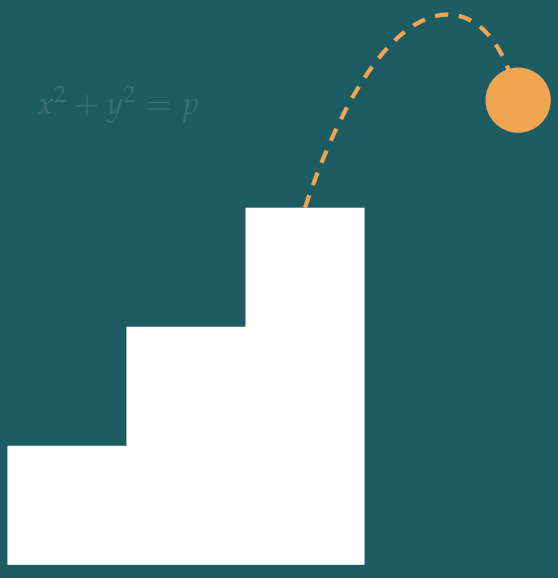
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2001 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2001 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2001 Worked Solutions

Worked solutions to all 42 questions of the 2001 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2001, Question 1

Pure • Geometric Inequalities

STEP I 2001, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q1

The key idea

Three points on three different sides of a square must use a pair of opposite sides, and the third side joins those two. Put coordinates on the square so that each point has a single free parameter between 0 and 1. Then argue by contradiction: if all three sides of the triangle were longer than the bound, each length condition would squeeze one parameter, and the squeezes together would push a point off the square.

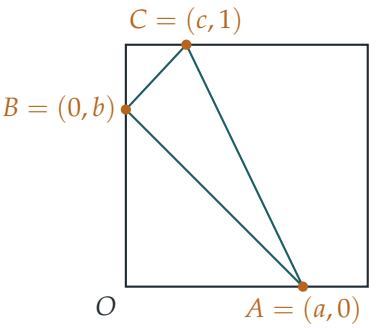
Solution

Work in centimetres. Four sides and three points on different sides means the points miss exactly one side of the square, so two of the points lie on a pair of opposite sides and the third lies on a side adjacent to both. Choose axes with the square as $0 \le x \le 1$, $0 \le y \le 1$, so that

$$A = (a, 0) \text{ on the bottom}, \qquad B = (0, b) \text{ on the left}, \qquad C = (c, 1) \text{ on the top},$$

with $a, b, c \in [0, 1]$. (Rotating or reflecting the square doesn't change any lengths, so this is no loss of generality.) Then

$$AB^2 = a^2 + b^2, \qquad BC^2 = c^2 + (1-b)^2, \qquad CA^2 = (c-a)^2 + 1.$$



Let $k = \sqrt{6} - \sqrt{2}$ and $t = 2 - \sqrt{3}$. Two small facts make the numbers behave:

$$k^2 = 8 - 2\sqrt{12} = 8 - 4\sqrt{3}, \qquad t^2 = 7 - 4\sqrt{3}, \qquad \text{so} \qquad k^2 - 1 = t^2.$$

Also $2t = 4 - 2\sqrt{3} = (\sqrt{3} - 1)^2$, so $\sqrt{2t} = \sqrt{3} - 1$, and therefore

$$\sqrt{2t} + t = (\sqrt{3} - 1) + (2 - \sqrt{3}) = 1.$$

Suppose, for contradiction, that all three sides of the triangle are longer than k. By the symmetry of reflecting the square in the line $y = \frac{1}{2}$ (which swaps the roles of A and C, and of a and c), we may assume $c \geq a$.

From $BC > k$. Since $c \leq 1$,

$$k^2 < BC^2 = c^2 + (1 - b)^2 \leq 1 + (1 - b)^2,$$

so $(1 - b)^2 > k^2 - 1 = t^2$. As $1 - b \geq 0$ and $t > 0$, this gives $1 - b > t$, that is, $0 \leq b < 1 - t$.

From $AB > k$. Since $0 \leq b < 1 - t$ we have $b^2 < (1 - t)^2$, so

$$a^2 > k^2 - b^2 > k^2 - (1 - t)^2 = (1 + t^2) - (1 - 2t + t^2) = 2t.$$

As $a \geq 0$, this gives $a > \sqrt{2t} = \sqrt{3} - 1$.

> **?** **Ask yourself** We've pinned down a; what does the third side CA add?

The side CA crosses the square from bottom to top, so it has vertical extent 1 and must make up the rest of its length horizontally.

From $CA > k$. We have $(c - a)^2 + 1 > k^2$, so $(c - a)^2 > t^2$, and since $c \geq a$ this means $c - a > t$. Hence

$$c > a + t > \sqrt{2t} + t = 1.$$

But C lies on the top side, so $c \leq 1$. This contradiction shows that the three sides can't all be longer than k: at least one side of triangle ABC has length at most $(\sqrt{6} - \sqrt{2})$ cm, as required.

The bound is attained

Every inequality above is tight when $c = 1$ and $a = b = \sqrt{3} - 1$, that is, $A = (\sqrt{3} - 1, 0)$, $B = (0, \sqrt{3} - 1)$, $C = (1, 1)$. Then

$$AB^2 = 2(\sqrt{3} - 1)^2 = 8 - 4\sqrt{3}, \qquad BC^2 = CA^2 = 1 + (2 - \sqrt{3})^2 = 8 - 4\sqrt{3},$$

so ABC is equilateral with side exactly k (with C at a corner, counted as lying on the top side). So $\sqrt{6} - \sqrt{2}$ is the best possible constant. In trigonometric terms, $k = \sec 15^\circ$: this is the largest equilateral triangle that fits in a unit square, tilted at 15° to the sides.

Checks

- $k = \sqrt{6} - \sqrt{2} \approx 1.0353$ is a little more than the side of the square, which is sensible: a triangle with a vertex on each of three sides can have all sides longer than 1, but only just.
- A random search over 2×10^6 triples $(a, b, c) \in [0, 1]^3$ found the shortest side never exceeding 1.0337, below k; the equilateral example above has all three sides equal to 1.03528 to machine precision.
- $\sec 15^\circ = 4/(\sqrt{6} + \sqrt{2}) = \sqrt{6} - \sqrt{2}$, matching the equilateral picture.

Common slips

- Assuming the triangle is equilateral, or that one vertex is at a corner, without proof. The extremal case suggests the answer but doesn't prove the inequality for every configuration.
- Forgetting to justify the set-up: the key structural fact is that three distinct sides must include an opposite pair.
- Taking square roots of $(1 - b)^2 > t^2$ or $(c - a)^2 > t^2$ without saying which sign is involved.

Beyond STEP

This is a small *max–min* problem: we are maximising the shortest side over a compact set of configurations, and the extremal configuration makes all the constraints equal at once, a pattern that runs through optimisation and game theory. Because $[0, 1]^3$ is closed and bounded and the shortest side is a continuous function, a maximum must exist, a fact proved in a first analysis course (the extreme value theorem). Abbott's *Understanding Analysis* covers this well, and Siklos's *Advanced Problems in Mathematics* has more STEP geometry in the same spirit.

WORKED SOLUTION

STEP I 2001, Question 2

Pure • Algebraic Inequalities

STEP I 2001, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q2

The key idea

With an inequality you can't multiply through by something whose sign you don't know, so the safe move in (i) is to collect everything on one side and factorise: then the answer is read off from a sign diagram. In (ii) the square roots tempt us to square, which is legitimate only when both sides are known to be non-negative. Keeping track of that condition is the whole question.

Solution

Part (i)

We can't multiply by x, since x may be negative. Instead we multiply by x^2, which is strictly positive because $x \neq 0$, so the inequality keeps its direction:

$$\begin{aligned} 1 + 2x - x^2 > \frac{2}{x} &\iff x^2 + 2x^3 - x^4 > 2x \\ &\iff 0 > x^4 - 2x^3 - x^2 + 2x \\ &\iff x\left(x^3 - 2x^2 - x + 2\right) < 0. \end{aligned}$$

The cubic factorises by grouping:

$$x^3 - 2x^2 - x + 2 = x^2(x-2) - (x-2) = (x-2)(x^2-1) = (x-2)(x-1)(x+1).$$

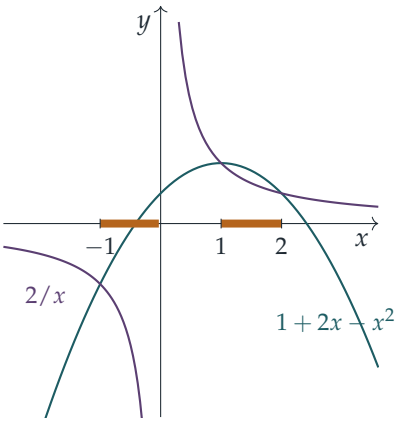
So we need

$$(x+1)\,x\,(x-1)(x-2) < 0.$$

This is a quartic with positive leading coefficient and four simple roots $-1, 0, 1, 2$, so its sign alternates as x crosses each root: positive for $x > 2$, negative on $(1, 2)$, positive on $(0, 1)$, negative on $(-1, 0)$ and positive for $x < -1$. The roots themselves give 0, which is not < 0 (and $x = 0$ is excluded anyway). Hence

$$\boxed{-1 < x < 0 \quad \text{or} \quad 1 < x < 2.}$$

The picture agrees: the parabola $y = 1 + 2x - x^2$ lies above the hyperbola $y = 2/x$ exactly between their crossings at $x = -1$, 1, 2, and on the negative side only until the hyperbola plunges to $-\infty$ at $x = 0$.



Part (ii)

First note that $x \geq -\frac{10}{3}$ gives $x + 4 \geq \frac{2}{3} > 0$ as well as $3x + 10 \geq 0$, so both square roots are defined throughout.

Both sides are non-negative, and for $a, b \geq 0$ we have $a > b \iff a^2 > b^2$. So we may square without gaining or losing solutions:

$$\begin{aligned}\sqrt{3x+10} > 2 + \sqrt{x+4} &\iff 3x + 10 > 4 + 4\sqrt{x+4} + (x+4)\\ &\iff 2x + 2 > 4\sqrt{x+4}\\ &\iff x + 1 > 2\sqrt{x+4}.\end{aligned}$$

Ask yourself Can we simply square again?

Not straight away. The right-hand side $2\sqrt{x+4}$ is positive, but $x + 1$ might be negative, and then squaring could turn a false inequality into a true one. So we split into cases.

- If $x + 1 \leq 0$, the left-hand side is at most 0, while the right-hand side is strictly positive. The inequality fails.
- If $x + 1 > 0$, both sides are positive, so squaring is again reversible:

$$\begin{aligned}x + 1 > 2\sqrt{x+4} &\iff x^2 + 2x + 1 > 4x + 16\\ &\iff x^2 - 2x - 15 > 0\\ &\iff (x-5)(x+3) > 0,\end{aligned}$$

  which holds when $x > 5$ or $x < -3$. Combined with $x > -1$, this leaves $x > 5$.

Therefore

$$\boxed{x > 5.}$$

Checks

- At $x = 5$: $\sqrt{25} = 5$ and $2 + \sqrt{9} = 5$, so equality holds there, as it should at the boundary. At $x = 12$: $\sqrt{46} \approx 6.78 > 2 + 4 = 6$. At $x = 0$: $\sqrt{10} \approx 3.16 < 4$.
- In (i), $x = 1.5$ gives $1.75 > 1.33$ and $x = -0.5$ gives $-0.25 > -4$, both true; $x = 0.5$ gives $1.75 < 4$, false.
- A computer tested both inequalities at 2×10^5 random points each, plus points within 10^{-6} of every boundary, and found no disagreement with the answers above.

Common slips

- Multiplying (i) by x without splitting into $x > 0$ and $x < 0$. This loses the interval $(-1, 0)$ or reverses the answer there.
- Squaring $x + 1 > 2\sqrt{x+4}$ without requiring $x + 1 > 0$. That gives the spurious interval $-\frac{10}{3} \leq x < -3$; try $x = -3.2$ in the original to see that it fails ($\sqrt{0.4} \approx 0.63 < 2 + \sqrt{0.8}$).

Beyond STEP

The principle in (ii), that an operation preserves an inequality only if it is an increasing function on the values involved, is the start of how analysts handle inequalities: $t \mapsto t^2$ is increasing on $[0, \infty)$ but not on $\mathbb{R}$. Sign diagrams for products generalise to the fact that a continuous function can change sign only by passing through zero, the intermediate value theorem. Abbott's *Understanding Analysis* treats both carefully.

WORKED SOLUTION

STEP I 2001, Question 3

Pure • Cubics and Discriminants

STEP I 2001, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q3

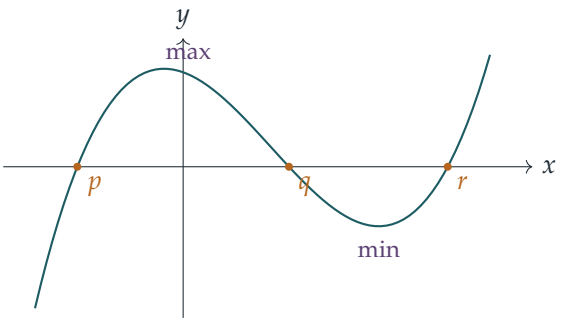
The key idea

A cubic with three distinct real roots must turn round between each pair of neighbouring roots, so its derivative, a quadratic, has two distinct real roots. Writing "two distinct real roots" as "discriminant positive" gives the first inequality. For the second part, the given product is a cubic whose coefficients are exactly the quantities in the target inequality, so the first part (or a completed square) does the work.

Solution

The sketch

Expanding, $f(x) = x^3 - (p+q+r)x^2 + \ldots$, so $f(x) \to +\infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$. The graph crosses the x-axis at p, q and r only, and since each root is simple, f changes sign at each: $f < 0$ for $x < p$, $f > 0$ on (p,q), $f < 0$ on (q,r) and $f > 0$ for $x > r$. So there is a local maximum somewhere between p and q and a local minimum somewhere between q and r. The y-intercept is $f(0) = -pqr$.



(The picture is drawn with $p < 0 < q < r$; the shape is the same wherever the roots lie, only the position of the y-axis changes.)

The first inequality

Expanding fully,

$$f(x) = x^3 - (p+q+r)x^2 + (qr+rp+pq)x - pqr,$$

so

$$f'(x) = 3x^2 - 2(p+q+r)x + (qr+rp+pq).$$

Since $f(p) = f(q) = 0$, Rolle's theorem gives a root of f' strictly between p and q; likewise there is a root strictly between q and r. These intervals don't overlap, so the quadratic $f'(x) = 0$ has two *distinct* real roots. (Without Rolle: $f'(p) = (p-q)(p-r) > 0$, $f'(q) = (q-p)(q-r) < 0$ and $f'(r) = (r-p)(r-q) > 0$, so f' changes sign on each interval.)

A quadratic has two distinct real roots exactly when its discriminant is positive, so

$$4(p+q+r)^2 - 12(qr+rp+pq) > 0, \qquad \text{that is,} \qquad (p+q+r)^2 > 3(qr+rp+pq),$$

as required.

The second inequality

Expanding,

$$(x^2+gx+h)(x-k) = x^3 + (g-k)x^2 + (h-gk)x - hk.$$

Compare this with the expansion of f: if the cubic has roots p, q, r, then

$$p+q+r = -(g-k), \qquad qr+rp+pq = h-gk,$$

so the target inequality $(g-k)^2 > 3(h-gk)$ is exactly the first inequality for the roots of this cubic.

Sufficient. Suppose $g^2 > 4h$. Then x^2+gx+h has two distinct real roots $\alpha < \beta$, and the cubic has real roots α, β, k. If k is different from both, the three roots are distinct and the first part gives $(g-k)^2 > 3(h-gk)$ at once.

Ask yourself What if k happens to equal α or β, so the roots are not distinct?

The first part assumed $p < q < r$, so we need a little more. The identity

$$(p+q+r)^2 - 3(qr+rp+pq) = \tfrac{1}{2}\left[(p-q)^2 + (q-r)^2 + (r-p)^2\right]$$

shows the left-hand side is positive unless $p = q = r$. Since $\alpha \neq \beta$, the three roots are never all equal, so the inequality holds in this case too.

A direct route avoids roots altogether. Multiplying out,

$$(g-k)^2 - 3(h-gk) = g^2 + gk + k^2 - 3h = \left(k + \tfrac{1}{2}g\right)^2 + \tfrac{3}{4}\left(g^2 - 4h\right),$$

which is a square plus a positive quantity when $g^2 > 4h$, and so positive. Either way, $g^2 > 4h$ is sufficient.

Not necessary. We need values with $g^2 \leq 4h$ for which the inequality still holds. Take $g = 0$, $h = 1$, $k = 2$. Then $g^2 = 0 \leq 4 = 4h$, but

$$(g-k)^2 = 4 > 3 = 3(h-gk).$$

So the inequality can hold without $g^2 > 4h$, and the condition is not necessary.

Ask yourself How can that be, when the first part needed three real roots?

The first part needed only that $f'(x) = 0$ has two distinct real roots, which is weaker. Here the cubic $(x^2+1)(x-2) = x^3 - 2x^2 + x - 2$ has only one real root, but $f'(x) = 3x^2 - 4x + 1 = (3x-1)(x-1)$, so it still has two stationary points. The inequality $(g-k)^2 > 3(h-gk)$ says precisely that the cubic has a distinct local maximum and minimum, whether or not it crosses the axis three times.

Checks

- With $p, q, r = 0, 1, 2$: $(p+q+r)^2 = 9 > 6 = 3(qr+rp+pq)$, and $f'(x) = 3x^2 - 6x + 2$ has roots $1 \pm 1/\sqrt{3}$, one in $(0,1)$ and one in $(1,2)$.
- The completed-square formula shows the boundary of the inequality is sharp: with $g = 2$, $h = 1$ (so $g^2 = 4h$) and $k = -1$ we get $(g-k)^2 = 9 = 3(h-gk)$, equality and not strict inequality. This is the cubic $(x+1)^3$, with all three roots equal.
- A computer checked both identities and both inequalities exactly on 20 000 random rational cases each, plus the example $g = 0$, $h = 1$, $k = 2$ (giving $4 > 3$).

Common slips

- Sketching the cubic the wrong way up: the coefficient of x^3 is $+1$, so the graph rises to the right.
- Forgetting the case where k is a root of $x^2 + gx + h$, where the first part does not apply as stated.
- Showing only that the inequality holds for one example with $g^2 \leq 4h$ and calling that "sufficient". One example proves *not necessary*; sufficiency needs a proof for all g, h, k.

Beyond STEP

The quantity $(p+q+r)^2 - 3(qr+rp+pq)$ is, up to a factor, the discriminant of the derivative, and it is a *symmetric function* of the roots: any such function can be written in terms of the coefficients, which is why the inequality could be translated from roots to g, h, k. The full discriminant of a cubic, $\prod(p-q)^2$, decides whether all three roots are real; this question uses its weaker cousin. Symmetric functions and discriminants lead on to Galois theory; Allenby's *Rings, Fields and Groups* is a good place to start.

WORKED SOLUTION

STEP I 2001, Question 4

Pure • Inverse Trigonometric Equations

STEP I 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q4

The key idea

The triple-angle formula comes straight from the addition formula for tan. After that, everything hinges on one fact: tan repeats with period π, so if $\tan 3\varphi = \tan 3\theta$ then φ can differ from θ by any multiple of $\pi/3$. The real work in (i) and (ii) is keeping track of the ranges of $\cos^{-1}$ and $\tan^{-1}$, which decide which of those infinitely many angles are allowed.

Solution

The triple-angle formula

Write $t = \tan\theta$. By the addition formula, $\tan 2\theta = \dfrac{2t}{1-t^2}$, and then

$$\tan 3\theta = \tan(2\theta+\theta) = \frac{\tan 2\theta + t}{1 - t\tan 2\theta} = \frac{\dfrac{2t}{1-t^2} + t}{1 - \dfrac{2t^2}{1-t^2}}$$

$$= \frac{2t + t(1-t^2)}{(1-t^2) - 2t^2} = \frac{3t - t^3}{1 - 3t^2},$$

where we multiplied top and bottom by $1 - t^2$. This is the required formula. (It holds whenever both sides are defined.)

The value of $\tan 3\theta$

Since $\cos\theta = 2/\sqrt{5}$ and $0 < \theta < \pi/2$, we have $\sin\theta > 0$, so

$$\sin\theta = \sqrt{1 - \tfrac{4}{5}} = \frac{1}{\sqrt{5}}, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{2}.$$

(Picture a right-angled triangle with sides 1, 2, $\sqrt{5}$.) Substituting $t = \frac{1}{2}$:

$$\tan 3\theta = \frac{\frac{3}{2} - \frac{1}{8}}{1 - \frac{3}{4}} = \frac{11/8}{1/4} = \frac{11}{2},$$

as required.

It will matter later that θ is fairly small: $\tan\theta = \frac{1}{2} < \frac{1}{\sqrt{3}} = \tan\frac{\pi}{6}$, so $0 < \theta < \pi/6$ and hence $0 < 3\theta < \pi/2$.

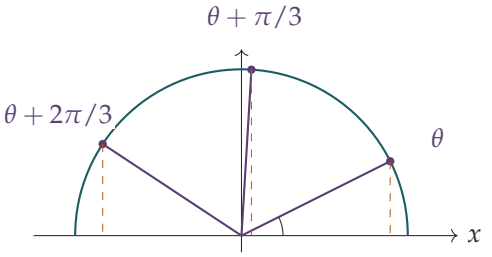
Part (i)

Let $\varphi = \cos^{-1} x$, so $0 \le \varphi \le \pi$ and $x = \cos\varphi$. The equation says $\tan 3\varphi = \frac{11}{2} = \tan 3\theta$. Because tan has period π and takes each value exactly once on each interval of length π,

$$3\varphi = 3\theta + k\pi \quad \Longleftrightarrow \quad \varphi = \theta + \frac{k\pi}{3}, \qquad k \in \mathbb{Z}.$$

Ask yourself How many of these angles are actually allowed?

We need $0 \le \varphi \le \pi$. Since $0 < \theta < \pi/6$, the values $k = 0, 1, 2$ give $\varphi = \theta,\ \theta + \frac{\pi}{3},\ \theta + \frac{2\pi}{3}$, all strictly inside $(0, \pi)$; $k = 3$ gives $\varphi > \pi$ and $k = -1$ gives $\varphi < 0$. So there are exactly three solutions.



The three values of x are the cosines of these angles (the feet of the dashed lines). Using $\cos\theta = 2/\sqrt{5}$, $\sin\theta = 1/\sqrt{5}$ and the addition formula:

$$x_1 = \cos\theta = \frac{2}{\sqrt{5}},$$

$$x_2 = \cos\theta\cos\tfrac{\pi}{3} - \sin\theta\sin\tfrac{\pi}{3} = \frac{2}{\sqrt{5}}\cdot\frac{1}{2} - \frac{1}{\sqrt{5}}\cdot\frac{\sqrt{3}}{2} = \frac{2-\sqrt{3}}{2\sqrt{5}},$$

$$x_3 = \cos\theta\cos\tfrac{2\pi}{3} - \sin\theta\sin\tfrac{2\pi}{3} = -\frac{2}{\sqrt{5}}\cdot\frac{1}{2} - \frac{1}{\sqrt{5}}\cdot\frac{\sqrt{3}}{2} = -\frac{2+\sqrt{3}}{2\sqrt{5}}.$$

None of these angles makes 3φ an odd multiple of $\pi/2$ (each 3φ is $3\theta + k\pi$ with $0 < 3\theta < \pi/2$), so $\tan 3\varphi$ is defined in every case. The solutions of (i) are

$$x = \frac{2}{\sqrt{5}}, \qquad x = \frac{2-\sqrt{3}}{2\sqrt{5}}, \qquad x = -\frac{2+\sqrt{3}}{2\sqrt{5}}.$$

(Rationalising, these are $\frac{2\sqrt{5}}{5}$, $\frac{(2-\sqrt{3})\sqrt{5}}{10}$ and $-\frac{(2+\sqrt{3})\sqrt{5}}{10}$.)

Another route. Putting $T = \tan\varphi$ into the triple-angle formula gives $2(3T - T^3) = 11(1 - 3T^2)$, that is,

$$2T^3 - 33T^2 - 6T + 11 = 0.$$

We know $T = \frac{1}{2}$ is a root, and factorising gives $(2T-1)(T^2 - 16T - 11) = 0$, so $T = \frac{1}{2}$ or $T = 8 \pm 5\sqrt{3}$. Then $x = \cos\varphi = \pm 1/\sqrt{1+T^2}$, with the sign of T (positive T means φ is acute; negative means obtuse). This gives the same three answers, and it is a nice check that $1 + (8+5\sqrt{3})^2 = 140 + 80\sqrt{3} = 1/x_2^2$.

Part (ii)

Let $\psi = \frac{1}{3}\tan^{-1} y$. Since $-\frac{\pi}{2} < \tan^{-1} y < \frac{\pi}{2}$, we have $-\frac{\pi}{6} < \psi < \frac{\pi}{6}$.

Ask yourself Does the condition $\cos\psi = 2/\sqrt{5}$ give one angle or two?

On $(-\pi/6, \pi/6)$, cos is even, so $\cos\psi = 2/\sqrt{5} = \cos\theta$ gives $\psi = \theta$ or $\psi = -\theta$, and both lie in the allowed range because $0 < \theta < \pi/6$. (There are no others: on $[0, \pi/6)$ the function cos is strictly decreasing, so it takes the value $2/\sqrt{5}$ only once there, and similarly on $(-\pi/6, 0]$.)

So $\tan^{-1} y = \pm 3\theta$. Since $0 < 3\theta < \pi/2$, both $\pm 3\theta$ are genuinely values of $\tan^{-1}$, and

$$y = \tan(\pm 3\theta) = \pm\frac{11}{2}.$$

The solutions of (ii) are $y = \frac{11}{2}$ and $y = -\frac{11}{2}$.

Checks

- $\theta = \cos^{-1}(2/\sqrt{5}) \approx 0.4636$, so $3\theta \approx 1.3909$ and $\tan 3\theta \approx 5.5000$.
- $x_2 = \frac{2-\sqrt{3}}{2\sqrt{5}} \approx 0.05992$ and $x_3 \approx -0.83451$; a calculator gives $\tan(3\cos^{-1} x) \approx 5.5$ for each, and a fine numerical scan of $[-1, 1]$ finds exactly three solutions.
- $\cos\left(\frac{1}{3}\tan^{-1}(\pm 5.5)\right) = \cos(\pm 0.4636) \approx 0.8944 = 2/\sqrt{5}$.

Common slips

- Giving only $x = 2/\sqrt{5}$ in (i), forgetting that tan repeats every π, so 3φ has three possible values in $[0, 3\pi]$.
- In (ii), giving only $y = 11/2$: cos is even, so $\psi = -\theta$ works too.
- Not checking ranges: the answers depend on $\theta < \pi/6$, which is why $\theta + 2\pi/3$ is allowed in (i) and why 3θ lies within the range of $\tan^{-1}$ in (ii).

Beyond STEP

Inverse trigonometric functions are a first taste of the problem of inverting a function that isn't one-to-one: we choose a *principal branch*, and equations like these are really about which branch we're on. In complex analysis the same issue appears for $\log z$ and $z^{1/3}$, where going round the origin moves you from one branch to the next, exactly like adding $2\pi/3$ to θ here. Needham's *Visual Complex Analysis* explains branches beautifully with pictures.

WORKED SOLUTION

STEP I 2001, Question 5

Pure • Differentiating Under the Integral

STEP I 2001, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q5

The key idea

Both integrals in (i) and (ii) yield to direct integration, the second after the substitution $u = 1 + tx$. The real point is the remark at the end: the integrand of (ii) is the t-derivative of the integrand of (i), and the answers are related in the same way. So differentiating with respect to the parameter t *inside* the integral seems to commute with integrating over x. Doing it once more, and then setting $t = 1$, gives the conjecture.

Solution

Throughout, $t > 0$ and $0 \leq x \leq 1$, so $1 + tx \geq 1 > 0$ and none of the integrands blows up.

Part (i)

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}(1+tx)^{-1} = -t(1+tx)^{-2}$, an antiderivative of $(1+tx)^{-2}$ is $-\dfrac{1}{t(1+tx)}$ (here we use $t \neq 0$). Hence

$$\int_0^1 \frac{1}{(1+tx)^2}\,\mathrm{d}x = \left[-\frac{1}{t(1+tx)}\right]_0^1 = \frac{1}{t}\left(1 - \frac{1}{1+t}\right) = \frac{1}{t}\cdot\frac{t}{1+t} = \frac{1}{1+t},$$

as required.

Part (ii)

The x in the numerator makes the direct approach awkward, so we put $u = 1 + tx$. Then $x = \dfrac{u-1}{t}$, $\mathrm{d}x = \dfrac{\mathrm{d}u}{t}$, and the limits $x = 0, 1$ become $u = 1, 1+t$:

$$\int_0^1 \frac{-2x}{(1+tx)^3}\,\mathrm{d}x = -\frac{2}{t^2}\int_1^{1+t}\frac{u-1}{u^3}\,\mathrm{d}u = -\frac{2}{t^2}\int_1^{1+t}\left(u^{-2}-u^{-3}\right)\mathrm{d}u$$
$$= -\frac{2}{t^2}\left[-\frac{1}{u}+\frac{1}{2u^2}\right]_1^{1+t} = -\frac{2}{t^2}\left(\frac{1}{2}-\frac{1}{1+t}+\frac{1}{2(1+t)^2}\right).$$

Ask yourself How do we tidy the bracket without expanding everything over a common denominator?

Writing $w = \dfrac{1}{1+t}$, the bracket is $\frac{1}{2} - w + \frac{1}{2}w^2 = \frac{1}{2}(1-w)^2$, a perfect square. And $1 - w = \dfrac{t}{1+t}$, so

$$\int_0^1 \frac{-2x}{(1+tx)^3}\,\mathrm{d}x = -\frac{2}{t^2}\cdot\frac{1}{2}\cdot\frac{t^2}{(1+t)^2} = -\frac{1}{(1+t)^2},$$

which is the required result.

The conjecture

Look at the integrands as functions of t with x held fixed:

$$\frac{\partial}{\partial t}(1+tx)^{-2} = -2x(1+tx)^{-3}.$$

So the integrand of (ii) is the t-derivative of the integrand of (i), just as the answer to (ii) is the derivative of the answer to (i). This suggests the rule

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^1 f(x,t)\,\mathrm{d}x = \int_0^1 \frac{\partial f}{\partial t}\,\mathrm{d}x.$$

Applying it once more,

$$\frac{\partial}{\partial t}\left(-2x(1+tx)^{-3}\right) = 6x^2(1+tx)^{-4}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(-(1+t)^{-2}\right) = 2(1+t)^{-3},$$

so we conjecture that, for $t > 0$,

$$\int_0^1 \frac{6x^2}{(1+tx)^4}\,\mathrm{d}x = \frac{2}{(1+t)^3}.$$

Putting $t = 1$, the conjectured value is

$$\int_0^1 \frac{6x^2}{(1+x)^4}\,\mathrm{d}x = \frac{2}{8} = \frac{1}{4}.$$

Checks

- At $t \to 0$ the integrand in (i) tends to 1 and the answer $\frac{1}{1+t}$ tends to 1; in (ii) the integrand tends to $-2x$, whose integral is -1, matching $-\frac{1}{(1+t)^2} \to -1$.
- We weren't asked to verify the conjecture, but it is quick to do: with $u = 1+x$ the integral becomes $6\int_1^2 (u^{-2} - 2u^{-3} + u^{-4})\,\mathrm{d}u = 6\left[-\frac{1}{u} + \frac{1}{u^2} - \frac{1}{3u^3}\right]_1^2 = 6\left(-\frac{7}{24} + \frac{1}{3}\right) = \frac{1}{4}$.
- Simpson's rule gives 0.250000 for the final integral, and agrees with $\frac{2}{(1+t)^3}$ to nine decimal places for 25 different values of t.

Common slips

- Forgetting the factor $\frac{1}{t}$ from $\mathrm{d}x = \mathrm{d}u/t$ in (ii): there are *two* factors of $\frac{1}{t}$, one from x and one from $\mathrm{d}x$.
- Not changing the limits to $u = 1$ and $u = 1 + t$ after substituting.
- Differentiating $-(1+t)^{-2}$ and getting $-2(1+t)^{-3}$: the two minus signs cancel, so the conjectured value is positive (as it must be, since the integrand is positive).

Beyond STEP

The rule we conjectured is *differentiation under the integral sign* (Leibniz's rule). It is a theorem of real analysis that it holds when f and $\partial f/\partial t$ are continuous on a closed rectangle, which is true here for $0 \le x \le 1$ and t in any interval $[a, b]$ with $a > 0$. Repeating the argument gives $\int_0^1 \frac{x^n}{(1+tx)^{n+2}}\,\mathrm{d}x = \frac{1}{(n+1)(1+t)^{n+1}}$ for every n. Richard Feynman famously used this trick to evaluate integrals that resisted other methods. For the theory behind swapping limits, derivatives and integrals, see Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP I 2001, Question 6

Pure • Surfaces of Revolution

STEP I 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.8, *Slicing a Spherical Loaf*: https://beyondstep.keithmcnulty.org/ch04/spherical-loaf.html

The key idea

Generate the sphere by rotating the semicircle $y = \sqrt{a^2 - x^2}$ about the x-axis. In the surface-area formula the integrand $y\sqrt{1 + (\mathrm{d}y/\mathrm{d}x)^2}$ turns out to be *constant*, so the crust on any slice depends only on its thickness and not on where the slice is. For the second part, we eat slices from one end (as the hint's limits $-a$ to $-a + t$ suggest), so what is left is a spherical cap; the ratio V/A becomes a quadratic in the cap's thickness, and we find its largest value.

Solution

The crust is proportional to the number of slices

Let the loaf have radius a and centre at the origin, so its surface is obtained by rotating

$$y = \sqrt{a^2 - x^2}, \qquad -a \le x \le a,$$

about the x-axis. Differentiating, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{x}{\sqrt{a^2 - x^2}} = -\dfrac{x}{y}$ for $|x| < a$, so

$$y\left(1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right)^{\frac{1}{2}} = y\left(\frac{y^2 + x^2}{y^2}\right)^{\frac{1}{2}} = y \cdot \frac{a}{y} = a,$$

using $x^2 + y^2 = a^2$ and $y > 0$. Hence the curved surface area of the sphere between the planes $x = x_1$ and $x = x_2$ (with $-a \le x_1 < x_2 \le a$) is

$$2\pi \int_{x_1}^{x_2} a \, \mathrm{d}x = 2\pi a \, (x_2 - x_1).$$

(The integrand is undefined only at the end-points $x = \pm a$, where it is still bounded, so the integral is fine.)

Now suppose the loaf is cut into N slices, each of thickness $h = 2a/N$. Every slice, wherever it lies, carries crust of area exactly $2\pi ah$. So if n slices remain, in any positions at all, the crust remaining is

$$A = n \cdot 2\pi ah,$$

which is proportional to n, the number of slices remaining. This is what we had to show. (It's Archimedes' old result: a sphere and its circumscribing cylinder have the same area between any two parallel planes.)

The crusty condition

Slices are eaten from one end, so after some have gone the remaining bread is the cap from $x = -a$ to $x = -a + t$ for some t with $0 < t \leq 2a$. From the first part, $A = 2\pi at$. For the volume, substitute $u = x + a$, so that $y^2 = a^2 - x^2 = (a - x)(a + x) = (2a - u)u$:

$$V = \pi \int_{-a}^{-a+t} (a^2 - x^2)\,\mathrm{d}x = \pi \int_0^t (2au - u^2)\,\mathrm{d}u = \pi\Big(at^2 - \frac{t^3}{3}\Big).$$

Therefore

$$\frac{V}{A} = \frac{\pi t^2\big(a - \frac{t}{3}\big)}{2\pi at} = \frac{t}{2} - \frac{t^2}{6a} =: f(t).$$

Ask yourself The ratio depends on how much has been eaten. Which value of t do we need to worry about?

The condition must hold after *any* number of slices have been eaten, so we need the largest value of f. Completing the square,

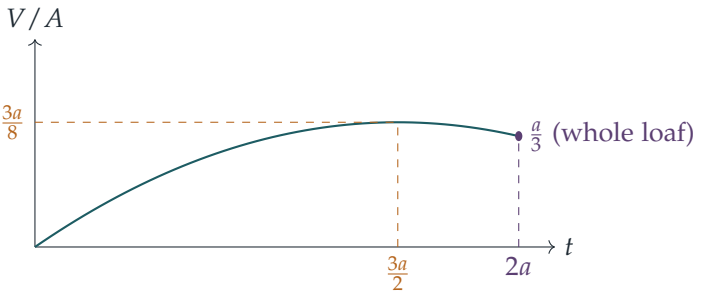
$$f(t) = \frac{3a}{8} - \frac{1}{6a}\Big(t - \frac{3a}{2}\Big)^2,$$

so f is largest when $t = \frac{3a}{2}$ (which lies in $(0, 2a]$), with maximum value $\frac{3a}{8}$.

Is $t = \frac{3}{2}a$ actually reached? It is if, for example, the loaf is cut into 4 slices and one is eaten (or into $4m$ slices and m are eaten). A loaf that is to be called crusty as a parallel-sliced loaf must satisfy the ruling for such a slicing, so it needs

$$\frac{3a}{8} < 1, \qquad \text{that is,} \qquad a < \frac{8}{3} = 2\tfrac{2}{3} \text{ metres},$$

as required. Conversely, if $a < \frac{8}{3}$ then $f(t) \leq \frac{3a}{8} < 1$ for every t, so the loaf is crusty however it is sliced.



The sketch shows $f(t)$ for $a = 2$: the ratio rises as we move towards the middle of the loaf and then falls, because near the far end we add a lot of crust but little bread.

Checks

- For the whole loaf, $t = 2a$ gives $f(2a) = a - \frac{2a}{3} = \frac{a}{3}$, which matches $\frac{4}{3}\pi a^3/(4\pi a^2)$.
- Take $a = \frac{8}{3}$ cut into 4 slices. With $1, 2, 3, 4$ slices left the ratio is $\frac{5}{9}, \frac{8}{9}, 1, \frac{8}{9}$, so after one slice is eaten the ratio is exactly 1, just failing the strict inequality.
- Numerically, quadrature of the area formula reproduced $2\pi a(x_2 - x_1)$ for random slices; and for $a = 2.66$ every slicing into up to 200 slices kept the ratio below 1.

Common slips

- Using only the whole loaf: $a/3 < 1$ gives $a < 3$, which is necessary but much weaker. The worst moment is when a quarter of the loaf's *width* has been eaten.
- Including the flat cut face in the crust. The crust is only the original curved surface, which is what the given formula measures.

Beyond STEP

The constant integrand is Archimedes' hat-box theorem: projecting a sphere sideways onto its circumscribing cylinder preserves area. That is the reason a point chosen uniformly on a sphere has a uniformly distributed height, a fact used constantly in probability and physics. Surface integrals like this one, and when maps preserve area, are developed in a first course on vector calculus; Needham, *Visual Complex Analysis*, has a beautiful geometric discussion of related maps of the sphere.

WORKED SOLUTION

STEP I 2001, Question 7

Pure • Concavity and Differential Equations

STEP I 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q7

The key idea

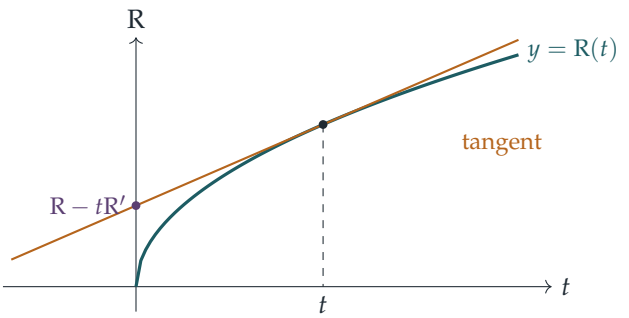
The three conditions say that R starts at the origin, rises, and bends downwards: its graph is *concave*. A concave graph lies below each of its tangents, and that single geometric fact gives the inequality in (i). Parts (ii) and (iii) are then separable differential equations, because $\mathrm{H} = \mathrm{R}'/\mathrm{R}$ is the derivative of $\ln \mathrm{R}$. Once we have R explicitly, we test it against the conditions, and the inequality from (i) gives a quick way to rule a model out.

Solution

First, a fact we use throughout: since $\mathrm{R}(0) = 0$ and R is strictly increasing for $t > 0$, we have $\mathrm{R}(t) > 0$ for $t > 0$. So $\mathrm{H}(t)$ is defined and $\mathrm{H}(t) > 0$ for $t > 0$.

Part (i)

The graph passes through the origin, has positive gradient, and the gradient decreases as t increases.



Fix $t > 0$. The tangent to the graph at time t has gradient $\mathrm{R}'(t)$, so it meets the vertical axis at height

$$\mathrm{R}(t) - t\,\mathrm{R}'(t).$$

The picture says this intercept is positive, because the curve, which passes through the origin, lies below the tangent. To make this rigorous, apply the mean value theorem to R on $[0, t]$: there is some c with $0 < c < t$ and

$$\mathrm{R}(t) - \mathrm{R}(0) = t\,\mathrm{R}'(c), \qquad \text{so} \qquad \mathrm{R}(t) = t\,\mathrm{R}'(c).$$

Since $\mathrm{R}'' < 0$, the function R' is strictly decreasing, and $c < t$ gives $\mathrm{R}'(c) > \mathrm{R}'(t)$. Hence

$$\mathrm{R}(t) > t\,\mathrm{R}'(t),$$

which is exactly the statement that the tangent's intercept is above the origin. Dividing by $\mathrm{R}'(t) > 0$,

$$t < \frac{\mathrm{R}(t)}{\mathrm{R}'(t)} = \frac{1}{\mathrm{H}(t)},$$

as required.

(Equivalently, the tangent meets the t-axis at $t - 1/\mathrm{H}(t)$, which is negative. In cosmology $1/\mathrm{H}$ is the "Hubble time", and this says that the universe is younger than its Hubble time in any decelerating model.)

Part (ii)

For $t > 0$, $\mathrm{R} > 0$, so we may write

$$\frac{\mathrm{d}}{\mathrm{d}t} \ln \mathrm{R}(t) = \frac{\mathrm{R}'(t)}{\mathrm{R}(t)} = \frac{a}{t}.$$

Integrating, $\ln \mathrm{R} = a \ln t + C$, so

$$\mathrm{R}(t) = k\,t^a \qquad (t > 0),$$

where $k = \mathrm{e}^C > 0$.

Now test the conditions, using $k > 0$ and $t^a > 0$:

- $\mathrm{R}'(t) = ka\,t^{a-1}$, which is positive for $t > 0$ if and only if $a > 0$.
- $\mathrm{R}''(t) = ka(a-1)\,t^{a-2}$. Given $a > 0$, this is negative if and only if $a < 1$.
- For $a > 0$, $kt^a \to 0$ as $t \to 0^+$, so $\mathrm{R}(0) = 0$ is consistent. (For $a \le 0$ it would fail anyway: kt^a tends to $k \neq 0$ or blows up.)

So the conditions $(*)$ hold exactly when

$$0 < a < 1.$$

This agrees with part (i): there $t < 1/\mathrm{H} = t/a$, which forces $a < 1$.

Part (iii)

Now $\mathrm{H}(t) = b/t^2$. First, $\mathrm{H}(t) > 0$ for $t > 0$ (from the opening remark), so we need $b > 0$. If $b \leq 0$ we are already done.

So suppose $b > 0$. By part (i), every $t > 0$ must satisfy

$$t < \frac{1}{\mathrm{H}(t)} = \frac{t^2}{b}, \qquad \text{that is,} \qquad t > b$$

(dividing by $t > 0$ and multiplying by $b > 0$). But this fails for every t with $0 < t \leq b$. So no value of b is consistent with $(*)$.

Ask yourself Is it legitimate to use part (i) like this?

Yes. Part (i) proved that *any* function satisfying $(*)$ has $t < 1/\mathrm{H}(t)$ for all $t > 0$. So if a proposed H breaks that inequality for even one t, no function with that H can satisfy $(*)$. This is a proof by contradiction.

For a direct confirmation, solve the equation as in (ii): $\ln \mathrm{R} = -b/t + C$, so $\mathrm{R}(t) = k\,\mathrm{e}^{-b/t}$ with $k > 0$. Then

$$\mathrm{R}'(t) = \frac{kb}{t^2}\,\mathrm{e}^{-b/t},$$

$$\mathrm{R}''(t) = kb\,\mathrm{e}^{-b/t}\left(\frac{b}{t^4} - \frac{2}{t^3}\right) = \frac{kb\,\mathrm{e}^{-b/t}}{t^4}\,(b - 2t).$$

With $b > 0$, $\mathrm{R}' > 0$ and $\mathrm{R}(t) \to 0$ as $t \to 0^+$, so the first two conditions hold. But $\mathrm{R}''(t) > 0$ for $0 < t < b/2$, so the graph bends the wrong way near the origin, and $(*)$ fails.

Checks

- In (ii), $a = \frac{1}{2}$ gives $\mathrm{R} = k\sqrt{t}$, the shape in the sketch: increasing and concave, and $t < 1/\mathrm{H} = 2t$. Taking $a = 1$ gives a straight line through the origin, where $t = 1/\mathrm{H}$ exactly: the boundary case of (i).
- In (iii) with $b = 1$, $k = 1$: numerically $\mathrm{R}''(0.25) \approx 2.34 > 0$, while $1/\mathrm{H}(0.5) = 0.25 < 0.5$, both breaking $(*)$.

The computer checked (i) on several concave test functions, the solutions and the derivative formulas in (ii) and (iii) by finite differences, and the sign change of R'' at $t = b/2$.

Common slips

- In (i), arguing only from the picture. "The curve lies below its tangent" needs a reason: the mean value theorem, plus R' decreasing.

- In (ii), forgetting that $a > 0$ is needed for $\mathrm{R}' > 0$ (and for $\mathrm{R}(0) = 0$), and giving only $a < 1$.
- In (iii), claiming concavity fails for large t. The sign of $b - 2t$ shows the failure is at *small* t.

Beyond STEP

Part (i) is a special case of a central fact about convex and concave functions: a differentiable function is concave exactly when its graph lies below all its tangents. This underlies Jensen's inequality and much of optimisation. The mean value theorem argument is standard first-year analysis; see Abbott, *Understanding Analysis*. In real cosmology, $a = \frac{2}{3}$ is the matter-dominated Einstein–de Sitter universe, for which the age is two-thirds of the Hubble time.

WORKED SOLUTION

STEP I 2001, Question 8

Pure • Second-Order Linear Equations

STEP I 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q8

The key idea

Each known solution, substituted into the equation, gives one equation linking the two unknown functions $p(x)$ and $q(x)$ at each value of x. Two solutions give two simultaneous linear equations, which we solve for p and q exactly as we would for two numbers. In the final part, the linearity proved in the middle part lets us replace the two given solutions with simpler combinations of them, and the trigonometric identities make those combinations very simple indeed.

Solution

Finding p and q

For $y = x$ we have $y' = 1$ and $y'' = 0$, so substituting into $(*)$ gives

$$p(x) + x\,q(x) = 0. \tag{1}$$

For $y = 1 - x^2$ we have $y' = -2x$ and $y'' = -2$, so

$$-2 - 2x\,p(x) + (1 - x^2)\,q(x) = 0. \tag{2}$$

From (1), $p(x) = -x\,q(x)$. Substituting this into (2),

$$-2 + 2x^2 q(x) + (1 - x^2)q(x) = 0 \quad \Longrightarrow \quad (1 + x^2)\,q(x) = 2.$$

Since $1 + x^2 > 0$ we may divide, so $q(x) = 2(1 + x^2)^{-1}$, and then (1) gives $p(x) = -x\,q(x) = -2x(1 + x^2)^{-1}$, as required. (We never divided by x, so this holds for every real x, including $x = 0$.)

Any combination $ax + b(1 - x^2)$ is a solution

Write $L[y] = y'' + p\,y' + q\,y$. Differentiation is linear, so if $y = ay_1 + by_2$ with a, b constant, then

$$\begin{aligned} L[ay_1 + by_2] &= a(y_1'' + p\,y_1' + q\,y_1) + b(y_2'' + p\,y_2' + q\,y_2) \\ &= a\,L[y_1] + b\,L[y_2]. \end{aligned}$$

Taking $y_1 = x$ and $y_2 = 1 - x^2$, both $L[y_1]$ and $L[y_2]$ are zero (that is what it means for them to satisfy $(*)$), so $L[ax + b(1 - x^2)] = 0$ for all constants a and b. If you prefer to see it directly: $y = ax + b(1 - x^2)$ has $y' = a - 2bx$ and $y'' = -2b$, and

$$\begin{aligned} y'' + p\,y' + q\,y &= -2b - \frac{2x(a - 2bx)}{1 + x^2} + \frac{2\big(ax + b - bx^2\big)}{1 + x^2} \\ &= \frac{-2b(1 + x^2) - 2ax + 4bx^2 + 2ax + 2b - 2bx^2}{1 + x^2} = 0. \end{aligned}$$

The final part

The argument we just gave used nothing about the particular p and q: for *any* equation of the form $(*)$, a combination $ay_1 + by_2$ of two solutions is again a solution. So if $\cos^2(\frac{1}{2}x^2)$ and $\sin^2(\frac{1}{2}x^2)$ both satisfy $(*)$, so do their sum and difference,

$$\cos^2(\tfrac{1}{2}x^2) + \sin^2(\tfrac{1}{2}x^2) = 1, \qquad \cos^2(\tfrac{1}{2}x^2) - \sin^2(\tfrac{1}{2}x^2) = \cos(x^2),$$

by $\cos^2\theta + \sin^2\theta = 1$ and $\cos^2\theta - \sin^2\theta = \cos 2\theta$ with $\theta = \frac{1}{2}x^2$.

Ask yourself Why is it fair to work with 1 and $\cos(x^2)$ instead of the functions we were given?

Because the step is reversible: the given functions are $\frac{1}{2}\big(1 + \cos x^2\big)$ and $\frac{1}{2}\big(1 - \cos x^2\big)$, so p and q make both of them solutions exactly when they make both 1 and $\cos(x^2)$ solutions. We lose no information, and the derivatives become much easier.

The solution $y = 1$. Here $y' = y'' = 0$, so $(*)$ reduces to $q(x) \cdot 1 = 0$, giving

$$q(x) = 0.$$

The solution $y = \cos(x^2)$. By the chain rule $y' = -2x\sin(x^2)$, and by the product rule

$$y'' = -2\sin(x^2) - 4x^2\cos(x^2).$$

With $q = 0$, $(*)$ becomes

$$-2\sin(x^2) - 4x^2\cos(x^2) - 2x\sin(x^2)\,p(x) = 0.$$

Wherever $x\sin(x^2) \neq 0$ we can divide, and

$$p(x) = -\frac{\sin(x^2) + 2x^2\cos(x^2)}{x\sin(x^2)} = -\frac{1}{x} - 2x\cot(x^2).$$

So $p(x) = -\dfrac{1}{x} - 2x\cot(x^2)$ and $q(x) = 0$.

At the points where $x\sin(x^2) = 0$ (that is, $x = 0$ and $x = \pm\sqrt{n\pi}$) the coefficient of p vanishes, so the equation says nothing about p there; indeed our p blows up at these points. This is not a slip: at such points both given functions have zero derivative, so they cannot be two "independent" solutions of a well-behaved equation of the form $(*)$, and p must be singular there.

Checks

- As an independent check of the first part, the *Wronskian* $W = y_1y_2' - y_2y_1'$ of two solutions satisfies $W' = -pW$. With $y_1 = x$, $y_2 = 1 - x^2$ we get $W = -2x^2 - (1 - x^2) = -(1 + x^2)$, so $-p = W'/W = 2x/(1 + x^2)$, agreeing with our p. For the final part, the solutions 1 and $\cos(x^2)$ give $W = -2x\sin(x^2)$, and $W'/W = 1/x + 2x\cot(x^2) = -p$ again.
- Numerically, at $x = 0.9$ the two given functions, substituted into $(*)$ with derivatives computed by finite differences, force $p(0.9) \approx -2.82465$ and $q(0.9) \approx 0$, matching $-1/0.9 - 1.8\cot(0.81) \approx -2.82465$.

Common slips

- Differentiating $\cos^2(\frac{1}{2}x^2)$ directly, twice, and getting lost in the algebra. Double-angle identities first, then differentiate.
- Dividing (1) by x to get $q = -p/x$; this works, but then you must deal with $x = 0$ separately. Eliminating p instead avoids the issue.
- Forgetting to say where the formula for p in the final part is valid.

Beyond STEP

This question is a glimpse of the theory of second-order linear differential equations: solutions form a two-dimensional vector space, any two independent solutions determine the equation, and the Wronskian (with Abel's formula $W' = -pW$) measures independence. Points where the Wronskian vanishes, like $x = \pm\sqrt{n\pi}$ here, are exactly where the coefficients must become singular. Boyce and DiPrima, *Elementary Differential Equations*, develops all of this carefully.

WORKED SOLUTION

STEP I 2001, Question 9

Mechanics • Interception and Relative Velocity

STEP I 2001, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q9

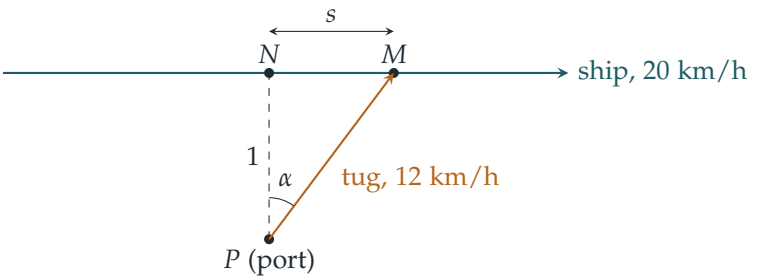
The key idea

Leaving as late as possible means the tug must go flat out in a straight line, so the only real choice is *where* on the ship's track to meet it. For each possible meeting point, work out the latest time the tug could leave and still be there in time. Then maximise that departure time over the meeting point. It's a one-variable optimisation once the set-up is right.

Solution

Setting up

Treat the ship and the tug as particles. Put the port P at the origin, and let the ship travel along the line $y = 1$ in the positive x direction (distances in km, times in hours). The foot of the perpendicular from P to the track is $N = (0, 1)$, the ship's closest point to the port.



Suppose the meeting takes place at the point $M = (s, 1)$. The ship's position along the track is a linear function of time, so it reaches M at time

$$t_M = \frac{s - c}{20},$$

where c is a constant fixed by where the ship starts. (The ship is still approaching, so c is less than any meeting position we will need.)

Ask yourself Why must the tug travel at full speed in a straight line?

If the tug is at M at time t_M having left at time T, it has covered at least the straight-line distance $PM = \sqrt{s^2+1}$ at no more than 12 km/h, so $t_M - T \geq \sqrt{s^2+1}/12$. Equality is attained by heading straight for M at top speed. Anything slower or less direct forces an earlier departure.

So the latest departure time that allows a meeting at M is

$$T(s) = \frac{s-c}{20} - \frac{\sqrt{s^2+1}}{12},$$

and we want the largest value of $T(s)$ over all s.

Maximising the departure time

Differentiating,

$$T'(s) = \frac{1}{20} - \frac{s}{12\sqrt{s^2+1}}.$$

Setting $T'(s) = 0$ gives

$$\frac{s}{\sqrt{s^2+1}} = \frac{12}{20} = \frac{3}{5}.$$

This needs $s > 0$. Squaring, $25s^2 = 9(s^2+1)$, so $s^2 = \frac{9}{16}$ and $s = \frac{3}{4}$.

This is a maximum, and a global one: $\sqrt{s^2+1}$ is convex, so T (a linear function minus a convex one) is concave. Equivalently, $\dfrac{s}{\sqrt{s^2+1}}$ is increasing in s, so $T'(s) > 0$ for $s < \frac{3}{4}$ and $T'(s) < 0$ for $s > \frac{3}{4}$. Note that the constant c has dropped out: the meeting point doesn't depend on where the ship started.

The tug therefore meets the ship $\frac{3}{4}$ km beyond N, and the distance it travels is

$$PM = \sqrt{\tfrac{9}{16}+1} = \sqrt{\tfrac{25}{16}} \quad\Longrightarrow\quad \boxed{PM = \tfrac{5}{4}\text{ km} = 1.25\text{ km}.}$$

The tug is at sea for $\frac{5/4}{12} = \frac{5}{48}$ hours, which is $6\frac{1}{4}$ minutes.

A geometric reading

The tug's heading makes angle α with PN, where $\sin\alpha = \frac{s}{\sqrt{s^2+1}} = \frac{3}{5}$, so $\sin\alpha$ is the ratio of the speeds, $12/20$. The tug's velocity is $(7.2, 9.6)$, and its velocity relative to the ship is $(7.2-20, 9.6) = (-12.8, 9.6)$. Their dot product is $-92.16 + 92.16 = 0$. At the critical departure time, the tug's velocity is perpendicular to its velocity relative to the ship. This is the standard condition for a slower pursuer that only just manages an interception.

Checks

- A brute-force search over a fine grid of meeting points, for several different starting positions of the ship, always puts the maximum of $T(s)$ at $s = 0.75$, so the tug's distance is 1.25 km.
- Stepping the motion numerically from the latest departure time with heading $(\frac{3}{5}, \frac{4}{5})$, the tug reaches the ship. Leaving just 0.002 hours later, no heading gets within 20 m of the ship.
- Sanity check: if the tug were as fast as the ship, $\sin\alpha$ would be 1, and the meeting point would run off to infinity. That makes sense, since a tug of equal speed can chase the ship indefinitely.

Common slips

- Meeting the ship at its closest point N (tug distance 1 km). That minimises the tug's *distance*, not its departure time: by going a little further forward, the tug gains time because the ship still has to sail there.
- Minimising the tug's travel time $\sqrt{s^2+1}/12$ without the $s/20$ term. The quantity to maximise is the departure time, which involves both boats.

Beyond STEP

Maximising a difference of arrival times is a small instance of an optimal-control or pursuit problem. The ratio-of-speeds condition $\sin\alpha = 12/20$ is the same geometry as the Mach cone behind a supersonic aircraft: the sine of the cone's half-angle is the ratio of the sound speed to the aircraft's speed, and the slow tug here plays the part of sound chasing a fast source. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative velocity and frames carefully, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation in mechanics.

WORKED SOLUTION

STEP I 2001, Question 10

Mechanics • Projectile Time of Flight

STEP I 2001, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q10

The key idea

Split the choice into two parts: the elevation fixes both the time of flight and the horizontal range, and the direction only decides *where* on a circle round the gun the shell comes down. So the shell can hit the line exactly when its range is at least d. The time of flight grows with the elevation but, beyond $45°$, the range shrinks, so we push the elevation up until the range has fallen to exactly d.

Solution

Time of flight and range for a given elevation

Model the shell as a particle moving under gravity alone (no air resistance), fired from ground level and landing on the plain. Fire it at elevation α, where $0 < \alpha \leq \frac{\pi}{2}$. Its vertical velocity component is $v\sin\alpha$ and its horizontal component is $v\cos\alpha$.

Vertically, its height after time t is $vt\sin\alpha - \frac{1}{2}gt^2$, which is zero again when

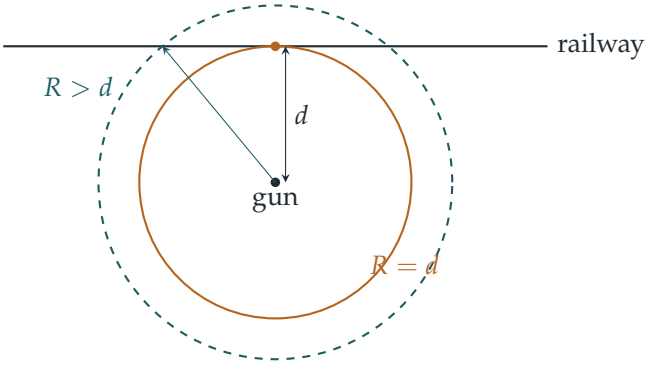
$$T = \frac{2v\sin\alpha}{g}.$$

In that time it travels a horizontal distance

$$R = v\cos\alpha \cdot \frac{2v\sin\alpha}{g} = \frac{v^2\sin 2\alpha}{g}.$$

When can the shell hit the line?

For a fixed elevation, turning the gun moves the landing point round a circle of radius R centred on the gun. The nearest point of the line is at distance d, and every other point of the line is further away. So the circle meets the line if and only if $R \geq d$. When $R = d$ the circle just touches the line, and the gun must point straight at the nearest point, perpendicular to the line.



(Seen from above. The dashed circle is a lower elevation, which reaches the line at two points. The solid circle is the optimal elevation, which just reaches it.)

So the gunner can hit the line at elevation α exactly when

$$\sin 2\alpha \geq \frac{gd}{v^2}.$$

Write $k = gd/v^2$. We're told $v^2 > gd$, so $0 < k < 1$ and some elevations do work. (The greatest range is v^2/g, at 45°, so this condition just says that the line is within range.)

Maximising the time of flight

Since $T = 2v\sin\alpha/g$ and $\sin\alpha$ is increasing on $(0, \frac{\pi}{2}]$, maximising T means choosing the *largest* allowed α.

For $0 < 2\alpha \leq \pi$, the condition $\sin 2\alpha \geq k$ holds exactly when

$$\arcsin k \leq 2\alpha \leq \pi - \arcsin k.$$

So the largest allowed elevation is $\alpha = \frac{\pi}{2} - \frac{1}{2}\arcsin k$, which is more than 45°. At this elevation $\sin 2\alpha = k$, and since 2α lies in $(\frac{\pi}{2}, \pi)$, $\cos 2\alpha$ is negative:

$$\cos 2\alpha = -\sqrt{1 - k^2}.$$

Ask yourself Wouldn't it be better to aim along the line at some distant point, so the shell flies further?

No. Flying further needs a *larger* range, and above 45° that means a *lower* elevation and so a shorter flight. Time of flight depends only on elevation, so the best we can do is the steepest shot that still reaches the line. Its range is exactly d, so the shell lands at the foot of the perpendicular from the gun to the line.

Now we square T and use $2\sin^2\alpha = 1 - \cos 2\alpha$:

$$\begin{aligned} g^2T^2 &= 4v^2\sin^2\alpha \\ &= 2v^2\left(1-\cos 2\alpha\right) \\ &= 2v^2 + 2v^2\sqrt{1-\frac{g^2d^2}{v^4}} \\ &= 2v^2 + 2\left(v^4 - g^2d^2\right)^{1/2}, \end{aligned}$$

taking v^2 inside the square root (allowed because $v^2 > 0$). This is the required result.

Checks

- As $d \to 0$ the formula gives $g^2T^2 \to 4v^2$, so $T \to 2v/g$. That's firing straight up, the longest possible flight, which is right when the line passes right by the gun.
- When $v^2 = gd$ the line is only just in range: the only possible shot is at 45°, and the formula gives $g^2T^2 = 2v^2$, which matches $T = 2v\sin 45^\circ/g = \sqrt{2}\,v/g$.
- Numerically, with $g = 9.8\,\mathrm{m\,s^{-2}}$, $d = 100\,\mathrm{m}$ and $v = 40\,\mathrm{m\,s^{-1}}$, the best elevation is about 71.1° and the formula gives $T \approx 7.72\,\mathrm{s}$. Stepping the motion numerically gives the same time, with the shell landing on the line. A brute-force search over all elevations agrees with the formula in hundreds of random cases.

Common slips

- Taking the lower root, $\alpha = \frac{1}{2}\arcsin k$. That shot also hits the line, but it gives the *shortest* time of flight, not the longest: in effect, taking the wrong sign for $\cos 2\alpha$.
- Assuming the shell must land at the nearest point of the line from the start. That turns out to be true, but it needs the argument above: any other point needs a larger range, and so a lower elevation.

Beyond STEP

This is a small constrained optimisation: maximise one quantity (flight time) subject to a constraint (reaching the line), and the optimum lies on the boundary of what's allowed. In first-year mechanics and optimisation courses you'll meet the general tools for this, such as Lagrange multipliers, and the related idea of the *envelope* of all trajectories (the "parabola of safety"), which bounds every point the gun can reach. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your projectile and particle mechanics.

WORKED SOLUTION

STEP I 2001, Question 11

Mechanics • Elastic Strings and Energy

STEP I 2001, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

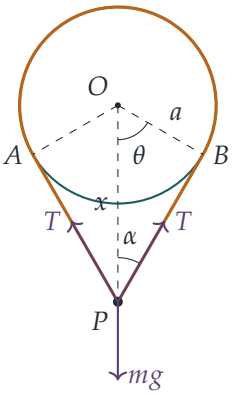
step.maths.org/questions/01-s1-q11

The key idea

Once the particle pulls the band down, the band stays on the cylinder over most of its length and leaves it *tangentially* on each side, running straight to the particle. So everything comes down to the geometry of two tangents from an external point to a circle. Because the cylinder is smooth, the tension is the same all the way round the band, so Hooke's law with the *total* extension gives it. Then part (i) is an energy question (the particle is momentarily at rest at its lowest point) and part (ii) is a force question (maximum speed means zero acceleration).

Solution

We model the particle as a point mass, the band as light and elastic (obeying Hooke's law), and the cylinder as smooth. Let O be the centre of the cross-section and suppose the particle P is a distance x below O, with $x \geq a$. The band leaves the circle at points A and B, where PA and PB are tangents.



The geometry and the tension

Triangle OAP has a right angle at A (a tangent is perpendicular to the radius), so

$$PA = PB = \sqrt{x^2 - a^2}, \qquad \cos\theta = \frac{a}{x}, \qquad \sin\alpha = \frac{a}{x},$$

where $\theta = \angle AOP$ and $\alpha = \angle APO$, the angle each straight piece of band makes with the vertical. The band is in contact with the circle everywhere except on the arc AB of angle 2θ nearest P, so its length is

$$L = (2\pi a - 2a\theta) + 2\sqrt{x^2 - a^2}.$$

The natural length is $2\pi a$, so the extension is

$$e(x) = 2\left(\sqrt{x^2 - a^2} - a\theta\right), \qquad \theta = \arccos\frac{a}{x}.$$

Since the cylinder is smooth, there is no friction along the band, so the tension T is the same throughout, and by Hooke's law $T = \dfrac{\lambda e}{2\pi a}$.

At $x = 2a$ we have $\cos\theta = \frac{1}{2}$, so $\theta = \frac{\pi}{3}$, $\alpha = \frac{\pi}{6}$ and $\sqrt{x^2 - a^2} = \sqrt{3}\,a$. Hence

$$e(2a) = 2\left(\sqrt{3}\,a - \frac{\pi a}{3}\right) = \frac{2a}{3}\left(3\sqrt{3} - \pi\right).$$

Part (i)

At release the particle is at the bottom of the cylinder, $x = a$, the band is unstretched and the particle is at rest. At $x = 2a$ it is at rest again (it falls no further), so it has no kinetic energy at either moment. The cylinder is smooth, so no energy is lost, and the loss of gravitational potential energy equals the gain in elastic energy. The particle has fallen a distance $2a - a = a$, and the elastic energy of a band of natural length $2\pi a$ with extension e is $\dfrac{\lambda e^2}{2 \cdot 2\pi a}$. So

$$mga = \frac{\lambda}{4\pi a} \cdot \frac{4a^2}{9}\left(3\sqrt{3} - \pi\right)^2 = \frac{\lambda a\left(3\sqrt{3} - \pi\right)^2}{9\pi},$$

which gives

$$\lambda = \frac{9\pi mg}{\left(3\sqrt{3} - \pi\right)^2},$$

as required.

Ask yourself Is it fair to use the elastic energy formula for the whole band, when part of it is wrapped round the cylinder?

Yes. Because the cylinder is smooth, the tension, and so the stretch per unit length, is the same along the whole band, wrapped or not. The band behaves exactly like a single elastic string of natural length $2\pi a$ with total extension e, and that is what the formula $\lambda e^2/(2l)$ describes.

Part (ii)

The speed is greatest when the acceleration is zero, that is, when the resultant force on the particle is zero. Horizontally the two tensions balance by symmetry. Resolving vertically at P,

$$2T\cos\alpha = mg.$$

At $x = 2a$, $\cos\alpha = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$, so $\sqrt{3}\,T = mg$, giving $T = \dfrac{mg}{\sqrt{3}}$. By Hooke's law with the extension found above,

$$T = \frac{\lambda}{2\pi a}\cdot\frac{2a}{3}\left(3\sqrt{3}-\pi\right) = \frac{\lambda\left(3\sqrt{3}-\pi\right)}{3\pi}.$$

Equating the two expressions for T,

$$\lambda = \frac{3\pi mg}{\sqrt{3}\left(3\sqrt{3}-\pi\right)} = \frac{\sqrt{3}\,\pi mg}{3\sqrt{3}-\pi}.$$

Ask yourself How do we know this is a maximum of the speed and not just a stationary point?

As x increases, e increases (each piece of band gets longer faster than the arc $a\theta$ shrinks) and $\cos\alpha = \sqrt{1-a^2/x^2}$ increases, so the upward force $2T\cos\alpha$ increases steadily from 0. The net downward force is positive above $x = 2a$ and negative below it, so the particle speeds up until $x = 2a$ and slows down afterwards.

Checks

- A neat consistency check: differentiating L gives $\dfrac{\mathrm{d}L}{\mathrm{d}x} = \dfrac{2x}{\sqrt{x^2-a^2}} - \dfrac{2a^2}{x\sqrt{x^2-a^2}} = \dfrac{2\sqrt{x^2-a^2}}{x} = 2\cos\alpha$. So the rate of change of elastic energy, $T\dfrac{\mathrm{d}L}{\mathrm{d}x}$, equals $2T\cos\alpha$, the upward force. The energy method of (i) and the force method of (ii) agree.
- Numerically, $\lambda \approx 6.70\,mg$ in (i) and $\lambda \approx 2.65\,mg$ in (ii). The band in (ii) is weaker, which makes sense: there the particle is still moving at $x = 2a$ and carries on past it, whereas in (i) a stiffer band has already brought it to rest.
- Stepping the equation of motion $m\ddot{x} = mg - 2T\cos\alpha$ numerically from rest at $x = a$ gives a lowest point at $x = 2.0000a$ for the first value of λ, and maximum speed at $x = 2.0000a$ for the second.

Common slips

- Forgetting that some of the band leaves the cylinder: the extension is $2\sqrt{x^2-a^2}$ *minus* the lost arc $2a\theta$, not just the length of the two straight pieces.

- Measuring the fall from the axis instead of from the release point: the particle falls a distance a, not $2a$.
- Using $\sin\alpha$ instead of $\cos\alpha$ when resolving vertically; α is measured from the vertical, and $\sin\alpha = a/x$.

Beyond STEP

The check that $\mathrm{d}(\text{elastic energy})/\mathrm{d}x$ equals the restoring force is an instance of the principle of virtual work: in a conservative system the force in any direction is minus the rate of change of potential energy in that direction. This viewpoint, where you describe a system by one coordinate and its potential energy, is the starting point of Lagrangian mechanics, which you will meet in the first or second year of a university mechanics course. For a careful treatment of energy, forces and potentials, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2001, Question 12

Probability and Statistics • Tree Diagrams

STEP I 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q12

The key idea

Everything depends on keeping careful track of *who knows what*. A student knows that someone has heard the rumour only if that person told them, or they told that person; nobody passes on where they got it from. Once we record each student's knowledge at every stage, the tree is small: every branching is a choice between two people, so it has only eight paths, and they all turn out to be equally likely.

Solution

Setting up

Write A, B, C, D for the four students. At any stage, the people whom student X *knows* have heard the rumour are: X themselves, whoever has told X the rumour, and whoever X has told it to. When X hears the rumour, X chooses uniformly at random among the people outside that set. If that set already contains all four students, the rumour stops with X.

At the start, A has told C, so A knows $\{A, C\}$ and C knows $\{A, C\}$.

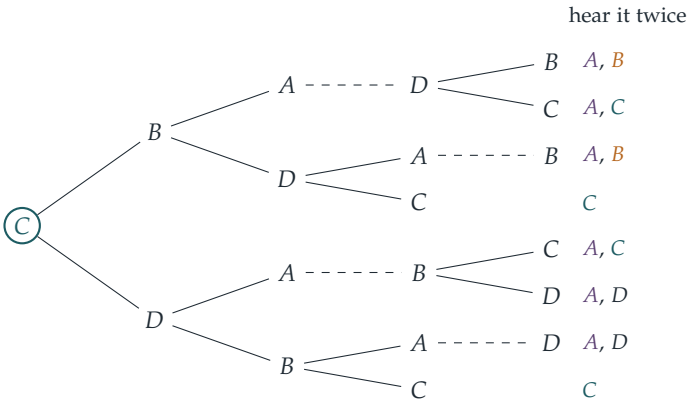
The tree

C omits A and tells B or D, each with probability $\frac{1}{2}$. Take the branch $C \to B$ first. Now C knows $\{A, B, C\}$ and B knows $\{B, C\}$ (not A, because C didn't say where the rumour came from). So B tells A or D, each with probability $\frac{1}{2}$.

- $B \to A$. Arthur now knows $\{A, B, C\}$, so he must tell D, and Arthur then knows everyone has heard. D knows only $\{A, D\}$, so D tells B or C with probability $\frac{1}{2}$ each. Either way the hearer completes their set: B already knew $\{A, B, C\}$, as did C, and each now learns of D. So the rumour stops.
- $B \to D$. Now B knows $\{B, C, D\}$ and D knows $\{B, D\}$, so D tells A or C.

- $D \to C$: C already knew $\{A, B, C\}$ and now learns of D, so the rumour stops. Arthur never rehears it.
- $D \to A$: Arthur knows $\{A, C, D\}$, so he tells B, who then knows all four. The rumour stops.

The branch $C \to D$ is the same with the roles of B and D swapped. Here is the whole tree. Every branch carries probability $\frac{1}{2}$ except the forced moves (probability 1), shown dashed.



Each path to a leaf contains exactly three choices with probability $\frac{1}{2}$ (the forced moves contribute a factor of 1), so each of the eight outcomes has probability $\frac{1}{8}$. The labels on the right show who hears the rumour twice on that path. (On every path a student hears it at most twice.)

Ask yourself Why does B, having just heard from C, not know that Arthur has heard it?

Because C doesn't say who told them. All B can deduce is that C has heard it. This is what keeps the rumour going: in the $C \to B \to A$ branch, Arthur is told something he already knew.

Arthur rehears it

Arthur rehears the rumour on every path except the two short ones, $C \to B \to D \to C$ and $C \to D \to B \to C$. So

$$\mathrm{P}(\text{Arthur rehears it}) = 1 - 2 \times \tfrac{1}{8} = \tfrac{3}{4},$$

as required.

Bertha and Chandra

Bertha hears it twice only on the paths $C \to B \to A \to D \to B$ and $C \to B \to D \to A \to B$, so

$$\mathrm{P}(\text{Bertha hears it twice}) = \tfrac{2}{8} = \tfrac{1}{4}.$$

Chandra hears it twice on four paths: two in each half of the tree ($C \to B \to A \to D \to C$, $C \to B \to D \to C$, $C \to D \to A \to B \to C$ and $C \to D \to B \to C$), so

$$\mathrm{P}(\text{Chandra hears it twice}) = \tfrac{4}{8} = \tfrac{1}{2}.$$

Chandra is more likely than Bertha to hear it again because Chandra heard it early: the people told later in the chain don't know that Chandra has heard it, so Chandra stays a possible target right to the end.

Checks

By symmetry between B and D, the probability that Delilah hears it twice should also be $\frac{1}{4}$, and the tree gives two such paths. The outcomes sum correctly: eight equally likely leaves, total probability 1. An exact enumeration of the process by computer gives $\frac{3}{4}$, $\frac{1}{4}$ and $\frac{1}{2}$, and a simulation of 200 000 rumours gave 0.750, 0.250 and 0.501.

Common slips

- Letting the hearer know everyone further back in the chain. Only the teller's identity is passed on, and with full knowledge the rumour would never return to Arthur.
- Forgetting that the *teller* also learns something: after telling Y, X knows that Y has heard it, which is what forces Arthur's moves.
- Stopping the tree as soon as Arthur rehears it. The later branches are needed for Bertha and Chandra.

Beyond STEP

The rumour is a random walk on the four students whose rules depend on the history, recorded here as each student's knowledge. Making the state rich enough that the next step depends only on the present state turns it into a *Markov chain*, the central object of a first course in random processes; rumour and epidemic models are a classic application. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains at exactly the right level.

WORKED SOLUTION

STEP I 2001, Question 13

Probability and Statistics • Conditional Poisson Distribution

STEP I 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q13

The key idea

Conditioning on the total number of breakages turns a Poisson question into a binomial one. If the four students break plates independently at the same rate, then once we know how many plates were broken altogether, each broken plate is equally likely to have been broken by any of the four, independently of the others. So the mathematician's share of the broken plates is binomial, and the unknown λ disappears.

Solution

Setting up

Let M be the number of plates broken by the mathematician, and let R be the total number broken by the other three students. We are told that each student's breakages are $\mathrm{Po}(\lambda)$, independently of the others. So $M \sim \mathrm{Po}(\lambda)$, and R, being a sum of three independent $\mathrm{Po}(\lambda)$ variables, is $\mathrm{Po}(3\lambda)$. Also M and R are independent, because R depends only on the other three students. The total is $T = M + R \sim \mathrm{Po}(4\lambda)$.

? **Ask yourself** Why is a sum of independent Poisson variables Poisson again?

For independent $X \sim \mathrm{Po}(\alpha)$ and $Y \sim \mathrm{Po}(\beta)$, sum over the ways of splitting n:

$$\mathrm{P}(X+Y=n) = \sum_{k=0}^{n} \frac{\mathrm{e}^{-\alpha}\alpha^k}{k!} \cdot \frac{\mathrm{e}^{-\beta}\beta^{n-k}}{(n-k)!} = \frac{\mathrm{e}^{-(\alpha+\beta)}}{n!} \sum_{k=0}^{n} \binom{n}{k} \alpha^k \beta^{n-k} = \frac{\mathrm{e}^{-(\alpha+\beta)}(\alpha+\beta)^n}{n!},$$

by the binomial theorem. So $X + Y \sim \mathrm{Po}(\alpha+\beta)$. Applying this twice gives $R \sim \mathrm{Po}(3\lambda)$, and once more gives $T \sim \mathrm{Po}(4\lambda)$.

The conditional distribution of M

For $m = 0, 1, \ldots, 5$, the event $\{M = m,\ T = 5\}$ is the same as $\{M = m,\ R = 5 - m\}$. By independence of M and R,

$$\begin{aligned}\mathrm{P}(M = m \mid T = 5) &= \frac{\mathrm{P}(M = m)\,\mathrm{P}(R = 5 - m)}{\mathrm{P}(T = 5)} \\ &= \frac{\dfrac{\mathrm{e}^{-\lambda}\lambda^m}{m!} \cdot \dfrac{\mathrm{e}^{-3\lambda}(3\lambda)^{5-m}}{(5-m)!}}{\dfrac{\mathrm{e}^{-4\lambda}(4\lambda)^5}{5!}} \\ &= \frac{5!}{m!\,(5-m)!} \cdot \frac{3^{5-m}}{4^5} = \binom{5}{m}\left(\tfrac{1}{4}\right)^m\left(\tfrac{3}{4}\right)^{5-m}.\end{aligned}$$

The exponentials cancel, and so does every power of λ (there are λ^5 on top and bottom). So, given $T = 5$, we have $M \sim \mathrm{B}(5, \frac{1}{4})$, whatever the value of λ.

The answer

Summing the binomial probabilities for $m = 3, 4, 5$:

$$\begin{aligned}\mathrm{P}(M \geq 3 \mid T = 5) &= \binom{5}{3}\frac{3^2}{4^5} + \binom{5}{4}\frac{3}{4^5} + \binom{5}{5}\frac{1}{4^5} \\ &= \frac{10 \times 9 + 5 \times 3 + 1}{1024} = \frac{106}{1024} = \frac{53}{512}.\end{aligned}$$

So the probability that the mathematician broke three or more of the five plates is $\dfrac{53}{512} \approx 0.104$.

Checks

- The six conditional probabilities $\binom{5}{m}3^{5-m}/4^5$ add up to $(1+3)^5/4^5 = 1$, as they must.
- Symmetry: by the same argument each student's expected share of the five plates is $5 \times \frac{1}{4} = \frac{5}{4}$, and the four shares add up to 5.
- We computed $\mathrm{P}(M \geq 3 \mid T = 5)$ directly from the four Poisson distributions (summing over every way of sharing five breakages among four students) for $\lambda = 0.3, 1, 2.7$ and 5, and got $0.103515625 = 53/512$ each time. A simulation of 4×10^5 rounds of washing up gave about 0.105 among the rounds with exactly five breakages.

Common slips

- Leaving λ in the answer. It must cancel; if it doesn't, the conditioning has gone wrong.

- Dividing by $\mathrm{P}(M+R=5)$ but treating R as $\mathrm{Po}(\lambda)$ instead of $\mathrm{Po}(3\lambda)$, which would amount to only two students.
- Forgetting the binomial coefficients, or treating the event as "exactly three" rather than "three or more".

Beyond STEP

This is a small instance of a general fact: if $X_1, \ldots, X_k$ are independent Poisson variables with means $\lambda_1, \ldots, \lambda_k$, then given their sum is n the vector $(X_1, \ldots, X_k)$ is multinomial with probabilities $\lambda_i / \sum \lambda_j$. Run backwards, it says that if each of a $\mathrm{Po}(\mu)$ number of events is independently labelled with one of k types, the counts of each type are independent Poisson variables. This "Poisson splitting" is a basic tool in the study of Poisson processes. Blitzstein and Hwang, *Introduction to Probability*, discusses both directions with many examples.

WORKED SOLUTION

STEP I 2001, Question 14

Probability and Statistics • Random Orderings

STEP I 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q14

This question has a full guided treatment in *Beyond STEP*, Section 12.12, *When to Stop Looking*: https://beyondstep.keithmcnulty.org/ch12/secretary.html

The key idea

Don't think about the whole interview order: think only about *where particular candidates sit*. The best of the first n is the best overall exactly when the top candidate is among the first n interviewed, and by symmetry the top candidate is equally likely to be in any of the N positions. Part (ii) then needs one extra case, or a neat complement.

Solution

Label the candidates by rank, so candidate 1 is the best overall, candidate 2 the second best, and so on. Since all $N!$ orders are equally likely, every candidate is equally likely to occupy any given position, and the set of candidates seen in the first n interviews is equally likely to be any of the $\binom{N}{n}$ subsets of size n. Throughout, $1 \leq n \leq N$.

Part (i)

Let A be the event that candidate 1 is among the first n interviewed. If A happens, candidate 1 is certainly the best of the first n; if it doesn't, the best of the first n is someone worse than candidate 1. So the event we want is exactly A.

Candidate 1 is equally likely to be in each of the N positions, and n of these are among the first n, so

$$\mathrm{P}(\text{best of first } n \text{ is best overall}) = \frac{n}{N}.$$

Part (ii)

Now take $N \geq 2$ so that there is a second-best candidate. The best of the first n is the best or second best overall if and only if

- candidate 1 is among the first n (event A), or
- candidate 1 is *not* among the first n, but candidate 2 is.

> **Ask yourself** In the second case, how do we know the best of the first n is candidate 2, and not someone even better?

The only candidate better than candidate 2 is candidate 1, who in this case was not interviewed early. Conversely, if neither 1 nor 2 is among the first n, the best of the first n is ranked 3 or worse. So the two cases above are exactly the event we want, and they are mutually exclusive.

The first case has probability $\frac{n}{N}$ from part (i). For the second, $\mathrm{P}(A') = \frac{N-n}{N}$, and given that candidate 1 is in one of the last $N-n$ positions, candidate 2 is equally likely to be in any of the remaining $N-1$ positions, n of which are among the first n. Hence

$$\begin{aligned}\mathrm{P}(\text{best of first } n \text{ is first or second overall}) &= \frac{n}{N} + \frac{N-n}{N} \cdot \frac{n}{N-1} \\ &= \frac{n\big[(N-1)+(N-n)\big]}{N(N-1)} \\ &= \frac{n(2N-n-1)}{N(N-1)}.\end{aligned}$$

Alternatively, use the complement: the event fails exactly when all n early candidates come from the $N-2$ candidates ranked 3 to N, which has probability

$$\frac{\binom{N-2}{n}}{\binom{N}{n}} = \frac{(N-n)(N-n-1)}{N(N-1)},$$

and $N(N-1) - (N-n)(N-n-1) = 2Nn - n^2 - n = n(2N-n-1)$, giving the same answer.

The case $N = 4$, $n = 2$

The formulas give $\frac{2}{4} = \frac{1}{2}$ for (i) and $\frac{2 \cdot 5}{4 \cdot 3} = \frac{5}{6}$ for (ii).

To list the possibilities, note that only *which* two candidates are interviewed first matters. Each of the $\binom{4}{2} = 6$ pairs arises from exactly $2! \times 2! = 4$ of the 24 orders (order the first two, then the last two), so the six pairs are equally likely:

First two interviewed	$\{1,2\}$	$\{1,3\}$	$\{1,4\}$	$\{2,3\}$	$\{2,4\}$	$\{3,4\}$
Best of these	1	1	1	2	2	3

The best of the first two is the best overall in 3 of the 6 cases, probability $\frac{1}{2}$, and is the best or second best in 5 of the 6 cases, probability $\frac{5}{6}$. Both agree with the formulas.

Checks

- $n = N$: we see everyone, so both probabilities should be 1. Indeed $\frac{N}{N} = 1$ and $\frac{N(N-1)}{N(N-1)} = 1$.
- $n = 1$: the first candidate is a uniformly random one, so (ii) should be $\frac{2}{N}$; the formula gives $\frac{2N-2}{N(N-1)} = \frac{2}{N}$.
- The answer to (ii) is at least the answer to (i), since $2N - n - 1 \geq N - 1$.
- A full enumeration of all $N!$ orders for $N \leq 7$ and every n agrees exactly with both formulas (for example $N = 5$, $n = 2$ gives $\frac{2}{5}$ and $\frac{7}{10}$).

Common slips

- Forgetting in (ii) that "second best overall" only counts when the best overall is *not* among the first n; simply adding $\frac{n}{N} + \frac{n}{N}$ double counts.
- Using $\frac{n}{N}$ for candidate 2 after conditioning on candidate 1's position: once candidate 1 is placed, only $N - 1$ positions remain.

Beyond STEP

This is the setting of the famous *secretary problem*: if you must accept or reject each candidate on the spot, the best strategy is to watch the first n and then accept the next candidate better than all of them, and an extension of the argument in part (i) shows that the optimal n is roughly N/e, succeeding with probability about $1/e$. The symmetry argument used here ("each candidate is equally likely to be anywhere") is the starting point for the study of random permutations. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of symmetry arguments like these.

WORKED SOLUTION

STEP II 2001, Question 1

Pure • Binomial Approximation

STEP II 2001, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q1

The key idea

The binomial series for $(1-x)^{1/2}$ converges quickly when x is small, so its first few terms give an excellent rational approximation. The skill is in choosing x: we want $1-x$ to be a square number times the number whose root we need, divided by a power of ten. Then the square root of $1-x$ is a simple multiple of the root we want. Since $99 = 9 \times 11$ and $9999 = 9 \times 1111$, the choices almost make themselves.

Solution

The quadratic approximation

By the binomial expansion, valid for $|x| < 1$,

$$\begin{aligned}(1-x)^{1/2} &= 1 + \tfrac{1}{2}(-x) + \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)}{2!}(-x)^2 + \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)\cdot\left(-\frac{3}{2}\right)}{3!}(-x)^3 + \cdots \\ &= 1 - \tfrac{1}{2}x - \tfrac{1}{8}x^2 - \tfrac{1}{16}x^3 - \cdots .\end{aligned}$$

So for small x,

$$\sqrt{1-x} \approx 1 - \frac{x}{2} - \frac{x^2}{8}.$$

We have also found the first neglected term, $-\frac{1}{16}x^3$, which we need for the errors. In fact every term after the first is negative (for $n \geq 1$ the product $\frac{1}{2}\cdot(-\frac{1}{2})\cdots(\frac{1}{2}-n+1)$ has $n-1$ negative factors, and $(-x)^n$ contributes $(-1)^n$, leaving one minus sign overall), so for $0 < x < 1$ the quadratic is an *overestimate*.

Part (i)

With $x = \frac{1}{100}$,

$$\sqrt{1-x} = \sqrt{\frac{99}{100}} = \frac{\sqrt{9\times 11}}{10} = \frac{3\sqrt{11}}{10}, \qquad \text{so} \qquad \sqrt{11} = \frac{10}{3}\sqrt{1-\tfrac{1}{100}}.$$

Using the quadratic approximation,

$$\sqrt{11} \approx \frac{10}{3}\left(1 - \frac{1}{200} - \frac{1}{80\,000}\right) = \frac{10}{3} \cdot \frac{80\,000 - 400 - 1}{80\,000}$$
$$= \frac{10}{3} \cdot \frac{79\,599}{80\,000} = \frac{79\,599}{24\,000},$$

as required.

Ask yourself The first neglected term is $-\frac{1}{16}x^3$. Is that the error in $\sqrt{11}$?

Not quite: that is the error in our approximation to $\sqrt{1-x}$, and we then multiplied by $\frac{10}{3}$. The error in $\sqrt{11}$ is therefore approximately

$$\frac{10}{3} \times \frac{1}{16}\left(\frac{1}{100}\right)^3 = \frac{10}{48 \times 10^6} = \frac{1}{4\,800\,000} \approx 2.08 \times 10^{-7}.$$

So, to 1 significant figure, the error is 2×10^{-7}, and $\frac{79\,599}{24\,000}$ is slightly larger than $\sqrt{11}$.

Part (ii)

We copy the trick of part (i). Since $1111 \times 9 = 9999 = 10^4 - 1$, take $x = \frac{1}{10\,000}$:

$$\sqrt{1-x} = \sqrt{\frac{9999}{10\,000}} = \frac{3\sqrt{1111}}{100}, \qquad \text{so} \qquad \sqrt{1111} = \frac{100}{3}\sqrt{1 - \tfrac{1}{10\,000}}.$$

Then

$$\sqrt{1111} \approx \frac{100}{3}\left(1 - \frac{1}{20\,000} - \frac{1}{8 \times 10^8}\right) = \frac{100}{3} \cdot \frac{800\,000\,000 - 40\,000 - 1}{8 \times 10^8}$$
$$= \frac{100}{3} \cdot \frac{799\,959\,999}{8 \times 10^8} = \frac{799\,959\,999}{24\,000\,000} = \frac{266\,653\,333}{8\,000\,000},$$

where in the last step we cancelled a factor of 3 (the digit sum of 799 959 999 is 75, which is divisible by 3).

The error is approximately $\frac{100}{3}$ times the first neglected term:

$$\frac{100}{3} \times \frac{1}{16}\left(\frac{1}{10^4}\right)^3 = \frac{100}{48 \times 10^{12}} \approx 2.08 \times 10^{-12},$$

which is about 2×10^{-12}, as the question asks. So a suitable rational number is

$$\sqrt{1111} \approx \frac{266\,653\,333}{8\,000\,000} \left(= \frac{799\,959\,999}{24\,000\,000}\right).$$

Checks

- $79\,599/24\,000 = 3.316625$, while $\sqrt{11} = 3.31662479\ldots$; the difference is 2.096×10^{-7}, matching the estimate 2.08×10^{-7} and confirming that the approximation is too big.

- $266\,653\,333/8\,000\,000 = 33.331666625$, while $\sqrt{1111} = 33.33166662291\ldots$; we computed the difference exactly to high precision and found 2.0835×10^{-12}.
- The estimate improves in part (ii) because the next neglected term, $-\frac{5}{128}x^4$, is smaller relative to the x^3 term when x is smaller.

Common slips

- Forgetting to multiply the neglected term by $\frac{10}{3}$ (or $\frac{100}{3}$), giving an error ten or a hundred times too small.
- Sign errors in the x^2 coefficient: with $(-x)^2$ the term is $\frac{\frac{1}{2}\cdot(-\frac{1}{2})}{2}x^2 = -\frac{1}{8}x^2$, not $+\frac{1}{8}x^2$.
- Trying x values that make $1 - x$ equal to 1111 divided by a power of ten. Such values are not small, so the series converges far too slowly to help.

Beyond STEP

The claim that the error is "about the first neglected term" can be made precise with Taylor's theorem and the Lagrange form of the remainder, which gives an exact formula for the error with an unknown point in $(0, x)$. From it we can get firm upper and lower bounds, not just estimates. Making these error bounds rigorous is a standard part of a first course in analysis; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2001, Question 2

Pure • Integer Part Sums

STEP II 2001, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q2

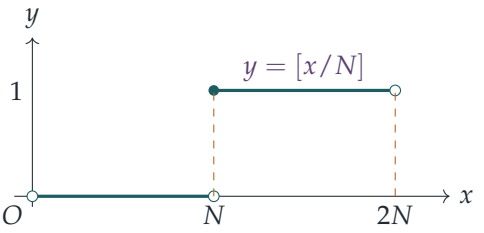
The key idea

The awkward-looking exponent $[k/N]$ takes only three values as k runs from 1 to $2N$, and the sketch shows exactly where it jumps. Once we know that, the sign $(-1)^{[k/N]}$ is constant on each of three blocks of k, so each sum splits into an arithmetic series (part (i)) or a geometric series (part (ii)) on each block. The only care needed is at the endpoints of the blocks.

Solution

The sketch

For $0 < x < N$ we have $0 < x/N < 1$, so $[x/N] = 0$. For $N \leq x < 2N$ we have $1 \leq x/N < 2$, so $[x/N] = 1$. The graph is a step: the jump happens at $x = N$, where the function already takes the value 1, and the right-hand end $x = 2N$ is not included (there the value would be 2).



Filled dots are included and open dots are excluded.

For the sums we need the integer values $k = 1, 2, \ldots, 2N$. Reading off the graph (and

adding the endpoint $k = 2N$):

$$[k/N] = \begin{cases} 0, & 1 \le k \le N-1, \\ 1, & N \le k \le 2N-1, \\ 2, & k = 2N. \end{cases}$$

So $(-1)^{[k/N]}$ is $+1$ for $k \le N-1$, -1 for $N \le k \le 2N-1$, and $+1$ again for the single term $k = 2N$. (When $N = 1$ the first block is empty; every formula below still holds, since an empty sum is 0.)

> **Ask yourself** Why can't we just say the sign is $+$ for the first N terms and $-$ for the next N?

Because the jumps sit exactly at multiples of N: the term $k = N$ already has $[k/N] = 1$, and the last term $k = 2N$ has $[k/N] = 2$, so it is positive. Getting these two endpoints right is the whole question.

Part (i)

Splitting into the three blocks,

$$\sum_{k=1}^{2N} (-1)^{[k/N]} k = \sum_{k=1}^{N-1} k \; - \sum_{k=N}^{2N-1} k \; + \; 2N.$$

The first sum is $\frac{1}{2}(N-1)N$. The second has N terms, from N up to $2N-1$, so it equals N times the average of its first and last terms: $\frac{1}{2}N(3N-1)$. Hence

$$\begin{aligned} \sum_{k=1}^{2N} (-1)^{[k/N]} k &= \tfrac{1}{2}\left(N^2 - N\right) - \tfrac{1}{2}\left(3N^2 - N\right) + 2N \\ &= -N^2 + 2N = 2N - N^2, \end{aligned}$$

as required.

Part (ii)

The same split gives

$$S_N = \sum_{k=1}^{N-1} 2^{-k} \; - \sum_{k=N}^{2N-1} 2^{-k} \; + \; 2^{-2N}.$$

Both sums are geometric with ratio $\frac{1}{2}$. Using $\sum_{k=a}^{b} 2^{-k} = 2^{-a}\left(1 - 2^{-(b-a+1)}\right) \big/ \frac{1}{2} = 2^{1-a} - 2^{-b}$,

$$\sum_{k=1}^{N-1} 2^{-k} = 1 - 2^{1-N},$$

$$\sum_{k=N}^{2N-1} 2^{-k} = 2^{1-N} - 2^{1-2N}.$$

Therefore

$$\begin{aligned} S_N &= 1 - 2^{1-N} - 2^{1-N} + 2^{1-2N} + 2^{-2N} \\ &= 1 - 4 \cdot 2^{-N} + 3 \cdot 2^{-2N}. \end{aligned}$$

Writing $t = 2^{-N}$, this is $1 - 4t + 3t^2$, which factorises neatly:

$$\boxed{S_N = 1 - \frac{4}{2^N} + \frac{3}{4^N} = \left(1 - \frac{1}{2^N}\right)\left(1 - \frac{3}{2^N}\right).}$$

As $N \to \infty$, $2^{-N} \to 0$ and $4^{-N} \to 0$, so

$$\lim_{N \to \infty} S_N = 1.$$

This makes sense: as N grows, the positive block $\frac{1}{2} + \frac{1}{4} + \cdots$ gets longer and its sum approaches 1, while the negative block starts so far along that its total, roughly 2^{1-N}, is negligible.

Checks

- $N = 1$: the terms are $k = 1$ with $[1] = 1$ and $k = 2$ with $[2] = 2$. Part (i) gives $-1 + 2 = 1 = 2 - 1$, and part (ii) gives $-\frac{1}{2} + \frac{1}{4} = -\frac{1}{4}$, matching $(1 - \frac{1}{2})(1 - \frac{3}{2}) = -\frac{1}{4}$.
- $N = 2$: the signs are $+, -, -, +$ for $k = 1, 2, 3, 4$. Then $1 - 2 - 3 + 4 = 0 = 4 - 4$, and $\frac{1}{2} - \frac{1}{4} - \frac{1}{8} + \frac{1}{16} = \frac{3}{16} = \frac{3}{4} \cdot \frac{1}{4}$.
- We checked both formulas exactly, with fractions, for every N from 1 to 59. For example $S_{50} \approx 0.999\,999\,999\,999\,996$.

Common slips

- Putting $k = N$ in the positive block, or forgetting that the final term $k = 2N$ has exponent 2 and so is positive.
- Miscounting the terms in $\sum_{k=N}^{2N-1}$: there are N of them, not $N - 1$.

Beyond STEP

The integer-part (floor) function turns up all over number theory: the classic example is Legendre's formula, which says the power of a prime p dividing $n!$ is $\sum_{j \geq 1} [n / p^j]$. Sums like part (ii), where the sign pattern changes in blocks, are also a gentle introduction to rearranging and grouping the terms of series, which analysis courses make rigorous. Davenport's *The Higher Arithmetic* is a lovely place to see the floor function at work.

WORKED SOLUTION

STEP II 2001, Question 3

Pure • Euclidean Geometry

STEP II 2001, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q3

The key idea

Put the cuboid in coordinates with one vertex at the origin and the edges along the axes. Then every vertex has an obvious position vector, the diagonals are just differences of these, and the angle between them comes straight from the scalar product. The volume-to-surface-area ratio is a simple rational function of λ, and the final part links the two results by turning the condition on R into a condition on λ.

Solution

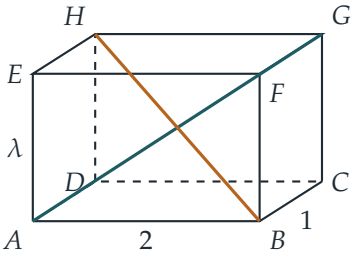
The angle between the diagonals

Since the edges AE, BF, CG, DH are perpendicular to the faces, all the angles at the vertices are right angles, so we may take axes along AB, AD and AE with A at the origin:

$$A = (0,0,0), \quad B = (2,0,0), \quad C = (2,1,0), \quad D = (0,1,0),$$

and the top face is the bottom face lifted by λ:

$$E = (0,0,\lambda), \quad F = (2,0,\lambda), \quad G = (2,1,\lambda), \quad H = (0,1,\lambda).$$



The two diagonals have direction vectors

$$\overrightarrow{AG} = \begin{pmatrix} 2 \\ 1 \\ \lambda \end{pmatrix}, \qquad \overrightarrow{BH} = \begin{pmatrix} -2 \\ 1 \\ \lambda \end{pmatrix},$$

each of length $\sqrt{5+\lambda^2}$, and

$$\overrightarrow{AG} \cdot \overrightarrow{BH} = -4 + 1 + \lambda^2 = \lambda^2 - 3.$$

If θ is the angle between these two vectors, then

$$\cos\theta = \frac{\lambda^2 - 3}{5 + \lambda^2}.$$

Ask yourself Where does the modulus sign come from?

Two lines that cross make two angles, θ and $\pi - \theta$, whose cosines are negatives of each other. The acute one (or the right angle, when $\lambda^2 = 3$) is the one with non-negative cosine, so

$$\cos\alpha = |\cos\theta| = \left|\frac{3 - \lambda^2}{5 + \lambda^2}\right|,$$

as required.

The ratio R

The volume is $2 \times 1 \times \lambda = 2\lambda$. The surface area consists of two faces of each of the areas 2×1, $2 \times \lambda$ and $1 \times \lambda$, so it is $2(2 + 3\lambda)$. Hence

$$R = \frac{2\lambda}{2(2+3\lambda)} = \frac{\lambda}{2+3\lambda}.$$

Since $\lambda > 0$, the denominator is positive, and

$$R < \tfrac{1}{3} \iff 3\lambda < 2 + 3\lambda \iff 0 < 2,$$

which is true. So $R < \frac{1}{3}$ for every possible λ. (As $\lambda \to \infty$, $R \to \frac{1}{3}$, so the bound cannot be improved.)

The final part

Again because $2 + 3\lambda > 0$, we may multiply through:

$$R \geq \tfrac{1}{4} \iff 4\lambda \geq 2 + 3\lambda \iff \lambda \geq 2.$$

So suppose $\lambda \geq 2$. Then $\lambda^2 \geq 4 > 3$, so $\lambda^2 - 3 > 0$ and the modulus can be dropped:

$$\cos\alpha = \frac{\lambda^2 - 3}{\lambda^2 + 5} = 1 - \frac{8}{\lambda^2 + 5}.$$

Writing it this way shows that $\cos\alpha$ increases as λ^2 increases. With $\lambda^2 \geq 4$ we have $\lambda^2 + 5 \geq 9$, so

$$\cos\alpha \geq 1 - \frac{8}{9} = \frac{1}{9}.$$

Both α and $\arccos\frac{1}{9}$ lie in $[0, \frac{\pi}{2}]$, where cosine is strictly decreasing, so $\cos\alpha \geq \frac{1}{9}$ gives

$$\alpha \leq \arccos\tfrac{1}{9},$$

which is what we had to prove. Equality holds exactly when $\lambda = 2$, i.e. when $R = \frac{1}{4}$.

Checks

- When $\lambda^2 = 3$ the formula gives $\cos\alpha = 0$. Check: $\overrightarrow{AG} \cdot \overrightarrow{BH} = -4 + 1 + 3 = 0$, so the diagonals really are perpendicular.
- As $\lambda \to \infty$ the cuboid becomes a tall thin prism, and both diagonals become nearly vertical, so $\alpha \to 0$; the formula gives $\cos\alpha \to 1$, as it should.
- At $\lambda = 2$: volume 4, surface area $2(2 + 4 + 2) = 16$, so $R = \frac{1}{4}$, and $\cos\alpha = \frac{1}{9}$, so $\alpha \approx 83.6°$. A numerical check over 10^5 random values of λ confirmed the angle formula, $R < \frac{1}{3}$, and $\alpha \leq \arccos\frac{1}{9}$ whenever $R \geq \frac{1}{4}$.

Common slips

- Forgetting the modulus: for $\lambda^2 < 3$ the vectors $\overrightarrow{AG}$ and $\overrightarrow{BH}$ make an obtuse angle, and the acute angle is its supplement.
- Getting the direction of the last inequality wrong: a *larger* cosine means a *smaller* angle, because cosine decreases on $[0, \frac{\pi}{2}]$.

Beyond STEP

Placing an object in coordinates and using scalar products to find angles is the basic tool of vector geometry, and in a first-year linear algebra course it generalises to inner products in any number of dimensions: the angle between two vectors in $\mathbb{R}^n$ is *defined* by $\cos\theta = \mathbf{u}\cdot\mathbf{v}/(|\mathbf{u}||\mathbf{v}|)$, with the Cauchy–Schwarz inequality guaranteeing that this lies in $[-1, 1]$. A nice exercise is to show that a long diagonal of an n-dimensional unit cube makes an angle with each edge whose cosine is $1/\sqrt{n}$, so the angle tends to a right angle as n grows. Strang's *Introduction to Linear Algebra* develops these ideas clearly.

WORKED SOLUTION

STEP II 2001, Question 4

Pure • Multiple Angle Equations

STEP II 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q4

The key idea

Every term of f has a factor $\sin x$: $\sin 2x = 2\sin x \cos x$, and the hint does the same for $\sin 3x$. Pulling out $\sin x$ leaves a quadratic in $\cos x$, and the condition on P, Q, R is exactly "this quadratic has negative discriminant". The second part is the same function with $P = 1$, $Q = 1$, $R = -1$ and x replaced by $2nx$. Now the quadratic does have roots, and the work is in finding which root sits closest to the top of the allowed range.

Solution

The first result

Write $c = \cos x$. Using $\sin 2x = 2\sin x \cos x$ and the given identity for $\sin 3x$,

$$\begin{aligned} \mathrm{f}(x) &= P\sin x + 2Q\sin x\cos x + R\sin x\,(4\cos^2 x - 1) \\ &= \sin x\left(4Rc^2 + 2Qc + (P - R)\right). \end{aligned}$$

So $\mathrm{f}(x) = 0$ exactly when $\sin x = 0$ or $4Rc^2 + 2Qc + (P - R) = 0$.

First, $R \neq 0$: if $R = 0$ the condition would read $Q^2 < 0$, which is impossible. So the bracket is a genuine quadratic in c, and its discriminant is

$$(2Q)^2 - 4\cdot 4R\,(P - R) = 4\big(Q^2 - 4R(P - R)\big) < 0$$

by hypothesis. A real quadratic with negative discriminant has no real roots, so the bracket is never zero for any real c, and in particular not for $c = \cos x$. Hence $\mathrm{f}(x) = 0$ forces $\sin x = 0$, that is, $x = m\pi$ with m an integer. Conversely, every $x = m\pi$ is a zero of f, so these are precisely the zeros, as required.

The largest root of g

Put $y = 2nx$. Then $\mathrm{g}(x) = \sin y + \sin 2y - \sin 3y$, which is $\mathrm{f}(y)$ with $P = 1$, $Q = 1$, $R = -1$. Here $Q^2 = 1$ is not less than $4R(P - R) = -8$, so the first result does not

apply, and the quadratic has real roots. With $c = \cos y$, the factorisation above gives

$$g(x) = \sin y\,(-4c^2 + 2c + 2) = -2\sin y\,(2c+1)(c-1).$$

So $g(x) = 0$ exactly when

$$\sin y = 0, \qquad \cos y = 1, \qquad \text{or} \qquad \cos y = -\tfrac{1}{2}.$$

The case $\cos y = 1$ is already contained in $\sin y = 0$. So the roots in y are

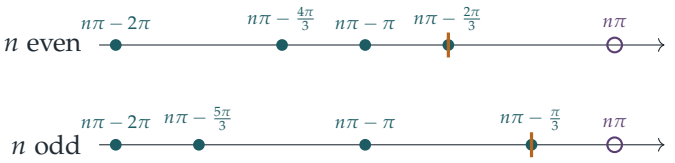
$$y = k\pi \quad \text{or} \quad y = 2k\pi \pm \tfrac{2\pi}{3} \qquad (k \in \mathbb{Z}).$$

Within each period $[2k\pi, 2k\pi + 2\pi)$ the roots sit at $2k\pi$ plus 0, $\frac{2\pi}{3}$, π, $\frac{4\pi}{3}$.

The range $0 < x < \frac{1}{2}\pi$ becomes $0 < y < n\pi$, and the right-hand end $y = n\pi$ is itself a root but is excluded. So we want the largest root strictly below $n\pi$, and only the position of $n\pi$ within its period matters.

Ask yourself Why does the parity of n matter?

The pattern of roots repeats every 2π, not every π. If n is even, $n\pi$ is a multiple of 2π; if n is odd, $n\pi$ is an odd multiple of π. In the two cases the roots just below $n\pi$ are spaced differently.



(Roots of g in terms of y over the last 2π below $n\pi$, drawn to scale. The open circle marks the excluded end and the amber tick marks the largest root.)

n even. Then $n\pi = 2k\pi$ with $k = \frac{n}{2} \geq 1$, and the roots in the last period below it are, measured from $n\pi$,

$$n\pi - 2\pi, \quad n\pi - \tfrac{4\pi}{3}, \quad n\pi - \pi, \quad n\pi - \tfrac{2\pi}{3}.$$

The last of these is $2(k-1)\pi + \frac{4\pi}{3}$, a root because its cosine is $-\frac{1}{2}$. There is no root in $(n\pi - \frac{2\pi}{3}, n\pi)$: no multiple of π lies there, and $\cos y = -\frac{1}{2}$ would need $y \equiv \frac{2\pi}{3}$ or $\frac{4\pi}{3}$ $(\text{mod } 2\pi)$, whereas here y lies strictly between $\frac{4\pi}{3}$ and 2π mod 2π. Also $n\pi - \frac{2\pi}{3} > 0$ since $n \geq 2$. So the largest root has $y = n\pi - \frac{2\pi}{3}$, and

$$x = \frac{y}{2n} = \frac{(3n-2)\pi}{6n} \qquad (n \text{ even}).$$

n odd. Then $n\pi = 2k\pi + \pi$ with $k = \frac{n-1}{2} \geq 0$. Measured within the period starting at $2k\pi = n\pi - \pi$, the roots below $n\pi$ are at offsets 0 and $\frac{2\pi}{3}$, so the candidates are $n\pi - \pi$ and $n\pi - \frac{\pi}{3}$. The interval $(n\pi - \frac{\pi}{3}, n\pi)$ corresponds to y strictly between $\frac{2\pi}{3}$ and π mod 2π, which contains no root. And $n\pi - \frac{\pi}{3} \geq \frac{2\pi}{3} > 0$. So the largest root has $y = n\pi - \frac{\pi}{3}$, and

$$x = \frac{(3n-1)\pi}{6n} \qquad (n \text{ odd}).$$

Checks

For $n = 1$, $\mathrm{g}(x) = \sin 2x + \sin 4x - \sin 6x$ on $(0, \frac{1}{2}\pi)$: the formula gives $x = \frac{\pi}{3}$, and indeed $\sin\frac{2\pi}{3} + \sin\frac{4\pi}{3} - \sin 2\pi = 0$. For $n = 2$ the formula also gives $\frac{\pi}{3}$ ($y = \frac{4\pi}{3}$). As $n \to \infty$ both formulae tend to $\frac{1}{2}\pi$, as they should, since the roots crowd together. We scanned g numerically on a fine grid for $n = 1, \ldots, 12$. Each predicted value (for example $x = \frac{4\pi}{9} \approx 1.3963$ for $n = 3$) is a root, and g keeps one sign between it and $\frac{1}{2}\pi$. For the first part, 2000 random triples satisfying $Q^2 < 4R(P - R)$ all gave a bracket with no zero on $[-1, 1]$.

Common slips

- Giving $x = \frac{1}{2}\pi$ itself: it is a root of g for every n, but it is outside the open interval.
- Forgetting the roots from $\sin y = 0$ and keeping only $\cos y = -\frac{1}{2}$. For even n, $y = n\pi - \pi$ is a root, though not the largest.
- Writing $\cos y = -\frac{1}{2} \Rightarrow y = \frac{2\pi}{3} + 2k\pi$ only, and missing the $\frac{4\pi}{3}$ family. That gives the wrong answer for even n.

Beyond STEP

The factorisation works because $\sin mx / \sin x$ is always a polynomial in $\cos x$: these are the Chebyshev polynomials of the second kind, $U_{m-1}(\cos x)$. So any finite sine series $\sum a_m \sin mx$ is $\sin x$ times a polynomial in $\cos x$, and counting its zeros becomes counting polynomial roots in $[-1, 1]$. This idea underlies much of approximation theory and Fourier analysis. Needham's *Visual Complex Analysis* gives a lovely view of multiple-angle formulae through $\mathrm{e}^{\mathrm{i}x}$.

WORKED SOLUTION

STEP II 2001, Question 5

Pure • Stationary Points and Integration

STEP II 2001, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q5

The key idea

Stationary points come straight from the gradient: they sit where $\frac{dy}{dx} = 0$, and their nature is decided by how the sign of the gradient changes. Because the exponential factor is always positive, that sign depends only on the polynomial factor $x(1-x^2)$, which is the same for both curves. For the heights, the first gradient integrates exactly (spot $u = x^2$); the second doesn't, but we don't need its value, only a comparison of two integrals.

Solution

The curve C_1

Since $e^{-x^2} > 0$ for every x, the gradient is zero exactly when $x(1-x^2) = x(1-x)(1+x) = 0$, that is, at $x = -1$, 0 and 1. Also the gradient has the same sign as $x(1-x)(1+x)$:

	$x < -1$	$-1 < x < 0$	$0 < x < 1$	$x > 1$
sign of $\frac{dy}{dx}$	$+$	$-$	$+$	$-$

At $x = 0$ the gradient changes from negative to positive, so the origin (which is on the curve) is a minimum. At $x = 1$ it changes from positive to negative, so this is a maximum; likewise $x = -1$ is a maximum.

To find the heights we integrate. Since the curve passes through the origin,

$$y = \int_0^x t(1-t^2)\, e^{-t^2}\, dt.$$

? **Ask yourself** How do we integrate $t(1-t^2)e^{-t^2}$?

The factor t next to a function of t^2 suggests $u = t^2$, $\mathrm{d}u = 2t\,\mathrm{d}t$:

$$y = \tfrac{1}{2}\int_0^{x^2} (1-u)\,\mathrm{e}^{-u}\,\mathrm{d}u.$$

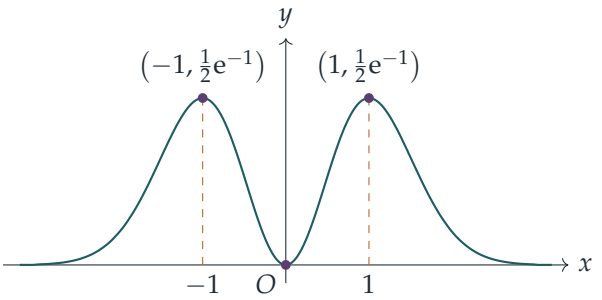
Now notice that $\frac{\mathrm{d}}{\mathrm{d}u}\left(u\mathrm{e}^{-u}\right) = \mathrm{e}^{-u} - u\mathrm{e}^{-u} = (1-u)\mathrm{e}^{-u}$ (integration by parts gives the same thing). Hence

$$y = \tfrac{1}{2}\Big[u\,\mathrm{e}^{-u}\Big]_0^{x^2} = \tfrac{1}{2}\,x^2\mathrm{e}^{-x^2}.$$

At $x = 1$ this gives $y = \frac{1}{2}\mathrm{e}^{-1}$, so the maximum is at $\left(1, \frac{1}{2}\mathrm{e}^{-1}\right)$, as required. Since y is an even function of x, the other stationary point is the maximum

$$\left(-1, \tfrac{1}{2}\mathrm{e}^{-1}\right).$$

Sketch. From $y = \frac{1}{2}x^2\mathrm{e}^{-x^2}$: the curve is symmetric about the y-axis, $y \geq 0$ with equality only at the origin, and $y \to 0$ as $x \to \pm\infty$ because the exponential decays faster than x^2 grows. So it is a "double hump": it touches the x-axis at the origin, rises to two equal maxima at $x = \pm 1$ (height $\frac{1}{2}\mathrm{e}^{-1} \approx 0.18$), then dies away towards the x-axis on both sides. (Vertical scale exaggerated below.)



The curve C_2

Again $\mathrm{e}^{-x^3} > 0$ for every x, so the gradient vanishes exactly at $x = -1, 0, 1$ and has the same sign as $x(1-x)(1+x)$, giving exactly the same sign table as before. So the origin (on the curve by hypothesis) is a minimum, and the stationary point with $x = 1$ is a maximum. (The one at $x = -1$ is a maximum too, though we aren't asked about it.)

Its height is

$$k = \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^3}\,\mathrm{d}x.$$

This has no elementary antiderivative, so instead we compare it with the C_1 integral, which we know equals $\frac{1}{2}\mathrm{e}^{-1}$.

Ask yourself How can we compare k with $\frac{1}{2}\mathrm{e}^{-1}$ without evaluating k?

Compare the integrands. For $0 < x < 1$ we have $x^3 < x^2$ (multiply $x < 1$ by $x^2 > 0$), so $-x^3 > -x^2$ and, as e^t is increasing, $\mathrm{e}^{-x^3} > \mathrm{e}^{-x^2}$. Also $x(1-x^2) > 0$ there, so multiplying preserves the inequality:

$$x(1-x^2)\,\mathrm{e}^{-x^3} > x(1-x^2)\,\mathrm{e}^{-x^2} \qquad \text{for } 0 < x < 1.$$

Both integrands are continuous on $[0,1]$ and the inequality is strict on the open interval, so integrating over $[0,1]$ gives a strict inequality:

$$k = \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^3}\,\mathrm{d}x > \int_0^1 x(1-x^2)\,\mathrm{e}^{-x^2}\,\mathrm{d}x = \tfrac{1}{2}\mathrm{e}^{-1},$$

the last equality being the height of C_1 at $x = 1$ found above. So the maximum is at $(1,k)$ with $k > \frac{1}{2}\mathrm{e}^{-1}$, as required.

Checks

Differentiating $\frac{1}{2}x^2\mathrm{e}^{-x^2}$ by the product rule gives $x\mathrm{e}^{-x^2} - x^3\mathrm{e}^{-x^2} = x(1-x^2)\mathrm{e}^{-x^2}$, and it is 0 at $x = 0$, so C_1 is right. Second derivatives confirm the classification: for both curves $y''(0) = 1 > 0$ and $y''(1) = -2\mathrm{e}^{-1} < 0$. Numerically (Simpson's rule), $k \approx 0.2034$, while $\frac{1}{2}\mathrm{e}^{-1} \approx 0.1839$.

Common slips

- Forgetting the constant of integration: the condition "passes through the origin" is what fixes it, and it matters for the heights.
- Claiming $k > \frac{1}{2}\mathrm{e}^{-1}$ because $\mathrm{e}^{-x^3} > \mathrm{e}^{-x^2}$ without noting that $x(1-x^2)$ is positive on $(0,1)$; multiplying by a negative factor would reverse the inequality.
- Saying $x^3 < x^2$ "for all x": it is true only for $x < 1$, $x \neq 0$, which is why the interval $[0,1]$ is exactly what we need.

Beyond STEP

The second part is a comparison argument: we learn something about a solution we can't write down by comparing it with one we can. This idea runs right through the theory of differential equations, where comparison theorems bound solutions of awkward equations by solutions of simpler ones, and through analysis, where monotonicity of the integral is a basic tool. Boyce and DiPrima, *Elementary Differential Equations*, is a good place to see how much can be said about solutions qualitatively, without formulae.

WORKED SOLUTION

STEP II 2001, Question 6

Pure • Integration by Substitution

STEP II 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q6

The key idea

The awkward-looking $\tan^{-1}\left(\frac{1-x}{1+x}\right)$ is really just $\frac{\pi}{4} - \tan^{-1} x$ in disguise: it is the tangent subtraction formula read backwards. Once that is spotted, part (i) becomes an integration by parts that lands exactly on the opening integral. Part (ii) looks unrelated, but the map $y = \frac{1-x}{1+x}$ swaps 0 and 1 and is its own inverse, so substituting it turns (ii) back into (i).

Solution

The opening integral

Dividing x^4 by $1 + x^2$ (or writing $x^4 = (x^4 - 1) + 1 = (x^2 - 1)(x^2 + 1) + 1$) gives

$$\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}.$$

Hence

$$\int_0^1 \frac{x^4}{1+x^2}\,\mathrm{d}x = \left[\tfrac{1}{3}x^3 - x + \tan^{-1} x\right]_0^1 = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3},$$

as required.

Part (i)

Let $\theta = \tan^{-1} x$, so for $0 \le x \le 1$ we have $0 \le \theta \le \frac{\pi}{4}$. By the subtraction formula,

$$\tan\left(\frac{\pi}{4} - \theta\right) = \frac{\tan\frac{\pi}{4} - \tan\theta}{1 + \tan\frac{\pi}{4}\tan\theta} = \frac{1-x}{1+x}.$$

? **Ask yourself** Does $\tan\left(\frac{\pi}{4} - \theta\right) = \frac{1-x}{1+x}$ really let us write $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \theta$?

Only if $\frac{\pi}{4} - \theta$ lies in the range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ of $\tan^{-1}$. Here $0 \le \frac{\pi}{4} - \theta \le \frac{\pi}{4}$, so it does, and on the whole interval of integration

$$\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \tan^{-1} x.$$

Therefore

$$\int_0^1 x^3 \tan^{-1}\left(\frac{1-x}{1+x}\right)\mathrm{d}x = \frac{\pi}{4}\int_0^1 x^3\,\mathrm{d}x - \int_0^1 x^3 \tan^{-1} x\,\mathrm{d}x = \frac{\pi}{16} - J,$$

where $J = \int_0^1 x^3 \tan^{-1} x\,\mathrm{d}x$. Integrating by parts with $u = \tan^{-1} x$ and $\mathrm{d}v = x^3\,\mathrm{d}x$ (so $v = \frac{1}{4}x^4$ and $\mathrm{d}u = \frac{\mathrm{d}x}{1+x^2}$):

$$\begin{aligned} J &= \left[\tfrac{1}{4}x^4 \tan^{-1} x\right]_0^1 - \frac{1}{4}\int_0^1 \frac{x^4}{1+x^2}\,\mathrm{d}x \\ &= \frac{\pi}{16} - \frac{1}{4}\left(\frac{\pi}{4} - \frac{2}{3}\right) = \frac{1}{6}, \end{aligned}$$

using the opening result. Hence

$$\int_0^1 x^3 \tan^{-1}\left(\frac{1-x}{1+x}\right)\mathrm{d}x = \frac{\pi}{16} - \frac{1}{6}.$$

Part (ii)

? **Ask yourself** Where does the strange weight $\frac{(1-y)^3}{(1+y)^5}$ come from?

It is what $x^3\,\mathrm{d}x$ becomes under the same transformation. Substitute $y = \frac{1-x}{1+x}$, which solves to $x = \frac{1-y}{1+y}$: the map is its own inverse, and it sends $x = 0$ to $y = 1$ and $x = 1$ to $y = 0$. Then

$$1 - y = \frac{2x}{1+x}, \qquad 1 + y = \frac{2}{1+x}, \qquad \mathrm{d}y = -\frac{2}{(1+x)^2}\,\mathrm{d}x,$$

so

$$\frac{(1-y)^3}{(1+y)^5}\,\mathrm{d}y = \frac{8x^3}{(1+x)^3}\cdot\frac{(1+x)^5}{32}\cdot\left(-\frac{2}{(1+x)^2}\right)\mathrm{d}x = -\frac{x^3}{2}\,\mathrm{d}x,$$

and $\tan^{-1} y = \tan^{-1}\left(\frac{1-x}{1+x}\right)$. The minus sign is absorbed by reversing the limits (y from 0 to 1 means x from 1 to 0):

$$\begin{aligned} \int_0^1 \frac{(1-y)^3}{(1+y)^5}\tan^{-1} y\,\mathrm{d}y &= \int_1^0 \left(-\frac{x^3}{2}\right)\tan^{-1}\left(\frac{1-x}{1+x}\right)\mathrm{d}x \\ &= \frac{1}{2}\int_0^1 x^3 \tan^{-1}\left(\frac{1-x}{1+x}\right)\mathrm{d}x = \frac{1}{2}\left(\frac{\pi}{16} - \frac{1}{6}\right) = \frac{\pi}{32} - \frac{1}{12} \end{aligned}$$

by part (i).

Checks

- Both answers are positive, as they must be (each integrand is non-negative on $[0,1]$): $\frac{\pi}{16} - \frac{1}{6} \approx 0.02968$ and $\frac{\pi}{32} - \frac{1}{12} \approx 0.01484$. Simpson's rule on the original integrands agrees to 12 decimal places.
- The answer to (ii) is exactly half that of (i), which you can sanity-check numerically before doing any algebra.
- Put $x = 0$ in the identity: $\tan^{-1} 1 = \frac{\pi}{4}$. Put $x = 1$: $\tan^{-1} 0 = 0$. Both match $\frac{\pi}{4} - \tan^{-1} x$.

Common slips

- Forgetting dy in (ii), or mishandling the sign: the factor $-\frac{2}{(1+x)^2}$ and the reversed limits cancel each other's minus signs.
- Writing $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \tan^{-1} x$ without checking the range; for $x < -1$ it would be false (it is off by π).

Beyond STEP

The substitution $y = \frac{1-x}{1+x}$ is a Möbius transformation, a map of the form $\frac{ax+b}{cx+d}$. These maps form a group under composition, and this one has order two. The same trick, combined with the symmetry, gives the famous result $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x = \frac{\pi}{8}\ln 2$, which is worth trying next. Möbius maps are central in complex analysis; Needham's *Visual Complex Analysis* gives a beautiful geometric account of them.

WORKED SOLUTION

STEP II 2001, Question 7

Pure • Complex Numbers and Rotations

STEP II 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q7

The key idea

Walking round a regular polygon, every side has the same length and at every vertex we turn through the same *exterior* angle. In the Argand diagram a turn through angle θ is multiplication by $e^{i\theta}$, so each side is the previous one multiplied by a fixed complex number. Once the vertices are known, the centre comes from a perpendicular bisector, and the rotation is just one more multiplication by $e^{i\theta}$.

Solution

Finding Q and S

The interior angle at each vertex is $5\pi/6$, so at each vertex the direction of travel turns through the exterior angle $\pi - 5\pi/6 = \pi/6$. Since the polygon lies in the upper half-plane and we start along the positive real axis from O to P, the turns must be *anticlockwise* (a clockwise turn at P would take Q below the real axis).

Write $\omega = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The side $\overrightarrow{OP}$ is the complex number 2, and each later side is the previous one multiplied by ω (same length, turned by $\pi/6$). So

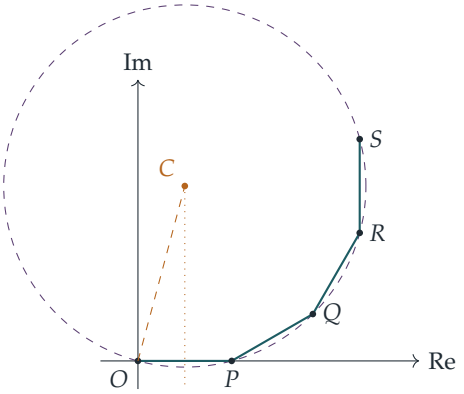
$$\begin{aligned}\overrightarrow{PQ} &= 2\omega = \sqrt{3} + i,\\ \overrightarrow{QR} &= 2\omega^2 = 2e^{i\pi/3} = 1 + \sqrt{3}\,i,\\ \overrightarrow{RS} &= 2\omega^3 = 2e^{i\pi/2} = 2i.\end{aligned}$$

Adding up,

$$\begin{aligned}Q &= 2 + (\sqrt{3} + i) = 2 + \sqrt{3} + i,\\ R &= Q + 1 + \sqrt{3}\,i = (3 + \sqrt{3}) + (1 + \sqrt{3})i,\\ S &= R + 2i = (3 + \sqrt{3}) + (3 + \sqrt{3})i.\end{aligned}$$

The first line is the required value of Q, and

$$S = (3 + \sqrt{3})(1 + i).$$



The centre C

The circle through O, P, Q is the circumcircle of the whole 12-gon, so C is the centre of the polygon. Since C is equidistant from O and $P = 2$, it lies on the perpendicular bisector $\operatorname{Re} z = 1$; write $C = 1 + h\mathrm{i}$. It is also equidistant from O and Q:

$$\begin{aligned} |C|^2 = |C - Q|^2 &\iff 1 + h^2 = (1 + \sqrt{3})^2 + (h - 1)^2 \\ &\iff 1 + h^2 = 4 + 2\sqrt{3} + h^2 - 2h + 1 \\ &\iff 2h = 4 + 2\sqrt{3}, \end{aligned}$$

so $h = 2 + \sqrt{3}$ and $C = 1 + (2 + \sqrt{3})\mathrm{i}$, in the first quadrant.

Ask yourself What is the argument of C exactly?

We need $\tan(\arg C) = 2 + \sqrt{3}$. Now

$$\tan \tfrac{5\pi}{12} = \tan\left(\tfrac{\pi}{4} + \tfrac{\pi}{6}\right) = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} = \frac{(\sqrt{3} + 1)^2}{2} = 2 + \sqrt{3},$$

and C is in the first quadrant, so $\arg C = 5\pi/12$. (Geometrically: the triangle OCP is isosceles with apex angle $2\pi/12$ at C, so its base angles are $\frac{1}{2}(\pi - \pi/6) = 5\pi/12$.)

The rotation

Rotating anticlockwise about O by angle θ multiplies every point by $\mathrm{e}^{\mathrm{i}\theta}$, so it moves C to argument $5\pi/12 + \theta$. For C to lie on the real axis we need $5\pi/12 + \theta$ to be a multiple of π; the smallest positive θ is

$$\theta = \pi - \tfrac{5\pi}{12} = \tfrac{7\pi}{12},$$

which puts C on the *negative* real axis.

Now write S in modulus–argument form: $1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$, so $S = (3 + \sqrt{3})\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4} =$

$(3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{\mathrm{i}\pi/4}$. The new position of S is

$$S' = S\,\mathrm{e}^{7\mathrm{i}\pi/12} = (3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{\mathrm{i}(3\pi/12+7\pi/12)} = (3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{5\mathrm{i}\pi/6}$$
$$= (3\sqrt{2}+\sqrt{6})\left(-\tfrac{\sqrt{3}}{2}+\tfrac{1}{2}\mathrm{i}\right) = -\tfrac{1}{2}(3\sqrt{2}+\sqrt{6})(\sqrt{3}-\mathrm{i}),$$

as required.

Checks

- $|C|^2 = 1+(2+\sqrt{3})^2 = 8+4\sqrt{3} = (\sqrt{6}+\sqrt{2})^2$, and $|C-S|^2 = (2+\sqrt{3})^2+1$ too, so S really is on the same circle.
- $|S'| = 3\sqrt{2}+\sqrt{6} = |S|$, as a rotation must preserve.
- Numerically $S \approx 4.732+4.732\mathrm{i}$ and $S' \approx -5.7956+3.3461\mathrm{i}$; the computer confirmed all the side lengths, the angles of $5\pi/6$, and that $7\pi/12$ is the first anticlockwise angle putting C on the real axis.

Common slips

- Turning by the interior angle $5\pi/6$ instead of the exterior angle $\pi/6$.
- Rotating by $-5\pi/12$ (clockwise) or by $\pi/2-5\pi/12$: the rotation is anticlockwise, and C first meets the real axis on its negative half.

Beyond STEP

Multiplication by $\mathrm{e}^{\mathrm{i}\theta}$ as rotation is the start of seeing complex functions as geometric transformations: $z \mapsto az+b$ gives all rotations, enlargements and translations of the plane, and these form a group. The vertices of a regular polygon are a geometric sum, $2(1+\omega+\cdots+\omega^{k-1})$, linking to the roots of unity. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

WORKED SOLUTION

STEP II 2001, Question 8

Pure • Periodic Functions and ODEs

STEP II 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q8

The key idea

A function with period 1 looks the same on every interval of length 1, so its integral over any such interval is the same number. That makes F grow by the same amount, $F(1)$, every time x goes up by 1. The differential equation in the last part is first order and linear, so an integrating factor turns its solution into an exponential of $-F(x)$, and the rest follows from the fact that $F(1)$ is positive.

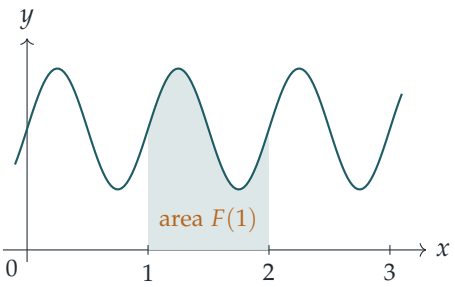
Solution

Part (i)

Take

$$f(x) = 2 + \sin(2\pi x).$$

Since sin has period 2π, $\sin(2\pi(x+1)) = \sin(2\pi x + 2\pi) = \sin(2\pi x)$, so $f(x+1) = f(x)$. Also $\sin(2\pi x) \geq -1$, so $f(x) \geq 1 > 0$ for every x. (A constant such as $f(x) = 1$ also works, but is less interesting.)



Every unit-width strip under the graph has the same area; that is the whole of part (ii).

Part (ii)

Define $G(x) = F(x+1) - F(x)$. Differentiating, using the chain rule on the first term,

$$G'(x) = f(x+1) - f(x) = 0$$

for all x, by the periodicity of f. So G is constant, and its value is $G(0) = F(1) - F(0) = F(1)$, since $F(0) = 0$. That is,

$$F(x+1) = F(x) + F(1) \quad \text{for all } x.$$

Now we use induction on n. For $n = 1$ the claim $F(1) = 1 \cdot F(1)$ is trivially true. If $F(k) = kF(1)$ for some positive integer k, then putting $x = k$ above,

$$F(k+1) = F(k) + F(1) = kF(1) + F(1) = (k+1)F(1).$$

Hence $F(n) = nF(1)$ for every positive integer n, as required.

(Equivalently, if f is continuous, $F(x) = \int_0^x f(t)\,\mathrm{d}t$ and $F(n)$ is the sum of n integrals over $[k, k+1]$, each equal to $\int_0^1 f(t)\,\mathrm{d}t$ after the substitution $t \mapsto t + k$. The argument above avoids having to write F as an integral at all.)

Part (iii)

The equation is linear in y, and the integrating factor is $\mathrm{e}^{\int f\,\mathrm{d}x} = \mathrm{e}^{F(x)}$, which is where F comes in. Indeed, by the product rule and $F' = f$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\,\mathrm{e}^{F(x)}\right) = \mathrm{e}^{F(x)}\left(\frac{\mathrm{d}y}{\mathrm{d}x} + f(x)\,y\right) = 0.$$

So $y\,\mathrm{e}^{F(x)}$ is constant. At $x = 0$ it equals $1 \cdot \mathrm{e}^{F(0)} = 1$, so

$$y = \mathrm{e}^{-F(x)}.$$

By part (ii), for each positive integer n,

$$y(n) = \mathrm{e}^{-F(n)} = \mathrm{e}^{-nF(1)} = \left(\mathrm{e}^{-F(1)}\right)^n.$$

Ask yourself How do we know $F(1) > 0$? We are not told anything about $F(1)$ directly.

Since $F'(x) = f(x) > 0$ for all x, F is strictly increasing, and so $F(1) > F(0) = 0$. (If you prefer integrals: $F(1)$ is the area under a positive curve between 0 and 1.)

Therefore $0 < \mathrm{e}^{-F(1)} < 1$, and a fixed number r with $0 < r < 1$ satisfies $r^n \to 0$ as $n \to \infty$. Hence $y(n) \to 0$ as $n \to \infty$, which is what we had to show.

Notice that we only claim this along the integers. Between integers, y is squeezed: since F is increasing, y is decreasing, so in fact $y(x) \to 0$ as $x \to \infty$ through all real values too.

Checks

With $f(x) = 2 + \sin(2\pi x)$ we have $F(1) = \int_0^1 (2 + \sin 2\pi t)\,\mathrm{d}t = 2$, since the sine integrates to zero over a full period. Simpson's rule gives $F(2), F(3), F(5), F(7)$ equal to $4, 6, 10, 14$ to better than 10^{-12}. Stepping the differential equation numerically (fourth-order Runge–Kutta) from $y(0) = 1$ gives $y(1) \approx 0.135335 = \mathrm{e}^{-2}$ and $y(4) \approx 3.3546 \times 10^{-4} = \mathrm{e}^{-8}$, matching $\mathrm{e}^{-nF(1)}$. We also checked $F(x+1) - F(x) = F(1)$ at non-integer x for the less symmetric example $f(x) = \mathrm{e}^{\cos 2\pi x} + 0.1$.

Common slips

- Writing $F(n) = nF(1)$ as "obvious" from periodicity without showing why the area over $[k, k+1]$ equals the area over $[0, 1]$. The function $G(x) = F(x+1) - F(x)$ or a substitution is needed.
- Forgetting to justify $F(1) > 0$. Without it, $\mathrm{e}^{-nF(1)}$ could stay at 1 or blow up.
- Getting the sign of the integrating factor wrong and ending with $y = \mathrm{e}^{F(x)}$, which grows instead of decaying.

Beyond STEP

This is the simplest case of *Floquet theory*: for a linear differential equation with periodic coefficients, the solution after one period is the starting value multiplied by a fixed factor (here $\mathrm{e}^{-F(1)}$), so the long-term behaviour is decided by whether that factor is less than or greater than 1. The same idea, with matrices in place of numbers, explains the stability of a pendulum whose support is shaken up and down. Boyce and DiPrima, *Elementary Differential Equations*, covers linear equations and integrating factors thoroughly and is a good next step.

WORKED SOLUTION

STEP II 2001, Question 9

Mechanics • Quadratic Air Resistance

STEP II 2001, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

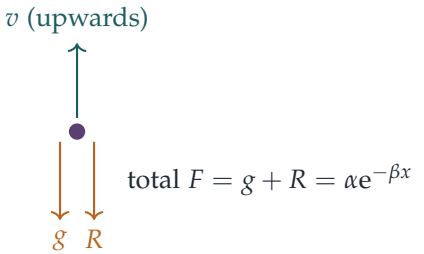
step.maths.org/questions/01-s2-q9

The key idea

The force is given as a function of *height*, not time, so the natural tool is the work–energy principle: the work done against F over the climb is exactly the kinetic energy lost. That links $\mathrm{e}^{-\beta x}$ to v^2, and the result shows that the air resistance depends only on the speed. Once we know the resistance law in terms of speed, the downward journey is a standard differential equation in v^2.

Solution

Model. The particle has unit mass and moves on a vertical line. On the way up, both gravity and air resistance act downwards, against the motion, so the total force F (magnitude $\alpha\mathrm{e}^{-\beta x}$) acts downwards. Take x upwards from the point of projection.



Energy expended

The energy expended in reaching height x is the work done against F:

$$\int_0^x \alpha\mathrm{e}^{-\beta s}\,\mathrm{d}s = \frac{\alpha}{\beta}\left(1-\mathrm{e}^{-\beta x}\right).$$

This energy comes from the particle's kinetic energy, so by the work–energy principle (unit mass)

$$\tfrac{1}{2}u^2 - \tfrac{1}{2}v^2 = \frac{\alpha}{\beta}\left(1 - \mathrm{e}^{-\beta x}\right). \qquad (*)$$

(Equivalently, Newton's second law in the form $v\frac{\mathrm{d}v}{\mathrm{d}x} = -\alpha\mathrm{e}^{-\beta x}$, integrated from $x = 0$, $v = u$, gives the same equation.)

The force in terms of speed

Multiply $(*)$ by β and rearrange:

$$\alpha - \alpha\mathrm{e}^{-\beta x} = \tfrac{1}{2}\beta u^2 - \tfrac{1}{2}\beta v^2 \quad \Longrightarrow \quad F = \alpha\mathrm{e}^{-\beta x} = \tfrac{1}{2}\beta v^2 + \alpha - \tfrac{1}{2}\beta u^2,$$

as required.

Ask yourself Why must $\alpha = \frac{1}{2}\beta u^2 + g$? Nothing in the question seems to force it.

Split F into its two parts: $F = g + R$, where R is the air resistance. Then

$$R = F - g = \tfrac{1}{2}\beta v^2 + \left(\alpha - \tfrac{1}{2}\beta u^2 - g\right).$$

So the resistance is a function of the speed alone: a v^2 term plus a constant. Air resistance is caused by motion through the air, so it must vanish when the particle is momentarily at rest, at the highest point. Putting $v = 0$ there, $R = 0$, so $F = g$ (only gravity acts) and

$$\alpha - \tfrac{1}{2}\beta u^2 = g, \qquad \text{that is,} \qquad \alpha = \tfrac{1}{2}\beta u^2 + g.$$

Consequently the constant term is zero and the resistance law is simply

$$R = \tfrac{1}{2}\beta v^2.$$

(For the record, the highest point is where $\alpha\mathrm{e}^{-\beta x} = g$, at height $\frac{1}{\beta}\ln(\alpha/g)$.)

The downward journey

Assume the same resistance law $R = \frac{1}{2}\beta v^2$ holds whatever the direction of motion; on the way down it acts *upwards*, opposing the motion, while gravity still acts downwards. Let y be the distance below the highest point and v the downward speed, with $v = 0$ at $y = 0$. Newton's second law (unit mass), downwards:

$$v\frac{\mathrm{d}v}{\mathrm{d}y} = g - \tfrac{1}{2}\beta v^2.$$

Since $v\frac{\mathrm{d}v}{\mathrm{d}y} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}y}(v^2)$, put $w = v^2$:

$$\frac{\mathrm{d}w}{\mathrm{d}y} = 2g - \beta w, \qquad w = 0 \text{ at } y = 0.$$

This is linear with constant coefficients (or separate the variables): the solution is

$$w = \frac{2g}{\beta}\left(1 - \mathrm{e}^{-\beta y}\right),$$

which indeed satisfies $w(0) = 0$ and $w' = 2g\mathrm{e}^{-\beta y} = 2g - \beta w$. Therefore the air resistance at distance y below the highest point is

$$R = \tfrac{1}{2}\beta v^2 = g\left(1 - \mathrm{e}^{-\beta y}\right).$$

Checks

As $y \to \infty$, $R \to g$: the particle approaches terminal speed $\sqrt{2g/\beta}$, where resistance balances gravity. At $y = 0$, $R = 0$, as it should be at rest. On the way up, at $x = 0$ the formula gives $F = \alpha = g + \frac{1}{2}\beta u^2$, gravity plus the resistance at speed u. Numerically, with $g = 9.8$, $\beta = 0.1$, $u = 20$ (so $\alpha = 29.8$), stepping the equations of motion with a Runge–Kutta method reproduces $F = \alpha\mathrm{e}^{-\beta x}$ on the way up and $R = g(1 - \mathrm{e}^{-\beta y})$ on the way down to about 10^{-12}, and the particle peaks at height ≈ 11.12, matching $\frac{1}{\beta}\ln(\alpha/g)$.

Common slips

- Treating F as the air resistance alone: it is the *total* force, so gravity must be subtracted to get R.
- Keeping the same sign for the resistance on the way down: it always opposes the motion, so it now acts upwards and the equation becomes $v\frac{\mathrm{d}v}{\mathrm{d}y} = g - \frac{1}{2}\beta v^2$.
- Answering in terms of v, or of u and α; the question wants R in terms of y, g and β only.

Beyond STEP

Quadratic drag, $R \propto v^2$, is the realistic model for a fast-moving object in air, and this question shows a neat trick: in terms of distance, the equation for v^2 is *linear*, even though the equation for v in terms of time is not. Undergraduate mechanics also treats the upward-then-downward problem asymmetrically: the particle returns to its start with a speed less than u, and the descent takes longer than the ascent. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses velocity-dependent forces and terminal velocity in detail.

WORKED SOLUTION

STEP II 2001, Question 10

Mechanics • Collisions and Restitution

STEP II 2001, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q10

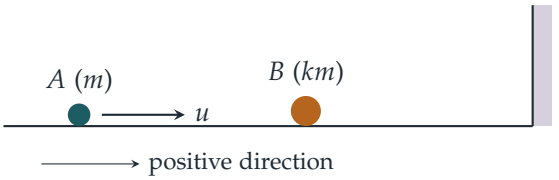
The key idea

This is two collisions and a race. The first collision is the standard pair of equations: conservation of momentum and Newton's experimental law. After B bounces off the wall, the only question is whether B, now coming back, can meet A. That happens exactly when A's velocity towards the wall is greater than B's, which turns into the inequality for k. For the energy part we write the kinetic energy as a function of k and compare it with $mu^2/27$. The comparison turns out to be a perfect square.

Solution

The second collision

Model A and B as particles on a smooth surface, so the only horizontal impulses come from the collisions. Take the direction from A towards the wall as positive. Let v_A and v_B be the velocities of A and B just after their first collision.



First collision. Momentum is conserved and the separation speed is e_1 times the approach speed:

$$mv_A + km\,v_B = mu,$$
$$v_B - v_A = e_1 u.$$

Adding k times the second equation to the first gives $(1+k)v_B = (1+e_1)u$. So

$$v_B = \frac{(1+e_1)u}{1+k}, \qquad v_A = v_B - e_1 u = \frac{(1-ke_1)u}{1+k}.$$

Since $v_B > 0$, B moves on towards the wall. A may be moving forwards, at rest, or moving backwards, depending on the sign of $1-ke_1$.

The wall. B hits the wall at speed v_B and rebounds at speed $e_2 v_B$, so its velocity becomes $-e_2 v_B$. Meanwhile A keeps moving with constant velocity v_A. Note that A cannot reach the wall first, because $v_A < v_B$ (the difference is $e_1 u > 0$) and A starts behind B.

Ask yourself After the rebound A and B might be moving in the same direction. How do we decide whether they meet?

After the rebound B is between A and the wall and nothing else acts on either particle. So the gap between them closes if and only if A's velocity is greater than B's:

$$v_A > -e_2 v_B \iff v_A + e_2 v_B > 0.$$

If $v_A > 0$ they are moving towards each other. If $v_A \le 0$, B is chasing A and catches it only if it is faster. The single inequality covers both cases. Substituting, and multiplying by the positive quantity $(1+k)/u$, we get

$$1 - ke_1 + e_2(1+e_1) > 0 \iff k < \frac{1+e_2(1+e_1)}{e_1},$$

dividing by $e_1 > 0$ in the last step. So there is a second collision provided this inequality holds, as required. (If equality holds, A and B move off with the same velocity and never meet.)

The kinetic energy

With $e_1 = \frac{1}{3}$ and $e_2 = \frac{1}{2}$,

$$v_A = \frac{(3-k)u}{3(1+k)}, \qquad e_2 v_B = \frac{1}{2}\cdot\frac{\frac{4}{3}u}{1+k} = \frac{2u}{3(1+k)}.$$

Just after B rebounds, A still has speed $|v_A|$ and B has speed $e_2 v_B$. So the total kinetic energy is

$$\begin{aligned} T &= \tfrac{1}{2}mv_A^2 + \tfrac{1}{2}km\,(e_2 v_B)^2 \\ &= \frac{mu^2}{18(1+k)^2}\left[(3-k)^2 + 4k\right] = \frac{mu^2\,(k^2-2k+9)}{18(1+k)^2}. \end{aligned}$$

Now compare with $mu^2/27$, using the common denominator $54(1+k)^2$:

$$\begin{aligned} T - \frac{mu^2}{27} &= \frac{mu^2}{54(1+k)^2}\left[3(k^2-2k+9) - 2(1+k)^2\right] \\ &= \frac{mu^2}{54(1+k)^2}\,(k^2-10k+25) = \frac{mu^2\,(k-5)^2}{54(1+k)^2}. \end{aligned}$$

Ask yourself What is the condition $k < 5$ for, if a square is never negative?

The right-hand side is never negative, and it is zero only when $k = 5$. So for $k < 5$ we have $(k-5)^2 > 0$ and therefore

$$T > \frac{mu^2}{27},$$

as required. The value 5 is not a coincidence. With these coefficients the first part's bound is $\left(1 + \frac{1}{2} \cdot \frac{4}{3}\right) \Big/ \frac{1}{3} = 5$. So $k < 5$ is exactly the condition for a second collision. At $k = 5$ both particles move at $-u/9$ after the rebound, and the energy is exactly $mu^2/27$.

Checks

- Take $e_1 = 1$ and $k = 1$ (equal masses, perfectly elastic). Then $v_A = 0$ and $v_B = u$, the familiar exchange of velocities. The bound becomes $k < 1 + 2e_2$, which $k = 1$ satisfies whenever $e_2 > 0$: a returning B will certainly hit a stationary A.
- For $k = 5$ the formula gives $v_A = -\frac{u}{9}$ and $-e_2 v_B = -\frac{1}{2} \cdot \frac{2u}{9} = -\frac{u}{9}$, equal velocities, as claimed.
- Numerically, with exact fractions for k from 0.001 to 4.999, the ratio $T/(mu^2/27)$ has minimum 1.00000001 near $k = 5$ and is always above 1. A random test of 20 000 triples (k, e_1, e_2) found no case where the inequality and a direct comparison of the post-rebound velocities disagree.

Common slips

- Requiring $v_A > 0$. A can be moving backwards and still be caught by B. The correct condition compares the two velocities after the rebound.
- Forgetting that the wall reverses B's direction. Its velocity becomes $-e_2 v_B$, not $e_2 v_B$.
- Using B's speed before the wall in the energy. The question wants the energy just after the rebound, so B's speed is $e_2 v_B$.

Beyond STEP

Newton's experimental law is a model for what happens in the short, violent deformation during an impact. The coefficient of restitution summarises how much of the relative kinetic energy is lost. In a first-year mechanics course, collisions are often analysed in the centre-of-mass frame. In that frame a collision simply reverses the relative velocity and multiplies it by e, and the energy lost is $\frac{1}{2}\mu(1-e^2)w^2$, where μ is the reduced mass and w the approach speed. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP II 2001, Question 11

Mechanics • Projectile Motion

STEP II 2001, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

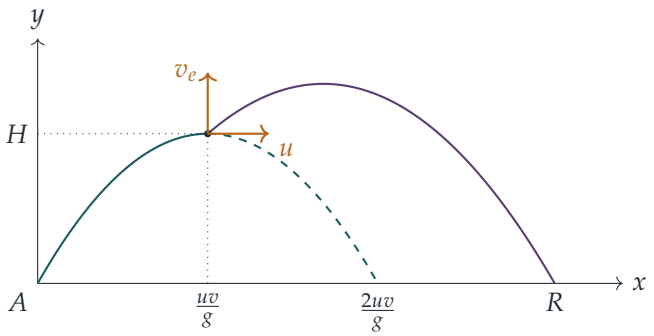
step.maths.org/questions/01-s2-q11

The key idea

Split the flight at the explosion. Before it, we have an ordinary projectile, and the highest point is where the vertical velocity is zero. After it, fragment P is a new projectile launched from that height with velocity (u, v_e), so the time to land comes from a quadratic. For the second part, the range is an increasing function of v_e, so "$R > D$" turns into "v_e exceeds the value that gives $R = D$ exactly".

Solution

We model the missile and fragment P as particles moving under gravity alone, with the ground horizontal. Take A as origin, x horizontal and y vertically upwards.



The dashed curve is where the missile would have landed with no explosion; the purple curve is the path of P.

The range of P

Up to the explosion. The vertical velocity is $v - gt$, which vanishes at $t_1 = v/g$. At that moment the missile has travelled horizontally

$$x_1 = ut_1 = \frac{uv}{g},$$

and its height, from $v^2 = 2gH$ with final vertical speed zero, is

$$H = \frac{v^2}{2g}.$$

After the explosion. Fragment P starts at height H with horizontal velocity u and vertical velocity v_e upwards. Measuring time t from the explosion, its height is

$$y = H + v_e t - \tfrac{1}{2}gt^2.$$

It lands when $y = 0$:

$$\tfrac{1}{2}gt^2 - v_e t - \frac{v^2}{2g} = 0 \quad \Longleftrightarrow \quad g^2t^2 - 2gv_e t - v^2 = 0.$$

Treating this as a quadratic in gt,

$$gt = v_e \pm \sqrt{v_e^2 + v^2}.$$

The product of the roots is $-v^2 < 0$, so one root is negative. That root is the time *before* the explosion at which the parabola would cross the ground, and it is irrelevant. So the flight time after the explosion is

$$t_2 = \frac{v_e + \sqrt{v_e^2 + v^2}}{g}.$$

The horizontal velocity stays u throughout, so the total horizontal distance is

$$R = \frac{uv}{g} + ut_2 = \frac{u}{g}\left(v + v_e + \sqrt{v_e^2 + v^2}\right),$$

that is,

$$\frac{gR}{u} = v + v_e + \sqrt{v_e^2 + v^2},$$

as required.

The condition for $R > D$

Write

$$f(v_e) = v_e + \sqrt{v_e^2 + v^2}, \qquad k = \frac{gD}{u} - v = \frac{gD - uv}{u}.$$

Then $R > D$ is equivalent to $f(v_e) > k$. Since $D > 2uv/g$, we have $gD > 2uv$, so $k > v > 0$.

Ask yourself Why is it enough to find the single value of v_e that makes $R = D$?

Because f is strictly increasing:

$$f'(v_e) = 1 + \frac{v_e}{\sqrt{v_e^2 + v^2}} > 0,$$

since $|v_e| < \sqrt{v_e^2 + v^2}$ when $v \neq 0$. (Physically, a harder upward kick keeps P in the air longer.) So if $f(v_e^*) = k$, then $f(v_e) > k$ exactly when $v_e > v_e^*$.

To find v_e^*, solve $\sqrt{v_e^2 + v^2} = k - v_e$. Squaring gives

$$v_e^2 + v^2 = k^2 - 2kv_e + v_e^2 \quad \Longrightarrow \quad v_e^* = \frac{k^2 - v^2}{2k} = \frac{(k-v)(k+v)}{2k}.$$

Squaring could introduce a false root, so we check that $k - v_e^* \geq 0$: indeed $k - v_e^* = \dfrac{k^2 + v^2}{2k} > 0$, so v_e^* really does satisfy $f(v_e^*) = k$.

Now substitute for k:

$$k - v = \frac{gD - 2uv}{u}, \qquad k + v = \frac{gD}{u}, \qquad 2k = \frac{2(gD - uv)}{u},$$

so

$$v_e^* = \frac{(gD - 2uv)\, gD/u^2}{2(gD - uv)/u} = \frac{gD}{2u}\left(\frac{gD - 2uv}{gD - uv}\right).$$

Hence, since f is increasing, $R > D$ whenever

$$v_e > \frac{gD}{2u}\left(\frac{gD - 2uv}{gD - uv}\right),$$

which is the required condition. In fact the argument shows it is also necessary. Note that $v_e^* > 0$ because $k > v$, so a genuine upward kick is needed.

Checks

- If $v_e = 0$ the formula gives $gR/u = 2v$, so $R = 2uv/g$: the usual range of an unexploded projectile, as it should be. This is also why the question assumes $D > 2uv/g$, since otherwise no kick is needed.
- With $u = v = 10$, $g = 10$ and $D = 30$, the threshold is $v_e^* = \frac{300}{20} \cdot \frac{100}{200} = 7.5$, and then $gR/u = 10 + 7.5 + 12.5 = 30$, so $R = 30 = D$ exactly.
- A numerical step-by-step simulation of the two-stage flight agreed with the range formula to about 10^{-9}, and 10^5 random cases confirmed that $R > D$ exactly when $v_e > v_e^*$.

Common slips

- Forgetting the horizontal distance uv/g covered before the explosion, and measuring R from the point below the explosion instead of from A.

- Taking the time to land from the top as v/g again, which ignores the new upward velocity v_e.
- Squaring $\sqrt{v_e^2 + v^2} = k - v_e$ without checking that the right-hand side is non-negative, or without saying why one equality value turns into an inequality.

Beyond STEP

The explosion conserves the momentum of the whole system, so the centre of mass of the two fragments carries on along the original parabola even though P does not. That principle, that internal forces cannot change the motion of the centre of mass, is the starting point for rocket motion and collisions in a first-year mechanics course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both carefully.

WORKED SOLUTION

STEP II 2001, Question 12

Probability and Statistics • Poisson Approximation

STEP II 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q12

The key idea

The syndicate is guaranteed exactly one winning ticket, so everything depends on how many *other* punters picked the winning number. That count is binomial with a huge number of trials and a tiny success probability, which is exactly the situation where a Poisson distribution takes over. The expected share is then W times the expectation of $1/(1+X)$, a sum that can be done in closed form by absorbing the $k+1$ into the factorial.

Solution

Setting up

The syndicate bought N tickets, so the other punters bought $3.8N - N = 2.8N$ tickets. Let X be the number of these non-syndicate tickets carrying the winning number.

Each non-syndicate punter chooses a number uniformly at random, independently of the others and of the draw, so whatever the winning number is, each of their tickets matches it with probability $1/N$, independently. Hence

$$X \sim \mathrm{B}\left(2.8N,\ \tfrac{1}{N}\right).$$

Since N is very large, $2.8N$ is large and $1/N$ is small while the mean $2.8N \cdot \frac{1}{N} = 2.8$ stays fixed, so X is approximately Poisson with mean $\lambda = 2.8$:

$$\mathrm{P}(X = k) \approx \frac{\mathrm{e}^{-2.8}\,(2.8)^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The syndicate holds exactly one ticket with each number, so it always holds exactly one winning ticket, and the total number of winning tickets is $T = 1 + X$.

The most probable number of winning tickets

Write $p_k = \mathrm{P}(X = k)$. Comparing neighbouring probabilities,

$$\frac{p_k}{p_{k-1}} = \frac{\lambda}{k} = \frac{2.8}{k},$$

which is greater than 1 for $k \le 2$ and less than 1 for $k \ge 3$. So $p_0 < p_1 < p_2$ and $p_2 > p_3 > p_4 > \cdots$, and the most probable value of X is 2. (The same holds for the exact binomial: there $p_k/p_{k-1} = \frac{(2.8N-k+1)}{k(N-1)}$, which exceeds 1 exactly when $k < 2.8 + \frac{1}{N}$.)

Therefore the most probable number of winning tickets is

$$T = 1 + 2 = 3.$$

The expected net loss

The jackpot is shared equally among the $T = 1 + X$ winning tickets, and the syndicate holds one of them, so it receives $\dfrac{W}{1+X}$. It paid £N, so its net loss is

$$L = N - \frac{W}{1+X}, \qquad \text{and} \qquad \mathrm{E}(L) = N - W\,\mathrm{E}\!\left(\frac{1}{1+X}\right).$$

Ask yourself How do we sum $\sum \frac{1}{k+1} \cdot \frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}$?

The trick is that $(k+1)\cdot k! = (k+1)!$, so the awkward factor $\frac{1}{k+1}$ merges into the factorial. Multiplying and dividing by λ turns the sum into a Poisson series with one term missing:

$$\begin{aligned}\mathrm{E}\!\left(\frac{1}{1+X}\right) &\approx \sum_{k=0}^{\infty} \frac{1}{k+1}\cdot\frac{\mathrm{e}^{-\lambda}\lambda^k}{k!} = \frac{\mathrm{e}^{-\lambda}}{\lambda}\sum_{k=0}^{\infty}\frac{\lambda^{k+1}}{(k+1)!} \\ &= \frac{\mathrm{e}^{-\lambda}}{\lambda}\sum_{j=1}^{\infty}\frac{\lambda^j}{j!} = \frac{\mathrm{e}^{-\lambda}}{\lambda}\left(\mathrm{e}^{\lambda}-1\right) = \frac{1-\mathrm{e}^{-\lambda}}{\lambda},\end{aligned}$$

where we put $j = k+1$ and used $\sum_{j\ge 0}\lambda^j/j! = \mathrm{e}^{\lambda}$, removing the $j = 0$ term, which is 1. With $\lambda = 2.8 = \frac{14}{5}$, this is $\frac{5}{14}\left(1-\mathrm{e}^{-2.8}\right)$, and so

$$\mathrm{E}(L) \approx N - \frac{5\left(1-\mathrm{e}^{-2.8}\right)}{14}\,W,$$

as required.

Checks

- The same trick works for the exact binomial $X \sim \mathrm{B}(n,p)$: $\mathrm{E}\left(\frac{1}{1+X}\right) = \dfrac{1-(1-p)^{n+1}}{(n+1)p}$, which tends to $(1-\mathrm{e}^{-\lambda})/\lambda$ as $n \to \infty$ with $np = \lambda$. With $N = 1000$ the exact value is 0.33536, against the Poisson value $\frac{5}{14}(1-\mathrm{e}^{-2.8}) \approx 0.33542$.

- A simulation of 10^5 draws with $N = 200$ gave an average share of $0.335W$, and 3 winning tickets was the most frequent outcome.
- Limiting case: if nobody else played ($\lambda \to 0$), $(1 - \mathrm{e}^{-\lambda})/\lambda \to 1$ and the syndicate collects the whole jackpot, as it should.

Common slips

- Using $3.8N$ rather than $2.8N$ for the other punters: the syndicate's own tickets are not random, and contribute exactly one winner.
- Writing $\mathrm{E}\left(\frac{W}{1+X}\right) = \frac{W}{1+\mathrm{E}(X)} = \frac{W}{3.8}$. Expectation does not pass through a non-linear function; in fact $\frac{1}{3.8} \approx 0.263$ is noticeably smaller than 0.335 (Jensen's inequality says it must be no larger, since $1/(1+x)$ is convex).

Beyond STEP

The approximation of $\mathrm{B}(n, \lambda/n)$ by $\mathrm{Po}(\lambda)$ is the "law of rare events", and first-year courses prove it by showing the probabilities, or the probability generating functions, converge. The calculation of $\mathrm{E}\big(1/(1+X)\big)$ is really an integral of the generating function, $\int_0^1 G_X(s)\,\mathrm{d}s$, a neat use of generating functions beyond computing means and variances. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP II 2001, Question 13

Probability and Statistics • Continuous Distributions and Estimation

STEP II 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q13

The key idea

We are handed the survival function $\mathrm{P}(T > t)$ directly, so the median comes from setting it equal to $\frac{1}{2}$, and the mean from differentiating to get the density and then integrating. The experiment gives a sample median (the fifth of nine ordered lifetimes) and a sample mean, and matching these to the formulas gives two equations in α and k. Dividing one by the other removes k, leaving an equation for α alone. The final probability is a conditional probability, and a pleasant surprise is waiting there.

Solution

The median and the mean

Write $S(t) = \mathrm{P}(T > t) = (1+kt)^{-\alpha}$ for $t \geq 0$. For this to fall from 1 to 0 we need $k > 0$.

Median. The median M satisfies $\mathrm{P}(T > M) = \frac{1}{2}$, so

$$(1+kM)^{\alpha} = 2 \quad \Longrightarrow \quad 1 + kM = 2^{1/\alpha} \quad \Longrightarrow \quad M = \frac{2^{1/\alpha} - 1}{k}.$$

Mean. The cumulative distribution function is $F(t) = 1 - (1+kt)^{-\alpha}$, so the probability density function is

$$f(t) = F'(t) = \alpha k\,(1+kt)^{-\alpha-1}, \qquad t \geq 0.$$

Then, integrating by parts with $u = t$ and $\mathrm{d}v/\mathrm{d}t = \alpha k(1+kt)^{-\alpha-1}$, so that $v = -(1+kt)^{-\alpha}$,

$$\begin{aligned} m = \int_0^{\infty} t\,\alpha k(1+kt)^{-\alpha-1}\,\mathrm{d}t &= \Big[-t(1+kt)^{-\alpha}\Big]_0^{\infty} + \int_0^{\infty} (1+kt)^{-\alpha}\,\mathrm{d}t \\ &= 0 + \left[-\frac{(1+kt)^{1-\alpha}}{k(\alpha-1)}\right]_0^{\infty} = \frac{1}{k(\alpha-1)}. \end{aligned}$$

Both limits at infinity use $\alpha > 1$: $t(1+kt)^{-\alpha}$ behaves like $t^{1-\alpha}/k^{\alpha} \to 0$, and $(1+kt)^{1-\alpha} \to 0$. (If $\alpha \leq 1$ the mean would be infinite, which is why the question insists on $\alpha > 1$.) So

$$M = \frac{2^{1/\alpha} - 1}{k}, \qquad m = \frac{1}{k(\alpha - 1)}.$$

Estimating α and k

With nine bulbs, the fifth failure is the middle one of the nine ordered lifetimes: it is the *sample median*. So we estimate $M \approx 1000$ and, from the sample mean, $m \approx 2400$.

Ask yourself We have two equations and two unknowns, but k appears in both. How can we get at α on its own?

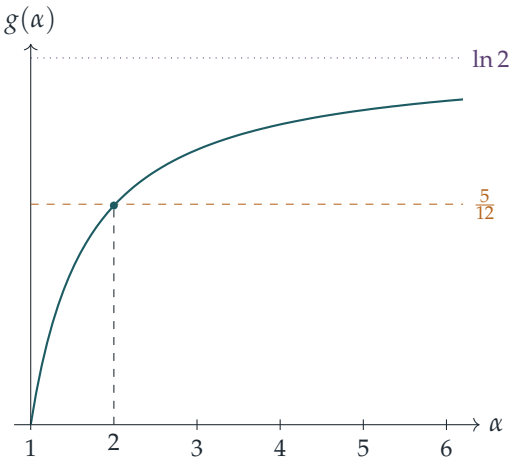
Divide: k cancels, since both M and m are proportional to $1/k$. That gives

$$\frac{M}{m} = \left(2^{1/\alpha} - 1\right)(\alpha - 1) \approx \frac{1000}{2400} = \frac{5}{12} \approx 0.4167.$$

Call the left-hand side $g(\alpha)$. At $\alpha = 2$,

$$g(2) = \left(\sqrt{2} - 1\right)(2 - 1) = \sqrt{2} - 1 \approx 0.4142,$$

which is within 0.003 of $5/12$. To see that no other value fits as well, note that $g(1.9) \approx 0.396$ and $g(2.1) \approx 0.430$, so the value $5/12$ lies between them. In fact g increases steadily from $g(1) = 0$ towards $\ln 2 \approx 0.693$ as $\alpha \to \infty$ (see the graph), so it takes the value $5/12$ only once, at an α just above 2. Hence $\alpha \approx 2$.



Now put $\alpha = 2$ back into either formula. From the mean,

$$k = \frac{1}{m(\alpha - 1)} \approx \frac{1}{2400},$$

and from the median, $k = (\sqrt{2} - 1)/1000 \approx 4.14 \times 10^{-4}$. The two agree closely, and

$$k \approx 4.2 \times 10^{-4} \text{ per hour.}$$

The conditional probability

A bulb that lasts M hours and then a further $m - M$ hours has lasted m hours in total. Since $m > M$, the event $\{T > m\}$ is contained in $\{T > M\}$, so

$$\mathrm{P}(T > m \mid T > M) = \frac{\mathrm{P}(T > m)}{\mathrm{P}(T > M)} = \frac{(1+km)^{-\alpha}}{1/2} = \frac{2}{(1+km)^{\alpha}}.$$

(Strictly, the bulb is found to be still working at M hours; for a continuous lifetime, $T \geq M$ and $T > M$ have the same probability.) Now the surprise: $km = 1/(\alpha - 1)$ exactly, whatever k is, so $1 + km = \alpha/(\alpha - 1)$ and

$$\mathrm{P}(T > m \mid T > M) = 2\left(\frac{\alpha - 1}{\alpha}\right)^{\alpha}.$$

With $\alpha = 2$ this is $2 \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{2}$. So the estimated probability is

$$\mathrm{P}(\text{lasts a further } m - M \text{ hours}) \approx \tfrac{1}{2}.$$

The estimate of k is not needed at all for this last step.

Checks

- At the median, $S(M) = (2^{1/\alpha})^{-\alpha} = \frac{1}{2}$, as it should be. Numerical integration of $S(t)$ over $[0, \infty)$ for random $\alpha > 1$ and $k > 0$ reproduces $m = 1/(k(\alpha - 1))$ (recall that for a non-negative variable $\mathrm{E}(T) = \int_0^{\infty} \mathrm{P}(T > t)\,\mathrm{d}t$, which is exactly the second integral in our working).
- Solving $g(\alpha) = 5/12$ exactly gives $\alpha \approx 2.015$, and then the conditional probability is $2(1 - 1/\alpha)^{\alpha} \approx 0.502$, so rounding to $\alpha = 2$ costs almost nothing.
- A simulation of 400 000 lifetimes with $\alpha = 2$, $k = 1/2400$ (taking $T = (U^{-1/2} - 1)/k$ for U uniform on $(0, 1)$) gave $\mathrm{P}(T > m \mid T > M) \approx 0.501$.

Common slips

- Treating $\mathrm{P}(T > t)$ as the cumulative distribution function and differentiating it without the minus sign, giving a negative "density".
- Answering $\mathrm{P}(T > m - M)$, as if the bulb started afresh at time M. This distribution is not memoryless (only the exponential is), so the condition $T > M$ matters.
- Using the fifth failure as an estimate of the mean rather than the median.

Beyond STEP

This is a Pareto-type (Lomax) distribution, a standard model for lifetimes and for incomes, with a heavy tail: $\mathrm{E}(T^r)$ is finite only for $r < \alpha$. What we did here is the *method*

of moments (strictly, matching a quantile and a moment): choose parameters so that the model's summary statistics equal the sample's. First-year statistics compares this with maximum likelihood and asks how accurate such estimates are from only nine bulbs. The ratio $\mathrm{P}(T > s + t \mid T > s)$ is the survival function used in reliability theory: here it *increases* with s, so old bulbs are more likely to keep going than new ones. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

WORKED SOLUTION

STEP II 2001, Question 14

Probability and Statistics • Minimum Variance Estimators

STEP II 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q14

The key idea

T is a weighted combination of two independent Bernoulli variables, so its mean and variance come straight from linearity of expectation and the rule $\mathrm{Var}(aX + cY) = a^2\,\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y)$ for independent X, Y. The weights are chosen so that the mean is p whatever λ is: T is an *unbiased estimator* of p, and the question is asking for the best one. For the final part, the central limit theorem makes $\overline{T}$ approximately normal, and a normal variable with a fixed mean is most concentrated when its variance is smallest.

Solution

Mean and variance of T

Here X and Y are Bernoulli variables with

$$\mathrm{E}(X) = p, \quad \mathrm{Var}(X) = p(1-p), \qquad \mathrm{E}(Y) = 2p, \quad \mathrm{Var}(Y) = 2p(1-2p).$$

(For a variable taking the value 1 with probability q and 0 otherwise, $\mathrm{E} = q$ and $\mathrm{E}(\cdot^2) = q$, so the variance is $q - q^2 = q(1-q)$.) By linearity of expectation,

$$\mathrm{E}(T) = \lambda p + \tfrac{1}{2}(1-\lambda)\cdot 2p = \lambda p + (1-\lambda)p = p,$$

as required. Since the coins are independent, X and Y are independent, so there is no covariance term and

$$\begin{aligned}\mathrm{Var}(T) &= \lambda^2\,\mathrm{Var}(X) + \tfrac{1}{4}(1-\lambda)^2\,\mathrm{Var}(Y)\\ &= \lambda^2 p(1-p) + \tfrac{1}{2}(1-\lambda)^2 p(1-2p).\end{aligned}$$

Minimising the variance

Write $V(\lambda) = \mathrm{Var}(T)$. This is a quadratic in λ with positive leading coefficient $p(1-p) + \tfrac{1}{2}p(1-2p)$ (both terms are positive since $0 < p < \tfrac{1}{2}$), so it has a single minimum,

where $V'(\lambda) = 0$:

$$V'(\lambda) = 2\lambda p(1-p) - (1-\lambda)p(1-2p) = 0.$$

Dividing by $p > 0$ and collecting the λ terms,

$$\lambda\big[2(1-p) + (1-2p)\big] = 1 - 2p \quad \Longrightarrow \quad \lambda = \frac{1-2p}{3-4p},$$

which is the stated value. (Here $3 - 4p > 1 > 0$, so there is no division by zero.)

We will also need the minimum value itself. With $a = p(1-p)$ and $c = \frac{1}{2}p(1-2p)$ we have $V = a\lambda^2 + c(1-\lambda)^2$, minimised at $\lambda = c/(a+c)$ (which is the value above), giving

$$V_{\min} = a\Big(\frac{c}{a+c}\Big)^2 + c\Big(\frac{a}{a+c}\Big)^2 = \frac{ac(a+c)}{(a+c)^2} = \frac{ac}{a+c}.$$

Now $a + c = \frac{1}{2}p\big[2(1-p) + (1-2p)\big] = \frac{1}{2}p(3-4p)$ and $ac = \frac{1}{2}p^2(1-p)(1-2p)$, so

$$V_{\min} = \frac{p(1-p)(1-2p)}{3-4p}.$$

The mean of n tosses

Let $T_1, \ldots, T_n$ be the values of T on the n tosses; they are independent, each with mean p and variance $V(\lambda)$. Then

$$\mathrm{E}(\overline{T}) = p, \qquad \mathrm{Var}(\overline{T}) = \frac{V(\lambda)}{n}.$$

Ask yourself T takes only four values. Why may we treat $\overline{T}$ as normal?

Because $\overline{T}$ is the mean of n independent, identically distributed variables with finite variance, and $n > 30$ is large, the central limit theorem says $\overline{T}$ is approximately $\mathrm{N}\big(p, V(\lambda)/n\big)$, whatever the shape of the distribution of a single T. Writing $\sigma_\lambda = \sqrt{V(\lambda)/n}$ and $Z = (\overline{T} - p)/\sigma_\lambda \approx \mathrm{N}(0,1)$,

$$\mathrm{P}\big(|\overline{T} - p| < b\big) = \mathrm{P}\Big(|Z| < \frac{b}{\sigma_\lambda}\Big) \approx \Phi\Big(\frac{b}{\sigma_\lambda}\Big) - \Phi\Big(-\frac{b}{\sigma_\lambda}\Big) = 2\Phi\Big(\frac{b}{\sigma_\lambda}\Big) - 1,$$

using the symmetry $\Phi(-z) = 1 - \Phi(z)$.

Since $b > 0$ is fixed and Φ is strictly increasing, this is largest when b/σ_λ is largest, that is, when σ_λ is smallest, which is when $V(\lambda)$ is minimised. That happens at $\lambda = (1-2p)/(3-4p)$, where

$$\sigma_\lambda^2 = \frac{V_{\min}}{n} = \frac{p(1-p)(1-2p)}{(3-4p)n} = s^2.$$

So the maximum value of $\mathrm{P}\big(|\overline{T} - p| < b\big)$ as λ varies is approximately $2\Phi(b/s) - 1$, as required.

Checks

- At $\lambda = 1$, $T = X$ and the formula gives $\operatorname{Var}(T) = p(1-p)$; at $\lambda = 0$, $T = \frac{1}{2}Y$ and it gives $\frac{1}{4} \cdot 2p(1-2p)$. Both are right.
- $V_{\min}$ is smaller than both of these: it is $p(1-p)$ times $\frac{1-2p}{3-4p} < 1$. Combining the two coins beats using either alone.
- Numerically, with $p = 0.3$, $n = 100$, $b = 0.03$: the best λ is $\frac{2}{9}$, $s \approx 0.0216$, and $2\Phi(b/s) - 1 \approx 0.835$. A simulation of 40 000 runs of 100 tosses gave 0.836, while $\lambda = 1$ (coin A alone) gave only about 0.487.

Common slips

- Writing $\operatorname{Var}(\frac{1}{2}(1-\lambda)Y) = \frac{1}{2}(1-\lambda)\operatorname{Var}(Y)$: constants come out of a variance squared.
- Forgetting to say *why* minimising the variance maximises the probability: it needs Φ increasing and the mean of $\overline{T}$ being p for every λ.

Beyond STEP

This is a first taste of estimation theory: among all unbiased estimators $\lambda X + \frac{1}{2}(1-\lambda)Y$ of p, we found the one with minimum variance. The answer weights each observation in inverse proportion to its variance, the same rule used in statistics to combine measurements of differing precision (inverse-variance weighting), and the minimum $ac/(a+c)$ is the "parallel resistors" formula $1/V_{\min} = 1/a + 1/c$. Grimmett and Welsh, *Probability: An Introduction*, covers the central limit theorem behind the final step.

WORKED SOLUTION

STEP III 2001, Question 1

Pure • Maclaurin Series

STEP III 2001, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q1

The key idea

The function $y = \ln\left(x + \sqrt{x^2+1}\right)$ is $\sinh^{-1} x$, and its first derivative is far simpler than the function itself. Squaring that derivative gives a relation with no square roots, $(x^2+1)(y')^2 = 1$, and differentiating it gives the $n = 0$ case of the recurrence. Each further differentiation moves the recurrence up one step, which is exactly what an induction needs. At $x = 0$ the x term disappears and the recurrence links $y^{(n+2)}(0)$ straight to $y^{(n)}(0)$, so all the Maclaurin coefficients come almost for free.

Solution

The derivative

Write $s = \sqrt{x^2+1}$, so that $\dfrac{\mathrm{d}s}{\mathrm{d}x} = \dfrac{x}{s}$. Note that $x + s > 0$ for every real x, since $s > \sqrt{x^2} = |x|$, so the logarithm is defined everywhere. By the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x+s}\left(1 + \frac{x}{s}\right) = \frac{1}{x+s} \cdot \frac{s+x}{s} = \frac{1}{\sqrt{x^2+1}},$$

as required.

The induction

Let $P(n)$ be the statement

$$(x^2+1)\, y^{(n+2)} + (2n+1)\, x\, y^{(n+1)} + n^2\, y^{(n)} = 0 \quad \text{for all real } x.$$

Base case, $n = 0$. We need $(x^2+1)y'' + xy' = 0$. From the first part, $(x^2+1)(y')^2 = 1$. Differentiating with respect to x,

$$2x(y')^2 + (x^2+1) \cdot 2y'y'' = 0.$$

Since $y' = 1/\sqrt{x^2+1}$ is never zero, we may divide by $2y'$, which gives $(x^2+1)y'' + xy' = 0$. That is $P(0)$. (Alternatively, differentiate $y' = (x^2+1)^{-1/2}$ directly: $y'' = -x(x^2+1)^{-3/2}$, so $(x^2+1)y'' = -x(x^2+1)^{-1/2} = -xy'$.)

Inductive step. Suppose $P(k)$ holds for some $k \geq 0$:

$$(x^2+1)\,y^{(k+2)} + (2k+1)\,x\,y^{(k+1)} + k^2\,y^{(k)} = 0.$$

This holds for all x, so we may differentiate it. Using the product rule on the first two terms,

$$2x\,y^{(k+2)} + (x^2+1)\,y^{(k+3)} + (2k+1)\,y^{(k+1)} + (2k+1)\,x\,y^{(k+2)} + k^2\,y^{(k+1)} = 0.$$

Collecting terms:

$$(x^2+1)\,y^{(k+3)} + (2k+3)\,x\,y^{(k+2)} + (k^2+2k+1)\,y^{(k+1)} = 0,$$

and since $2k+3 = 2(k+1)+1$ and $k^2+2k+1 = (k+1)^2$, this is precisely $P(k+1)$.

Ask yourself Why does the coefficient of $y^{(k+2)}$ go up by exactly 2?

Differentiating (x^2+1) in the first term throws off a $2x\,y^{(k+2)}$, which joins the $(2k+1)x\,y^{(k+2)}$ from the second term. Similarly the $(2k+1)$ from differentiating the x in the second term joins k^2, and k^2+2k+1 is a perfect square. The recurrence is built to reproduce itself.

Since $P(0)$ holds and $P(k) \Rightarrow P(k+1)$ for every $k \geq 0$, by induction $P(n)$ holds for all $n \geq 0$.

The Maclaurin series

Put $x = 0$ in $P(n)$. The middle term vanishes, leaving

$$y^{(n+2)}(0) = -n^2\,y^{(n)}(0) \qquad (n \geq 0).$$

We need two starting values: $y(0) = \ln(0+1) = 0$ and $y'(0) = 1/\sqrt{1} = 1$.

Even derivatives. From $y(0) = 0$ the recurrence gives $y''(0) = 0$, then $y^{(4)}(0) = 0$, and so on: every even-order derivative vanishes at 0. (This also follows from y being an odd function: $y(-x) = \ln\left(\sqrt{x^2+1} - x\right) = -\ln\left(\sqrt{x^2+1} + x\right)$, because $\left(\sqrt{x^2+1} - x\right)\left(\sqrt{x^2+1} + x\right) = 1$.)

Odd derivatives. Starting from $y'(0) = 1$,

$$\begin{aligned} y'''(0) &= -1^2 \cdot y'(0) = -1, \\ y^{(5)}(0) &= -3^2 \cdot y'''(0) = 9, \\ y^{(7)}(0) &= -5^2 \cdot y^{(5)}(0) = -225. \end{aligned}$$

Ask yourself Which value of n gives $y^{(5)}(0)$ from $y'''(0)$?

To get $y^{(n+2)}$ with $n+2=5$ we need $n=3$, so the factor is $-3^2=-9$, not -5^2. The factor uses the order of the *lower* derivative.

The Maclaurin series $y=\sum y^{(k)}(0)\,x^k/k!$ is therefore

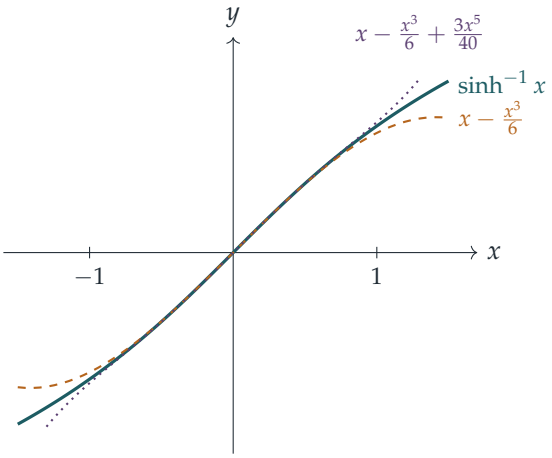
$$\begin{aligned} y &= x+\frac{-1}{3!}x^3+\frac{9}{5!}x^5+\frac{-225}{7!}x^7+\cdots \\ &= x-\frac{x^3}{6}+\frac{9}{120}x^5-\frac{225}{5040}x^7+\cdots \\ &= x-\frac{x^3}{6}+\frac{3x^5}{40}-\frac{5x^7}{112}+\cdots, \end{aligned}$$

which begins as required, and the next non-zero term is

$$-\frac{5x^7}{112}.$$

Another route. Integrating the binomial series $(1+x^2)^{-1/2}=1-\frac{1}{2}x^2+\frac{3}{8}x^4-\frac{5}{16}x^6+\cdots$ term by term (valid for $|x|<1$), with $y(0)=0$, gives $x-\frac{1}{6}x^3+\frac{3}{40}x^5-\frac{5}{112}x^7+\cdots$, which agrees.

The picture shows how the partial sums hug $y=\sinh^{-1}x$ near the origin, alternately overshooting and undershooting further out.



Checks

- At $x=0.1$, $\ln\left(0.1+\sqrt{1.01}\right)=0.0998340789$, while $x-\frac{x^3}{6}+\frac{3x^5}{40}-\frac{5x^7}{112}=0.0998340789$ as well; the difference is about 3×10^{-11}, the size of the next (x^9) term.
- The $n=0$ case is easy to test at $x=1$: $y'=1/\sqrt{2}$ and $y''=-1/(2\sqrt{2})$, so $2y''+y'=0$.

- The odd derivatives at 0 are $1, -1, 9, -225$, that is $(-1)^m\big(1 \cdot 3 \cdots (2m-1)\big)^2$: a pleasing pattern worth spotting.

Common slips

- Using the wrong n in $y^{(n+2)}(0) = -n^2 y^{(n)}(0)$, giving factors $-9, -25, -49$ instead of $-1, -9, -25$.
- Forgetting to divide by $k!$: the coefficient of x^7 is $-225/5040$, not -225.
- In the inductive step, missing the $2x\,y^{(k+2)}$ that comes from differentiating $x^2 + 1$.

Beyond STEP

A Taylor series is only useful where it converges, and this one converges exactly for $|x| \leq 1$, even though $\sinh^{-1} x$ is perfectly smooth on the whole real line. The reason lives in the complex plane: $1/\sqrt{1+z^2}$ blows up at $z = \pm\mathrm{i}$, at distance 1 from the origin, and that distance is the radius of convergence. Needham's *Visual Complex Analysis* explains this beautifully; for the real-variable story of Taylor's theorem and its remainder, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2001, Question 2

Pure • Hyperbolic Substitution and Area

STEP III 2001, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q2

The key idea

Both inequalities say "stay on one side of a rectangular-type hyperbola", so the first job is to sketch the two hyperbolas and find where they cross. The region turns out to be a bounded, four-fold symmetric patch, and its area is a pair of integrals of the form $\int \sqrt{x^2 \pm a^2}\,\mathrm{d}x$. Those are exactly the integrals that hyperbolic substitutions tame, and the first part is there so that we can turn the resulting $\cosh^{-1}$ (and $\sinh^{-1}$) values into logarithms.

Solution

The logarithmic form of $\cosh^{-1} x$

Let $x \geq 1$ and $y = \cosh^{-1} x$, which by definition is the *non-negative* number with $\cosh y = x$. Then

$$x = \frac{\mathrm{e}^y + \mathrm{e}^{-y}}{2} \quad \Longrightarrow \quad \mathrm{e}^{2y} - 2x\,\mathrm{e}^y + 1 = 0,$$

a quadratic in e^y, so

$$\mathrm{e}^y = x \pm \sqrt{x^2 - 1}.$$

? **Ask yourself** Which sign do we take, and why?

The two roots multiply to 1 (the constant term of the quadratic), and the root with the plus sign is at least 1. So the root with the minus sign is at most 1, which would give $y \leq 0$. Since $y \geq 0$ we must take the plus sign (when $x = 1$ the two roots coincide anyway), and so

$$\cosh^{-1} x = \ln\bigl(x + \sqrt{x^2 - 1}\bigr),$$

as required. Exactly the same argument with $x = \sinh y$ gives $\mathrm{e}^{2y} - 2x\,\mathrm{e}^y - 1 = 0$; now the roots multiply to -1, so only the plus root is positive, and $\sinh^{-1} x = \ln\bigl(x + \sqrt{x^2 + 1}\bigr)$. We shall use both.

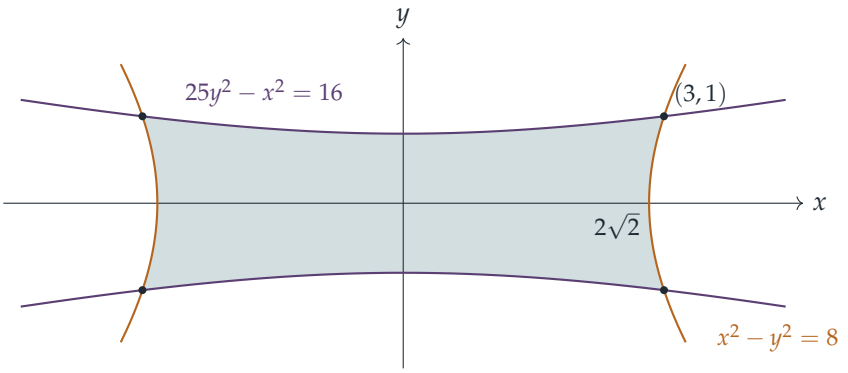
The region

Rewrite the inequalities as

$$x^2 - y^2 \leq 8 \qquad \text{and} \qquad 25y^2 - x^2 \leq 16.$$

The curve $x^2 - y^2 = 8$ is a hyperbola with branches opening left and right through $(\pm 2\sqrt{2}, 0)$; the first inequality keeps us *between* its branches, i.e. $|x| \leq \sqrt{y^2 + 8}$. The curve $25y^2 - x^2 = 16$ is a hyperbola with branches opening up and down through $(0, \pm\frac{4}{5})$; the second inequality keeps us between those branches, i.e. $|y| \leq \frac{1}{5}\sqrt{x^2 + 16}$.

The curves cross where $x^2 = y^2 + 8$ and $25y^2 = x^2 + 16 = y^2 + 24$, so $y^2 = 1$ and $x^2 = 9$: at the four points $(\pm 3, \pm 1)$.



The region (shaded) is bounded: if (x, y) is in it then $x^2 \leq y^2 + 8 \leq \frac{1}{25}(x^2 + 16) + 8$, so $x^2 \leq 9$, and then $y^2 \leq \frac{25}{25} = 1$. Both inequalities are unchanged by $x \mapsto -x$ and $y \mapsto -y$, so the region is symmetric in both axes and its area is four times the area A of the part in the first quadrant.

In the first quadrant we slice vertically.

- For $0 \leq x \leq 2\sqrt{2}$ we have $x^2 - 8 \leq 0 \leq y^2$, so the first inequality holds automatically, and y runs from 0 up to $\frac{1}{5}\sqrt{x^2 + 16}$.
- For $2\sqrt{2} \leq x \leq 3$, y runs from $\sqrt{x^2 - 8}$ up to $\frac{1}{5}\sqrt{x^2 + 16}$ (and the lower curve is below the upper one exactly when $x^2 \leq 9$, by the calculation of the crossing points).

Hence

$$A = \frac{1}{5}\int_0^3 \sqrt{x^2 + 16}\,\mathrm{d}x \; - \; \int_{2\sqrt{2}}^3 \sqrt{x^2 - 8}\,\mathrm{d}x = \tfrac{1}{5}I_1 - I_2.$$

The first integral

Substitute $x = 4\sinh s$, so $\mathrm{d}x = 4\cosh s\,\mathrm{d}s$ and $\sqrt{x^2 + 16} = 4\cosh s$ (positive, since $\cosh s > 0$). The limits are $s = 0$ and $s = S = \sinh^{-1}\frac{3}{4} = \ln\left(\frac{3}{4} + \frac{5}{4}\right) = \ln 2$. Using

$\cosh^2 s = \frac{1}{2}(\cosh 2s + 1)$,

$$I_1 = \int_0^S 16\cosh^2 s\,\mathrm{d}s = \int_0^S 8(\cosh 2s + 1)\,\mathrm{d}s = \Big[\,8\sinh s\cosh s + 8s\,\Big]_0^S$$
$$= 8\cdot\tfrac{3}{4}\cdot\tfrac{5}{4} + 8\ln 2 = \tfrac{15}{2} + 8\ln 2,$$

since $\sinh S = \frac{3}{4}$ and $\cosh S = \sqrt{1+\frac{9}{16}} = \frac{5}{4}$.

The second integral

Substitute $x = 2\sqrt{2}\cosh t$ with $t \geq 0$, so $\mathrm{d}x = 2\sqrt{2}\sinh t\,\mathrm{d}t$ and $\sqrt{x^2-8} = 2\sqrt{2}\sinh t$ (positive, since $t \geq 0$). The limits are $t = 0$ and

$$t = T = \cosh^{-1}\frac{3}{2\sqrt{2}} = \ln\Big(\frac{3}{2\sqrt{2}} + \sqrt{\tfrac{9}{8}-1}\Big) = \ln\frac{4}{2\sqrt{2}} = \ln\sqrt{2} = \tfrac{1}{2}\ln 2,$$

by the first part. Using $\sinh^2 t = \frac{1}{2}(\cosh 2t - 1)$,

$$I_2 = \int_0^T 8\sinh^2 t\,\mathrm{d}t = \int_0^T 4(\cosh 2t - 1)\,\mathrm{d}t = \Big[\,4\sinh t\cosh t - 4t\,\Big]_0^T$$
$$= 4\cdot\frac{1}{2\sqrt{2}}\cdot\frac{3}{2\sqrt{2}} - 2\ln 2 = \tfrac{3}{2} - 2\ln 2,$$

since $\cosh T = \frac{3}{2\sqrt{2}}$ and $\sinh T = \sqrt{\frac{9}{8}-1} = \frac{1}{2\sqrt{2}}$.

Putting it together

$$A = \tfrac{1}{5}\Big(\tfrac{15}{2} + 8\ln 2\Big) - \Big(\tfrac{3}{2} - 2\ln 2\Big) = \tfrac{3}{2} + \tfrac{8}{5}\ln 2 - \tfrac{3}{2} + 2\ln 2 = \tfrac{18}{5}\ln 2.$$

The algebraic parts cancel exactly, and the total area is

$$4A = \frac{72}{5}\ln 2,$$

as required.

Checks

- $\frac{72}{5}\ln 2 \approx 9.981$. Simpson's rule on the two integrals gives $I_1 \approx 13.0452$ and $I_2 \approx 0.11371$, matching $\frac{15}{2} + 8\ln 2$ and $\frac{3}{2} - 2\ln 2$; and simply counting points of a fine grid that satisfy both inequalities gives an area of about 9.98.
- Sanity check on size: the region fits inside the rectangle $[-3,3]\times[-1,1]$ of area 12, and it misses the corners of that rectangle only by thin slivers, so an area just under 10 is about right.
- $\cosh^{-1} 1 = \ln 1 = 0$, as it should be.

Common slips

- Reading the inequalities as "outside" the hyperbolas: $y^2 \geq x^2 - 8$ is the region *between* the branches of $x^2 - y^2 = 8$, which contains the y-axis.
- Forgetting that the first inequality only bites for $x > 2\sqrt{2}$, and so subtracting $\int_0^3 \sqrt{x^2 - 8}\,\mathrm{d}x$, which is not even real.
- Taking the minus sign in $\mathrm{e}^y = x \pm \sqrt{x^2 - 1}$, which gives $-\cosh^{-1} x$.

Beyond STEP

The two families $\int \sqrt{x^2 + a^2}\,\mathrm{d}x$ and $\int \sqrt{x^2 - a^2}\,\mathrm{d}x$ are handled by hyperbolic substitutions for the same reason $\int \sqrt{a^2 - x^2}\,\mathrm{d}x$ is handled by $x = a\sin\theta$: the identity $\cosh^2 - \sinh^2 = 1$ removes the square root. In fact the hyperbolic functions measure *area*: the parameter t in $(\cosh t, \sinh t)$ is twice the area of the sector of the unit hyperbola, just as θ is twice the area of a circular sector. Siklos, *Advanced Problems in Mathematics*, has many more questions of this flavour, and Bartle and Sherbert, *Introduction to Real Analysis*, shows how the exponential, logarithm and hence the hyperbolic functions are put on a rigorous footing.

WORKED SOLUTION

STEP III 2001, Question 3

Pure • Roots of Quadratics

STEP III 2001, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q3

The key idea

Translate statements about the roots into statements about the coefficients. With roots α and β we have $\alpha + \beta = b$ and $\alpha\beta = c$, and the discriminant tells us when the roots are real. For part (ii) it's easier to think about the graph of $f(x) = x^2 - bx + c$: both roots lie strictly between -1 and 1 exactly when the parabola is positive at ± 1 and its vertex lies between them.

Solution

Throughout, write $f(x) = x^2 - bx + c$, and when the roots are real call them $\alpha \leq \beta$, so that

$$\alpha + \beta = b, \qquad \alpha\beta = c, \qquad \alpha, \beta = \tfrac{1}{2}\left(b \mp \sqrt{b^2 - 4c}\right).$$

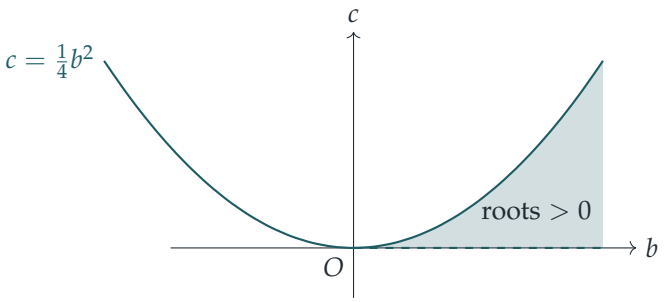
A repeated root counts as two equal real roots.

Part (i)

($\Rightarrow$) Suppose the roots are real and positive. Real roots need $b^2 - 4c \geq 0$. Then $b = \alpha + \beta > 0$ and $c = \alpha\beta > 0$, since both roots are positive. Together these give $b > 0$ and $b^2 \geq 4c > 0$.

($\Leftarrow$) Suppose $b > 0$ and $b^2 \geq 4c > 0$. Since $b^2 - 4c \geq 0$, the roots are real. Since $\alpha\beta = c > 0$, they are non-zero and have the same sign. Since $\alpha + \beta = b > 0$, that common sign can't be negative. So both roots are positive.

This proves the equivalence. In the b-c plane the conditions say $b > 0$ and $0 < c \leq \frac{1}{4}b^2$: the region in the first quadrant between the positive b-axis and the parabola $c = \frac{1}{4}b^2$. The parabola is included (repeated root $\frac{1}{2}b > 0$), and the b-axis is excluded (there $c = 0$ and one root is 0).



The solid curve (for $b > 0$) belongs to the region; the dashed segment of the b-axis and the origin do not.

Part (ii)

We need both roots real with $-1 < \alpha \leq \beta < 1$. We claim that, given real roots, this happens if and only if

$$f(1) = 1 - b + c > 0, \qquad f(-1) = 1 + b + c > 0, \qquad -1 < \tfrac{1}{2}b < 1.$$

($\Rightarrow$) If $-1 < \alpha \leq \beta < 1$ then $f(1) = (1-\alpha)(1-\beta) > 0$ and $f(-1) = (1+\alpha)(1+\beta) > 0$, as each bracket is positive. Also $\frac{1}{2}b = \frac{1}{2}(\alpha+\beta)$ is the average of two numbers in $(-1,1)$, so it lies in $(-1,1)$.

($\Leftarrow$) Because $f(x) = (x-\alpha)(x-\beta)$ is ≤ 0 exactly on $[\alpha,\beta]$, the condition $f(1) > 0$ says that 1 is not in $[\alpha,\beta]$. So either $1 < \alpha$ or $1 > \beta$. The vertex $\frac{1}{2}b$ lies in $[\alpha,\beta]$, and $\frac{1}{2}b < 1$, so $\alpha \leq \frac{1}{2}b < 1$. That rules out $1 < \alpha$, so $\beta < 1$. In the same way, $f(-1) > 0$ and $\frac{1}{2}b > -1$ give $\alpha > -1$.

? **Ask yourself** Why can't we just use $f(1) > 0$ and $f(-1) > 0$ on their own?

Both roots could lie on the same side of the interval. For example, $x^2 - 5x + 6 = 0$ has roots 2 and 3, yet $f(\pm 1) > 0$. The vertex condition $|b| < 2$ is what puts the roots between -1 and 1 rather than both beyond one end.

Combining these with the reality condition $b^2 \geq 4c$, the region is

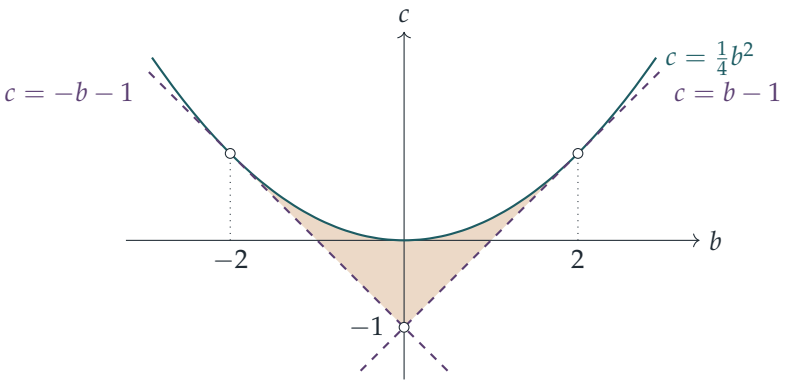
$$|b| < 2, \qquad c > b - 1, \qquad c > -b - 1, \qquad c \leq \tfrac{1}{4}b^2,$$

that is, $|b| < 2$ and $|b| - 1 < c \leq \frac{1}{4}b^2$.

To sketch it, note how the lines meet the parabola. Since

$$\tfrac{1}{4}b^2 - (b-1) = \tfrac{1}{4}(b-2)^2 \geq 0,$$

the line $c = b - 1$ lies below the parabola and touches it only at $(2,1)$. So it is the tangent there. By symmetry ($b \mapsto -b$), $c = -b - 1$ is the tangent at $(-2,1)$. The two lines cross at $(0,-1)$. The region is the curvilinear triangle bounded by these two tangents and the arc of the parabola between the points of contact.



The arc of the parabola with $|b| < 2$ is included (a repeated root $\frac{1}{2}b$ with $|\frac{1}{2}b| < 1$). The two dashed tangent segments are excluded, because on them one root is ± 1. The three corners $(\pm 2, 1)$ and $(0, -1)$ are excluded too.

Checks

- $(b, c) = (0, -\frac{1}{4})$ is inside the part (ii) region: the roots are $\pm\frac{1}{2}$. $(b, c) = (0, -1)$ is the excluded corner: the roots are exactly ± 1.
- $(b, c) = (2.7, 1.8)$ satisfies $c > |b| - 1$ and $c \leq \frac{1}{4}b^2$ but has $|b| > 2$. Its roots are 1.2 and 1.5, so it is correctly left out by the vertex condition.
- We tested both regions numerically, comparing the exact roots with the stated conditions at over 115 000 rational points (b, c), including points on every boundary curve. There were no disagreements.

Common slips

- Forgetting the vertex condition $|b| < 2$ in part (ii), which wrongly adds the thin slivers between each tangent line and the parabola beyond $|b| = 2$.
- Getting the boundaries wrong: the parabola is included in both parts, but the b-axis in (i) and the tangent lines in (ii) are not.

Beyond STEP

The conditions on $f(\pm 1)$ and the vertex are a small case of a general question: when do all roots of a polynomial lie in a given region? For the unit disc this is answered by the Schur–Cohn test. For the left half-plane it is the Routh–Hurwitz criterion, which control engineers use to decide whether a system is stable. Part (ii) is exactly the stability region for the recurrence $u_{n+2} = bu_{n+1} - cu_n$ with real characteristic roots. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP III 2001, Question 4

Pure • Inverse Trigonometric Functions

STEP III 2001, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q4

The key idea

The expression $\dfrac{2x}{1+x^2}$ should ring a bell: it is the "t-formula" for $\sin 2\theta$ when $x = \tan\theta$. Substituting $x = \tan\theta$ turns $\sin^{-1}$ of the expression into $\sin^{-1}(\sin 2\theta)$, and the whole question becomes one about when $\sin^{-1}(\sin\phi)$ really does give back ϕ. The answer depends on whether ϕ lies in the principal range $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$, and that is exactly where the split at $x = \pm 1$ comes from.

Solution

Part (i)

First some basic features of $g(x) = \dfrac{2x}{1+x^2}$.

- The denominator is never zero, so g is defined and continuous everywhere.
- $g(-x) = -g(x)$, so g is odd and its graph has rotational symmetry about the origin.
- $g(0) = 0$; $g(x)$ has the same sign as x; and $g(x) \to 0$ as $x \to \pm\infty$, since the denominator grows like x^2 and the numerator only like x. So the x-axis is an asymptote, approached from above as $x \to \infty$ and from below as $x \to -\infty$.

By the quotient rule,

$$g'(x) = \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} = \frac{2(1-x^2)}{(1+x^2)^2}.$$

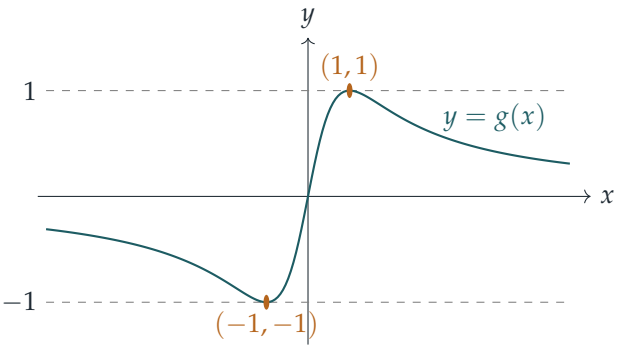
This is zero only at $x = \pm 1$. It is positive for $|x| < 1$ and negative for $|x| > 1$, so $x = 1$ gives a maximum, $g(1) = 1$, and $x = -1$ gives a minimum, $g(-1) = -1$. The gradient at the origin is $g'(0) = 2$.

We can confirm that these turning points are global without calculus:

$$1 - g(x) = \frac{(1-x)^2}{1+x^2} \geq 0, \qquad 1 + g(x) = \frac{(1+x)^2}{1+x^2} \geq 0,$$

so $-1 \leq g(x) \leq 1$ for every x, with equality only at $x = \mp 1$. Since g is continuous and takes the values -1 and 1, it takes every value in between (by the intermediate value theorem). Hence

$$\text{the range of } g \text{ is } \; -1 \leq g(x) \leq 1.$$



Part (ii)

By part (i), $\frac{2x}{1+x^2}$ always lies in $[-1, 1]$, the domain of $\sin^{-1}$, so f is defined for every real x.

Let $\theta = \tan^{-1} x$, so that $x = \tan\theta$ with $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. Then

$$\frac{2x}{1+x^2} = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2\tan\theta}{\sec^2\theta} = 2\sin\theta\cos\theta = \sin 2\theta,$$

and so

$$f(x) = \sin^{-1}(\sin 2\theta).$$

Ask yourself Isn't $\sin^{-1}(\sin 2\theta)$ just 2θ?

Only when 2θ is in the range of $\sin^{-1}$, namely $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$. By definition, $\sin^{-1} y$ is the *unique* angle in that interval whose sine is y. So to find $f(x)$ we must find the angle in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ with the same sine as 2θ.

Case $-1 \leq x \leq 1$**.** Since tan is increasing on $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, this means $-\frac{1}{4}\pi \leq \theta \leq \frac{1}{4}\pi$, so $-\frac{1}{2}\pi \leq 2\theta \leq \frac{1}{2}\pi$. Then 2θ is itself the angle in the principal range with sine $\sin 2\theta$, so

$$f(x) = 2\theta = 2\tan^{-1} x \qquad (-1 \leq x \leq 1).$$

Case $x \geq 1$**.** Now $\frac{1}{4}\pi \leq \theta < \frac{1}{2}\pi$, so $\frac{1}{2}\pi \leq 2\theta < \pi$, which is outside the principal range. But

$$\sin(\pi - 2\theta) = \sin 2\theta \qquad \text{and} \qquad 0 < \pi - 2\theta \leq \tfrac{1}{2}\pi,$$

so $\pi - 2\theta$ is the angle in the principal range with the right sine. Hence

$$f(x) = \pi - 2\theta = \pi - 2\tan^{-1} x \qquad (x \geq 1).$$

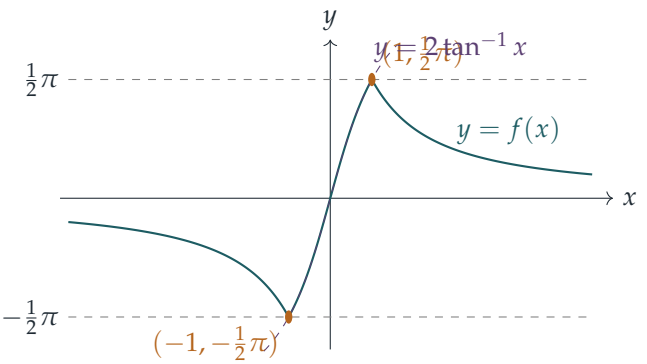
At $x = 1$ both formulae give $\frac{1}{2}\pi$, as they must. This completes the "show that".

For the sketch we also want $x \leq -1$. Since g is odd and $\sin^{-1}$ is odd, f is odd, so $f(x) = -f(-x) = -\pi - 2\tan^{-1} x$ for $x \leq -1$. (Equally: $-\pi < 2\theta \leq -\frac{1}{2}\pi$ there, and $-\pi - 2\theta$ is the angle in $[-\frac{1}{2}\pi, 0)$ with the same sine.)

Shape of the graph.

- On $[-1, 1]$, $f(x) = 2\tan^{-1} x$ increases from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, through the origin with gradient 2.
- For $x \geq 1$, $f(x) = \pi - 2\tan^{-1} x$ decreases from $\frac{1}{2}\pi$, and tends to $\pi - 2 \cdot \frac{1}{2}\pi = 0$ from above as $x \to \infty$. By oddness, for $x \leq -1$ it rises from $-\frac{1}{2}\pi$ towards 0 from below as $x \to -\infty$. So the x-axis is again an asymptote.
- The gradient is $\dfrac{2}{1+x^2}$ for $|x| < 1$ and $-\dfrac{2}{1+x^2}$ for $|x| > 1$. At $x = 1$ it jumps from $+1$ to -1 (and similarly at $x = -1$), so the graph has sharp corners at $(\pm 1, \pm\frac{1}{2}\pi)$: f is continuous but not differentiable there.

You can see the corners directly from the chain rule: $f'(x) = g'(x)/\sqrt{1 - g(x)^2}$, and part (i) gave $1 - g^2 = (1-x^2)^2/(1+x^2)^2$, whose square root is $|1-x^2|/(1+x^2)$. So $f'(x) = \dfrac{2(1-x^2)}{(1+x^2)\,|1-x^2|} = \pm\dfrac{2}{1+x^2}$, with the sign of $1 - x^2$.



The dashed curve shows $2\tan^{-1} x$ continuing past $x = 1$: it would climb above $\frac{1}{2}\pi$, which $\sin^{-1}$ can never do, so f "reflects" back down instead.

Checks

- At $x = \sqrt{3}$: $g(\sqrt{3}) = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$, so $f(\sqrt{3}) = \frac{1}{3}\pi$; and the formula gives $\pi - 2 \cdot \frac{1}{3}\pi = \frac{1}{3}\pi$. Agreed.

- At $x = 10$: $f(10) = \sin^{-1}(20/101) \approx 0.19934$, and $\pi - 2\tan^{-1} 10 \approx 0.19934$. A computer testing 10^5 random values of x in each of $[-1, 1]$, $[1, 10^4]$ and $[-10^4, -1]$ found agreement to 10^{-7}, and the one-sided gradients at $x = 1$ came out as $+1$ and -1.
- If f were $2\tan^{-1} x$ everywhere it would reach nearly π for large x, impossible for a value of $\sin^{-1}$; the reflection fixes this.

Common slips

- Writing $\sin^{-1}(\sin 2\theta) = 2\theta$ without checking that 2θ is in $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$.
- Drawing f with a smooth rounded top at $x = 1$. The graph has a corner there, because $\sqrt{1 - g^2}$ involves $|1 - x^2|$, not $1 - x^2$.
- Forgetting that f tends to 0 (not $\pm\pi$) as $x \to \pm\infty$.

Beyond STEP

Inverse trigonometric functions are a first example of choosing a *branch*: sin is not one-to-one, so we must restrict its domain before inverting, and identities like $\sin^{-1}(\sin\phi) = \phi$ then hold only on that branch. The same issue becomes central in complex analysis, where $\log z$ and $z^{1/2}$ need branch cuts. The corner at $x = 1$ is also a neat example of a function built from smooth pieces that is continuous but not differentiable at a point. For branches and multi-valued functions with plenty of pictures, see Needham, *Visual Complex Analysis*; for continuity and differentiability, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2001, Question 5

Pure • Normals to a Parabola

STEP III 2001, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q5

The key idea

Asking which normals pass through a fixed point (h, k) turns into asking how many real roots a cubic in the parameter t has. The cubic has no t^2 term, so it has exactly the shape $t^3 + pt + q$ from the opening part. The number of real roots changes only when two roots merge into a repeated root, so the boundary between "one normal" and "three normals" is where the cubic has a repeated root. That boundary is the locus we want.

Solution

The cubic

Let $f(x) = x^3 + px + q$. Then $f'(x) = 3x^2 + p \geq 0$ for all x when $p \geq 0$, and $f'(x) = 0$ can happen at most at the single point $x = 0$ (only when $p = 0$). So f is strictly increasing on $\mathbb{R}$: if $x_1 < x_2$ then $f(x_2) - f(x_1) = \int_{x_1}^{x_2} f'(x)\,\mathrm{d}x > 0$, because the integrand is non-negative and zero at no more than one point.

Also $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to +\infty$ as $x \to +\infty$. Since f is continuous, it takes the value 0 somewhere, and because it is strictly increasing it does so only once. So there is exactly one real solution.

The normals through (h, k)

On C, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \dfrac{2a}{2at} = \dfrac{1}{t}$, so the normal at the point with parameter t has gradient $-t$. (This also works at $t = 0$, where the tangent is vertical and the normal is the x-axis.) The normal is

$$y - 2at = -t\,(x - at^2), \qquad \text{that is} \qquad y + tx = 2at + at^3.$$

It passes through (h,k) exactly when $k+th = 2at + at^3$, i.e.

$$\boxed{at^3 + (2a-h)\,t - k = 0} \qquad \text{or, dividing by } a>0, \qquad t^3 + \frac{2a-h}{a}\,t - \frac{k}{a} = 0.$$

Ask yourself Could two different values of t give the same normal line, so that counting roots overcounts normals?

No. The normal at t has gradient $-t$, so different values of t give normals with different gradients, and these are different lines. So the number of normals through (h,k) is exactly the number of distinct real roots of the cubic.

If $h \le 2a$ then the coefficient $p = \dfrac{2a-h}{a}$ is ≥ 0, so by the first part the cubic has exactly one real root. Hence exactly one normal to C passes through (h,k), as required.

The locus of points with exactly two normals

Write the cubic as $t^3 + Pt + Q = 0$ with $P = \dfrac{2a-h}{a}$ and $Q = -\dfrac{k}{a}$. A real cubic has either one or three real roots counted with multiplicity. So it has exactly two *distinct* real roots exactly when it has a double root s and a different simple root.

The roots add up to 0, since there is no t^2 term, so the simple root is $-2s$. We need $s \neq 0$ so that $-2s \neq s$. Comparing coefficients,

$$t^3 + Pt + Q = (t-s)^2(t+2s) = t^3 - 3s^2 t + 2s^3,$$

so $P = -3s^2$ and $Q = 2s^3$. Translating back:

$$h = 2a + 3as^2, \qquad k = -2as^3, \qquad s \neq 0.$$

To eliminate s, note $(h-2a)^3 = 27a^3s^6$ and $k^2 = 4a^2s^6$, so $4(h-2a)^3 = 108a^3s^6 = 27a\,k^2$.

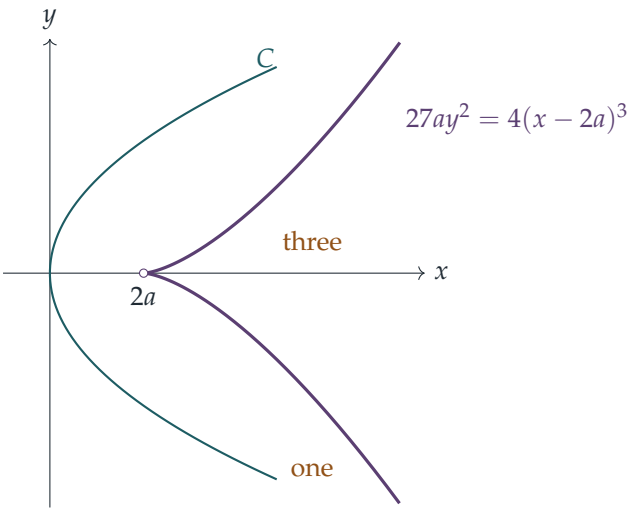
Conversely, suppose $27ak^2 = 4(h-2a)^3$ with $(h,k) \neq (2a,0)$. Then $h - 2a \ge 0$, and in fact $h > 2a$ (if $h = 2a$ then $k = 0$). Let s be the real number with $2as^3 = -k$. Then $4a^2s^6 = k^2 = \frac{4}{27a}(h-2a)^3$, so $(h-2a)^3 = (3as^2)^3$ and therefore $h - 2a = 3as^2$. So (h,k) has the form above with $s \neq 0$, and exactly two normals pass through it. (Equivalently, this is the repeated-root condition $4P^3 + 27Q^2 = 0$.)

The point $(2a,0)$ itself is excluded. There $P = Q = 0$, the cubic is $t^3 = 0$ with a triple root, and only one normal (the x-axis) passes through it.

So the required locus is

$$\boxed{27a\,y^2 = 4\,(x-2a)^3}, \qquad \text{excluding the point } (2a,0).$$

This is a semicubical parabola with a cusp at $(2a,0)$, symmetric about the x-axis and lying in $x \ge 2a$, with $y = \pm\dfrac{2}{3\sqrt{3a}}(x-2a)^{3/2}$. Inside it (the region containing the positive x-axis beyond $2a$) the cubic has three distinct real roots, so three normals pass through each point. Outside it only one normal does.



The sketch uses $a = 1$. The open circle marks the excluded cusp, and the labels give the number of normals through points in each region.

Checks

- Take $a = 1$ and $s = 1$, which gives the point $(5, -2)$. The cubic is $t^3 - 3t + 2 = (t-1)^2(t+2)$, so the normals at $(1, 2)$ and $(4, -4)$ both pass through $(5, -2)$. Indeed, the normal at $t = 1$ is $y + x = 3$ and the normal at $t = -2$ is $y - 2x = -12$. Both contain $(5, -2)$, and $27 \cdot 4 = 108 = 4 \cdot 27$.
- At $(5, 0.1)$ (with $a = 1$), $4P^3 + 27Q^2 = -108 + 0.27 < 0$, giving three normals. At $(5, 4)$ it is $-108 + 432 > 0$, giving one normal. This matches the sketch.
- We tested the normal equation and the one-root claims numerically on hundreds of random cases.

Common slips

- Forgetting to exclude the cusp $(2a, 0)$. The equation is satisfied there, but there is only one normal.
- Claiming $f'(x) > 0$ when $p \geq 0$. For $p = 0$, $f'(0) = 0$, so you need the slightly stronger argument that f is still strictly increasing.
- Writing the gradient of the normal as $1/t$ (the tangent's gradient) instead of $-t$.

Beyond STEP

The locus is the *evolute* of the parabola: the curve traced out by its centres of curvature, and also the envelope of its normals. Crossing the evolute changes the number of

normals from a point by two, which is a first example of the ideas behind catastrophe theory and the discriminant of a cubic. Siklos, *Advanced Problems in Mathematics*, has more STEP problems in this spirit. Bruce and Giblin, *Curves and Singularities*, develops evolutes and envelopes properly.

WORKED SOLUTION

STEP III 2001, Question 6

Pure • Vectors and Triangle Centres

STEP III 2001, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q6

The key idea

Everything is controlled by the normal vector $\mathbf{n} = \left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right)$ of the plane: a point $\mathbf{r}$ lies in the plane exactly when $\mathbf{r} \cdot \mathbf{n} = 1$, and the perpendiculars from O and M both run along $\mathbf{n}$. The orthocentre and circumcentre come out of two pieces of 3D geometry: OA, OB, OC are mutually perpendicular, and M is the centre of the box with edges OA, OB, OC, so it is equidistant from O, A, B and C. For the last part, the two perpendiculars are parallel, so a pair of similar triangles (or one line of vector algebra) does the rest.

Solution

Throughout, a, b, c are non-zero (otherwise the plane doesn't meet all three axes), so

$$A = (a, 0, 0), \qquad B = (0, b, 0), \qquad C = (0, 0, c),$$

and we write $S = \dfrac{1}{a^2} + \dfrac{1}{b^2} + \dfrac{1}{c^2} = |\mathbf{n}|^2 > 0$.

The centroid

Points of the line OM are $t\left(\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c\right)$. Substituting into the plane's equation gives $\frac{1}{2}t + \frac{1}{2}t + \frac{1}{2}t = 1$, so $t = \frac{2}{3}$, and the meeting point is

$$G = \left(\tfrac{1}{3}a, \tfrac{1}{3}b, \tfrac{1}{3}c\right) = \tfrac{1}{3}(A + B + C),$$

which is the centroid of triangle ABC. Note for later that $\overrightarrow{OG} = \frac{2}{3}\overrightarrow{OM}$.

The perpendicular from O: the orthocentre

The perpendicular from O is the line $\lambda\mathbf{n}$. It meets the plane where $\lambda\,\mathbf{n}\cdot\mathbf{n} = 1$, that is $\lambda = 1/S$, at

$$H = \frac{1}{S}\left(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}\right).$$

We show that $AH \perp BC$. Using $\overrightarrow{BC} = (0, -b, c)$,

$$\overrightarrow{AH}\cdot\overrightarrow{BC} = \left(\frac{1}{aS} - a\right)\cdot 0 + \frac{1}{bS}\cdot(-b) + \frac{1}{cS}\cdot c = -\frac{1}{S} + \frac{1}{S} = 0.$$

> **Ask yourself** Is there a reason this dot product had to vanish, rather than just happening to?

Yes. BC lies in the plane, so it is perpendicular to the normal OH; and BC lies in the yz-plane, so it is perpendicular to OA, which is along the x-axis. A line perpendicular to two non-parallel lines OA and OH is perpendicular to the whole plane OAH, and in particular to AH.

By the same argument with the letters permuted, $BH \perp CA$ and $CH \perp AB$. Since H lies in the plane of the triangle, $AH \perp BC$ says that H lies on the altitude from A, and similarly for the other two altitudes. So H is the orthocentre of triangle ABC.

The perpendicular from M: the circumcentre

The perpendicular from M is $M + \mu\mathbf{n}$. It meets the plane where $\frac{3}{2} + \mu S = 1$, so $\mu = -\dfrac{1}{2S}$, at

$$P = M - \frac{1}{2S}\,\mathbf{n}.$$

Now M is equidistant from all four of O, A, B, C: for example

$$MA^2 = \left(\tfrac{1}{2}a - a\right)^2 + \left(\tfrac{1}{2}b\right)^2 + \left(\tfrac{1}{2}c\right)^2 = \tfrac{1}{4}\left(a^2 + b^2 + c^2\right),$$

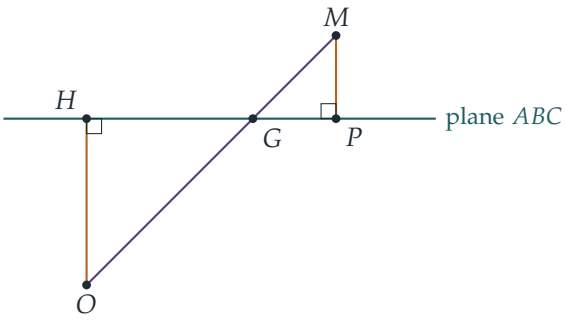
and MB^2, MC^2 (and MO^2) give the same value by symmetry. Since MP is perpendicular to the plane and PA lies in it, triangle MPA has a right angle at P, so by Pythagoras

$$PA^2 = MA^2 - MP^2,$$

and likewise $PB^2 = MB^2 - MP^2$ and $PC^2 = MC^2 - MP^2$. The right-hand sides are equal, so $PA = PB = PC$. The only point of the plane equidistant from the three vertices is the circumcentre, so P is the circumcentre of triangle ABC.

The final part

The perpendiculars OH and MP are both parallel to $\mathbf{n}$, so O, H, M, P lie in one plane, and the line OM crosses the triangle's plane at G. Also O and M are on opposite sides of the triangle's plane, since $O\cdot\mathbf{n} = 0 < 1$ and $M\cdot\mathbf{n} = \frac{3}{2} > 1$. Here is the cross-section through these four points:



Triangles OHG and MPG have right angles at H and P and equal vertically opposite angles at G, so they are similar, with ratio $OG : GM = 2 : 1$. Hence $HG : GP = 2 : 1$, and G lies on the segment HP.

The same conclusion drops out of the vectors directly. With $H = \frac{1}{S}\mathbf{n}$, $P = M - \frac{1}{2S}\mathbf{n}$ and $G = \frac{2}{3}M$,

$$\tfrac{1}{3}(H + 2P) = \tfrac{1}{3}\left(\tfrac{1}{S}\mathbf{n} + 2M - \tfrac{1}{S}\mathbf{n}\right) = \tfrac{2}{3}M = G.$$

So G lies on the segment from the orthocentre H to the circumcentre P and divides it in the ratio $HG : GP = 2 : 1$, as required. (If the triangle is equilateral then $H = P = G$ and there is nothing to prove.)

Ask yourself Does this prove the result for *every* triangle?

Not quite. For the argument to apply, a triangle has to be congruent to some triangle ABC of the type above. Its sides then satisfy $BC^2 = b^2 + c^2$, $CA^2 = c^2 + a^2$ and $AB^2 = a^2 + b^2$, so

$$AB^2 + AC^2 - BC^2 = 2a^2 > 0,$$

and similarly at B and C. By the cosine rule every angle is acute. Conversely, any acute triangle can be realised: solve these three equations for a^2, b^2, c^2, which are all positive. So the method proves the result for acute triangles. For a right-angled or obtuse triangle, here is a short planar argument that covers every case. Put the circumcentre at the origin, so the vertices $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all have the same length. Let $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$. Then

$$(\mathbf{h} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{c}) = (\mathbf{b} + \mathbf{c}) \cdot (\mathbf{b} - \mathbf{c}) = |\mathbf{b}|^2 - |\mathbf{c}|^2 = 0,$$

so $\mathbf{h}$ is on the altitude from $\mathbf{a}$, and by symmetry on the other two altitudes: it is the orthocentre. The centroid is $\frac{1}{3}\mathbf{h} = \frac{1}{3}(\mathbf{h} + 2 \cdot \mathbf{0})$, which is the same $2 : 1$ statement.

Checks

Take $a = 1$, $b = c = 2$, so $S = \frac{3}{2}$. Then $H = \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $P = \left(\frac{1}{6}, \frac{5}{6}, \frac{5}{6}\right)$ and $G = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$. Each satisfies $x + \frac{1}{2}y + \frac{1}{2}z = 1$. You can check that $\overrightarrow{AH} \cdot \overrightarrow{BC} = 0$ and $PA^2 = PB^2 = PC^2 = \frac{25}{12}$, and that $\frac{1}{3}(H + 2P) = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right) = G$. We also checked all of these facts in exact arithmetic for 3000 random choices of a, b, c of both signs.

Common slips

- Showing only $AH \perp BC$ and stopping. You need either all three altitudes, or a clear "by symmetry". You also need to say that H lies in the triangle's plane: a point off the plane can have $AH \perp BC$ without being on the altitude.
- Getting the ratio the wrong way round. The centroid is twice as far from the orthocentre as from the circumcentre.
- Claiming the "hence" covers every triangle. As shown above, it only reaches acute ones.

Beyond STEP

The line through the orthocentre, centroid and circumcentre is the *Euler line*. The fact $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$ (with the circumcentre at the origin) is the starting point for the nine-point circle, whose centre is also on the Euler line, halfway between H and the circumcentre. The trick of seeing a plane figure as a slice of a simple 3D one is a recurring idea in geometry. Coxeter's *Introduction to Geometry* and Coxeter and Greitzer's *Geometry Revisited* both treat the Euler line and the nine-point circle beautifully.

WORKED SOLUTION

STEP III 2001, Question 7

Pure • Orthogonal Trajectories

STEP III 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q7

The key idea

The differential equation is separable, so solving it shows directly what the family of curves is. For the orthogonal curve, perpendicular tangents mean the gradients multiply to -1. That gives a second separable equation, and when we integrate it the function from the opening sketch turns up. The first sketch is there so that we can draw the final curve almost for free.

Solution

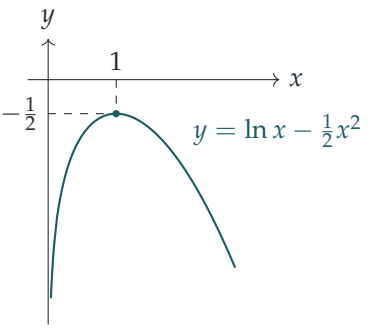
The graph of $\ln x - \frac{1}{2}x^2$

Let $f(x) = \ln x - \frac{1}{2}x^2$, defined for $x > 0$. Then

$$f'(x) = \frac{1}{x} - x = \frac{1 - x^2}{x}, \qquad f''(x) = -\frac{1}{x^2} - 1 < 0.$$

For $x > 0$ the only stationary point is at $x = 1$, where $f(1) = -\frac{1}{2}$. Since $f'' < 0$ everywhere, this is a maximum and the graph is concave throughout. So $f(x) \leq -\frac{1}{2} < 0$ for every $x > 0$, and the graph never meets the x-axis.

At the ends of the domain, $f(x) \to -\infty$ as $x \to 0^+$ (because $\ln x \to -\infty$), so the y-axis is a vertical asymptote. Also $f(x) \to -\infty$ as $x \to \infty$, because $\frac{1}{2}x^2$ grows much faster than $\ln x$.



The family of parabolas

The constant function $y = 0$ is a solution. Elsewhere, we separate the variables, working away from $x = \pm 1$, where the equation is undefined:

$$\int \frac{\mathrm{d}y}{y} = \int \frac{2x}{x^2 - 1}\,\mathrm{d}x \quad \Longrightarrow \quad \ln|y| = \ln|x^2 - 1| + c.$$

So $|y| = \mathrm{e}^c\,|x^2 - 1|$. Absorbing the sign into the constant, and including $y = 0$, the solutions are

$$y = A(x^2 - 1), \qquad A \text{ constant.}$$

Conversely, each such curve satisfies the equation, since

$$y' = 2Ax = \frac{2x \cdot A(x^2 - 1)}{x^2 - 1} = \frac{2xy}{x^2 - 1}.$$

For $A \neq 0$ each curve is a parabola. It passes through $(1, 0)$ and $(-1, 0)$ because $x^2 - 1 = 0$ there. Its gradient $2Ax$ is zero only at $x = 0$, so its vertex is $(0, -A)$, which lies on the y-axis. This is the required family.

The orthogonal curve

Take a point (x, y) with $x \neq \pm 1$ and $y \neq 0$. Exactly one parabola of the family passes through it, namely the one with $A = y/(x^2 - 1)$, and its gradient there is $\dfrac{2xy}{x^2 - 1}$. For the required curve to cross that parabola at right angles, the product of the two gradients must be -1, so the curve must satisfy

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x^2 - 1}{2xy} = \frac{1 - x^2}{2xy}.$$

Ask yourself Why is it legitimate to use the parabolas' differential equation, rather than the formula $y = A(x^2 - 1)$, to find the gradient?

The differential equation gives the gradient of the family member through (x, y) in terms of x and y alone, with no reference to A. That is exactly what we need, because the orthogonal curve meets a different parabola at each of its points.

Separating the variables again,

$$2y\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x} - x \quad\Longrightarrow\quad y^2 = \ln x - \tfrac{1}{2}x^2 + C,$$

taking $x > 0$ because the curve passes through $(1,1)$. The right-hand side is $f(x) + C$, the function we sketched at the start. At $(1,1)$ we need $1 = 0 - \frac{1}{2} + C$, so $C = \frac{3}{2}$ and the curve is

$$\boxed{y^2 = \ln x - \tfrac{1}{2}x^2 + \tfrac{3}{2}} \qquad \text{or equivalently} \qquad y^2 = f(x) + \tfrac{3}{2}.$$

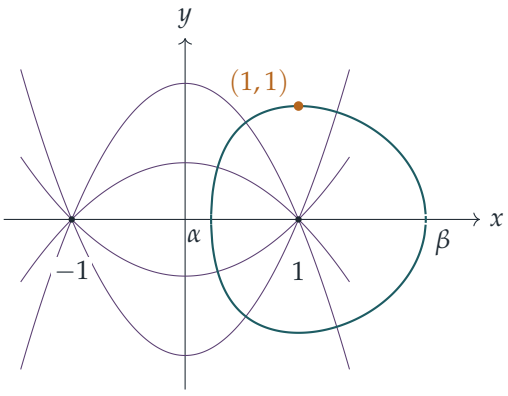
Strictly, a solution of the differential equation through $(1,1)$ is the upper half $y = \sqrt{f(x) + \frac{3}{2}}$. Its reflection in the x-axis also cuts every parabola orthogonally, so the natural answer is the whole closed curve.

Sketching the curve

Real points need $f(x) \geq -\frac{3}{2}$. Our first sketch shows that f rises from $-\infty$ to its maximum of $-\frac{1}{2}$ at $x = 1$ and then falls back to $-\infty$. So $f(x) = -\frac{3}{2}$ has exactly one root $\alpha \in (0,1)$ and one root $\beta > 1$. Numerically, $\alpha \approx 0.229$ and $\beta \approx 2.123$. Therefore:

- the curve lies in the strip $\alpha \leq x \leq \beta$ and is symmetric about the x-axis;
- y^2 is largest, equal to 1, at $x = 1$, giving the points $(1, \pm 1)$, where the tangent is horizontal;
- at $(\alpha, 0)$ and $(\beta, 0)$ we have $y^2 = 0$ while $\mathrm{d}(y^2)/\mathrm{d}x = f'(x) \neq 0$, so $\mathrm{d}y/\mathrm{d}x$ is infinite and the tangents are vertical.

So the curve is a closed oval that is lopsided: it rises steeply on the left and slopes off more gently on the right. Some of the parabolas are drawn in for comparison.



If we allowed $x < 0$ and replaced $\ln x$ by $\ln|x|$, the same working would also produce the mirror-image oval in the left half-plane. That oval does not pass through $(1,1)$, so it is not part of the answer.

Checks

- At $(1,1)$ the curve's gradient is $\frac{1-1}{2} = 0$. No parabola passes through $(1,1)$ itself, since every parabola meets the line $x = 1$ only at $(1,0)$. Close to $x = 1$ the parabolas are nearly vertical, which fits with a horizontal tangent on the curve.
- Numerically, at $x = 0.5$ the curve has $y = \sqrt{\ln 0.5 - 0.125 + 1.5} \approx 0.826$ and gradient $\frac{0.75}{2(0.5)(0.826)} \approx 0.908$. The parabola through that point has gradient $\frac{2(0.5)(0.826)}{-0.75} \approx -1.101$, and $0.908 \times (-1.101) \approx -1$.
- We confirmed orthogonality at several thousand random points of the oval, and computed the roots $\alpha \approx 0.2291$ and $\beta \approx 2.1226$ by bisection.

Common slips

- Using the gradient of one fixed parabola, such as $2Ax$ with a particular A, instead of the gradient of the family member through the general point (x, y).
- Forgetting that $y^2 = f(x) + \frac{3}{2}$ needs $f(x) \geq -\frac{3}{2}$, and so drawing an unbounded curve rather than a closed oval.

Beyond STEP

Finding a family of curves that meets a given family at right angles is the problem of *orthogonal trajectories*. In physics these appear as field lines and equipotentials, for example in electrostatics or two-dimensional fluid flow. The real and imaginary parts of an analytic complex function always give two such families, which is one route into conformal mapping. Boyce and DiPrima, *Elementary Differential Equations*, covers separable equations and orthogonal trajectories. Needham, *Visual Complex Analysis*, shows the complex-variable picture.

WORKED SOLUTION

STEP III 2001, Question 8

Pure • Complex Loci

STEP III 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q8

The key idea

Write $z = x + \mathrm{i}y$ and expand: both equations in (i) collapse to the same circle. For the argument condition in (ii), compute $\frac{z-2}{z}$ explicitly; an argument of $\frac{\pi}{4}$ means "real part equals imaginary part, and both are positive", which is the circle again plus an inequality. For (iii), substitute $z = 2/w$ into whichever form of the circle makes the algebra tidiest. The second form in (i) is there for exactly this purpose.

Solution

Part (i)

Let $z = x + \mathrm{i}y$. The first equation is

$$(x-1)^2 + (y-1)^2 = 2 \iff x^2 + y^2 - 2x - 2y = 0.$$

The second is

$$\begin{aligned} & (x-1)^2 + (y+1)^2 = 2\left[(x-1)^2 + y^2\right] \\ \iff\ & x^2 + y^2 - 2x + 2y + 2 = 2x^2 + 2y^2 - 4x + 2 \\ \iff\ & 0 = x^2 + y^2 - 2x - 2y. \end{aligned}$$

Each step is reversible, so both equations are equivalent to $x^2 + y^2 - 2x - 2y = 0$ and describe the same locus. That locus is the circle with centre $1 + \mathrm{i}$ and radius $\sqrt{2}$. It passes through 0, 2 and 2i (and $2 + 2\mathrm{i}$), and its lowest point is $1 + (1 - \sqrt{2})\mathrm{i}$, just below the real axis.

Part (ii)

For $z \neq 0$,

$$\frac{z-2}{z} = \frac{(z-2)\bar{z}}{|z|^2} = \frac{x^2 + y^2 - 2x + 2\mathrm{i}y}{x^2 + y^2},$$

since $(z-2)\bar{z} = |z|^2 - 2\bar{z} = x^2 + y^2 - 2x + 2\mathrm{i}y$.

Ask yourself What exactly does arg(something) $= \frac{\pi}{4}$ say about its real and imaginary parts?

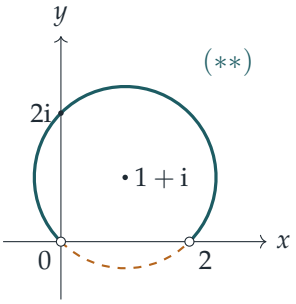
A nonzero complex number $a + \mathrm{i}b$ has argument $\frac{\pi}{4}$ exactly when it lies on the ray $a = b > 0$. Equality alone isn't enough: $a = b < 0$ gives argument $-\frac{3\pi}{4}$. The denominator $x^2 + y^2$ is positive, so $(**)$ holds if and only if

$$x^2 + y^2 - 2x = 2y \quad \text{and} \quad 2y > 0,$$

that is, if and only if z lies on the circle from (i) with $y > 0$. The condition $y > 0$ automatically rules out $z = 0$, where the quotient is undefined, and $z = 2$, where it is zero and has no argument.

So $(**)$ describes the part of the circle above the real axis: the major arc from 0 round through 2i and $2 + 2\mathrm{i}$ to 2, with both endpoints excluded. The short arc below the real axis gives argument $-\frac{3\pi}{4}$ instead.

This fits the geometry. $\arg\frac{z-2}{z}$ is the angle at z between the chords to 0 and to 2. The chord from 0 to 2 subtends a right angle at the centre $1 + \mathrm{i}$, so by the angle-at-the-centre theorem it subtends $\frac{\pi}{4}$ at every point of the major arc.



The whole circle is $(*)$; the solid arc (endpoints 0 and 2 excluded) is $(**)$, and the dashed arc is not part of $(**)$.

Part (iii)

Since $w = 2/z$ we have $z \neq 0$ and $z = 2/w$. Use the second form of the circle, $|z - (1-\mathrm{i})|^2 = 2|z-1|^2$. Substituting,

$$z - 1 = \frac{2-w}{w}, \qquad z - (1-\mathrm{i}) = \frac{2-(1-\mathrm{i})w}{w} = \frac{(1-\mathrm{i})\left[(1+\mathrm{i}) - w\right]}{w},$$

using $\frac{2}{1-\mathrm{i}} = 1 + \mathrm{i}$. Since $|1-\mathrm{i}|^2 = 2$, the equation becomes

$$\frac{2\,|w-(1+\mathrm{i})|^2}{|w|^2} = \frac{2\,|w-2|^2}{|w|^2} \iff |w-(1+\mathrm{i})| = |w-2|.$$

So w is equidistant from $1+\mathrm{i}$ and 2: the locus is the perpendicular bisector of the segment joining them. The midpoint is $\frac{3}{2}+\frac{1}{2}\mathrm{i}$ and the segment has direction $1-\mathrm{i}$, so the bisector has direction $1+\mathrm{i}$. With $w=u+\mathrm{i}v$, the locus is the line

$$v=u-1.$$

Every point of this line arises. Each w on it is nonzero, since 0 isn't on the line, and $z=2/w$ then satisfies $(*)$ by the reversible steps above. The only point of the circle with no image is $z=0$, which is "sent to infinity". A circle through the origin maps to a line under $w=2/z$, as expected.

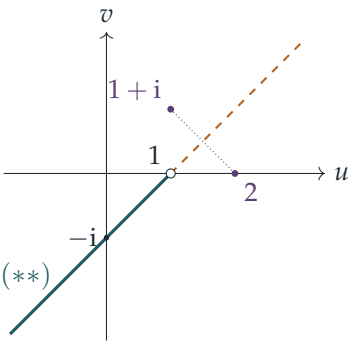
For the part corresponding to $(**)$, notice that

$$\frac{z-2}{z}=1-\frac{2}{z}=1-w,$$

so $(**)$ becomes $\arg(1-w)=\frac{\pi}{4}$, i.e. $1-w=r\,\mathrm{e}^{\mathrm{i}\pi/4}$ with $r>0$. Hence

$$w=1-\frac{r}{\sqrt{2}}(1+\mathrm{i}),\qquad r>0,$$

which is the half-line of $v=u-1$ starting at $w=1$ (excluded) and running down to the left: the points with $u<1$, equivalently $v<0$. This matches (ii): $\operatorname{Im} w=-2y/|z|^2$, so $y>0$ corresponds to $v<0$. For example, $z=2\mathrm{i}$ maps to $w=-\mathrm{i}$, and $z=2$ maps to the excluded endpoint $w=1$.



The whole line $v=u-1$ is the image of $(*)$; the solid half-line ($u<1$, with $w=1$ excluded) is the image of $(**)$. The dotted segment joins the two points that the line bisects.

Checks

- $z=2\mathrm{i}$ is on the circle and above the axis: $\frac{2\mathrm{i}-2}{2\mathrm{i}}=1+\mathrm{i}$, which has argument $\frac{\pi}{4}$. Its image $w=-\mathrm{i}$ satisfies $v=u-1$ with $u<1$.
- The top of the circle, $z=1+(1+\sqrt{2})\mathrm{i}$, maps to $w\approx 0.2929-0.7071\mathrm{i}$, and indeed $0.2929-1=-0.7071$.
- A point below the axis, $z=1+(1-\sqrt{2})\mathrm{i}$, gives $\arg\frac{z-2}{z}=-\frac{3\pi}{4}$. Its image has $u>1$, on the other half of the line.

Common slips

- Treating $\arg Z = \frac{\pi}{4}$ as just $\operatorname{Re} Z = \operatorname{Im} Z$, and so claiming the whole circle. You also need both parts positive.
- Forgetting to exclude $z = 0$ and $z = 2$ in (ii), or $w = 1$ in (iii).
- Expecting the image of a circle to be another circle. Under $w = 2/z$, circles through the origin go to straight lines.

Beyond STEP

The map $w = 2/z$ is a Möbius transformation. These maps send circles and lines to circles and lines, and a circle goes to a line exactly when it passes through the point sent to infinity. Here the endpoints 0 and 2 of the arc go to ∞ and 1. The quotient $\frac{z-2}{z-0}$, which measures the angle subtended by those endpoints, becomes $1 - w$: the direction from the image point to 1. That is why the arc in (ii) becomes a half-line in (iii). Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

WORKED SOLUTION

STEP III 2001, Question 9

Mechanics • Friction and SHM

STEP III 2001, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q9

The key idea

The beams are d apart along a line inclined at α to the horizontal, so the plank lies at angle α. As the plank slides, its weight shifts from B_1 to B_2, and the two normal reactions change *linearly* with the position of the centre of gravity. Since the two coefficients of friction differ, the total friction changes linearly too, and a restoring force proportional to displacement means simple harmonic motion. The condition $\mu_1 + \mu_2 = 2\tan\alpha$ is exactly what puts the centre of that motion halfway between the beams.

Solution

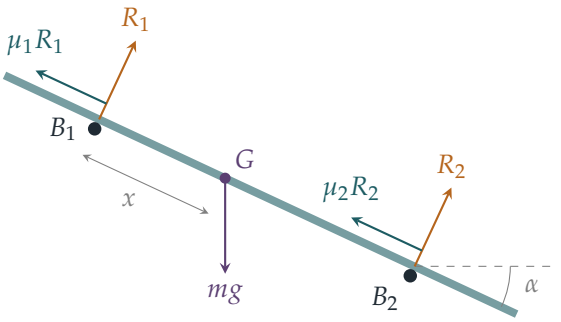
The model

Treat the plank as a rigid rod of mass m that stays in contact with both beams, which we treat as points. The line from B_1 to B_2 has length

$$\sqrt{d^2\sin^2\alpha + d^2\cos^2\alpha} = d$$

and makes angle α with the horizontal, so the plank lies along that line, sloping down from B_1 to B_2. Let x be the distance of the centre of gravity G from B_1, measured along the plank down the slope, so that $x = 0$ at release and G is over B_2 when $x = d$.

While $0 \le x \le d$, G lies between the beams. The forces are the weight mg at G; normal reactions R_1, R_2 perpendicular to the plank at B_1, B_2; and, while the plank moves down the slope, friction $\mu_1 R_1$ and $\mu_2 R_2$ up the slope.



The normal reactions

The plank does not move perpendicular to itself, so resolving perpendicular to the plank,

$$R_1 + R_2 = mg\cos\alpha.$$

It does not rotate either, so taking moments about G (the weight and both friction forces act along lines through G or along the plank, so they have no moment),

$$R_1 x = R_2(d - x).$$

Solving these two equations,

$$R_1 = \frac{mg\cos\alpha\,(d-x)}{d}, \qquad R_2 = \frac{mg\cos\alpha\,x}{d}.$$

Both are non-negative for $0 \le x \le d$, so the plank does stay on both beams.

The equation of motion

Resolving down the slope, while the plank moves down it,

$$\begin{aligned} m\ddot{x} &= mg\sin\alpha - \mu_1 R_1 - \mu_2 R_2 \\ &= mg\cos\alpha\left[\tan\alpha - \mu_1 - \frac{(\mu_2-\mu_1)x}{d}\right]. \end{aligned}$$

Now use the given condition: $\mu_1 + \mu_2 = 2\tan\alpha$ gives $\tan\alpha - \mu_1 = \frac{1}{2}(\mu_2 - \mu_1)$. Hence

$$\ddot{x} = -\omega^2\left(x - \tfrac{1}{2}d\right), \qquad \text{where} \qquad \omega^2 = \frac{g(\mu_2-\mu_1)\cos\alpha}{d} > 0,$$

the inequality holding because $\mu_2 > \mu_1$ and $\cos\alpha > 0$.

Ask yourself Does the plank actually start to move, and is the friction always pointing up the slope?

At $x = 0$, $\ddot{x} = \frac{1}{2}\omega^2 d > 0$, so the plank does start to slide down. The equation above is simple harmonic motion about $x = \frac{1}{2}d$, and with $x = 0$, $\dot{x} = 0$ at $t = 0$ its solution is

$$x = \tfrac{1}{2}d\,(1 - \cos\omega t), \qquad \dot{x} = \tfrac{1}{2}d\,\omega\sin\omega t.$$

For $0 < t < \pi/\omega$ we have $\dot{x} > 0$, so the plank really is moving down the slope throughout, and friction acting up the slope was the right assumption. Also $0 \le x \le d$ throughout, so G stays between the beams and the expressions for R_1, R_2 remain valid.

Coming to rest

The velocity is first zero again when $\omega t = \pi$, and then $x = \frac{1}{2}d(1 - \cos\pi) = d$: the centre of gravity is directly over B_2. At that instant $R_1 = 0$ and $R_2 = mg\cos\alpha$, so the plank rests on B_2 alone. It stays there, because friction at B_2 can supply up to $\mu_2 mg\cos\alpha$, while the component of weight down the slope is $mg\sin\alpha$, and

$$\mu_2 = \tfrac{1}{2}(\mu_1 + \mu_2) + \tfrac{1}{2}(\mu_2 - \mu_1) > \tfrac{1}{2}(\mu_1 + \mu_2) = \tan\alpha.$$

So the plank comes to rest with its centre of gravity over B_2 after time

$$\frac{\pi}{\omega} = \pi\left(\frac{d}{g(\mu_2 - \mu_1)\cos\alpha}\right)^{\frac{1}{2}},$$

as required.

Checks

- If $\mu_2 - \mu_1$ is small, the plank barely decelerates and the time is long, as the formula says. Each μ is then close to $\tan\alpha$, so the net force is close to zero everywhere.
- We stepped the equation of motion numerically for random α, d, μ_1 (with $\mu_2 = 2\tan\alpha - \mu_1$), stopping when the velocity first returned to zero. For example, with $\alpha = 0.7$, $d = 2$, $\mu_1 = 0.3$ and $g = 9.8$, the predicted time is about 1.56 seconds. In every case the stopping time and position agreed with π/ω and $x = d$ to within 10^{-5} (relative).

Common slips

- Taking $R_1 = R_2 = \frac{1}{2}mg\cos\alpha$. The reactions depend on where G is, and that dependence is what produces the oscillation.
- Stopping at "the velocity is zero" without checking that the plank then stays at rest: you need $\mu_2 \ge \tan\alpha$.

Beyond STEP

The motion is half a cycle of an oscillator whose "spring" comes from friction varying with load. Systems in which friction depends on position or velocity lead to rich behaviour, including the stick–slip oscillations behind squeaking hinges and bowed violin strings. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, rigid-body equilibrium and simple harmonic motion with care.

WORKED SOLUTION

STEP III 2001, Question 10

Mechanics • Relative Velocity

STEP III 2001, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q10

The key idea

Relative velocities are differences: the velocity of A relative to C is $\mathbf{v}_1 - \mathbf{u}$. Once both conditions are written as dot products, everything is measured from the midpoint $\frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$, and the two conditions become a difference of two squares. Read geometrically, the equations describe a circle and a line, and that is where the count of two velocities comes from.

Solution

The conditions

We treat the ships as particles moving in the horizontal plane of the sea, so all the velocities are two-dimensional vectors. The velocities of A and B relative to C are

$$\mathbf{a} = \mathbf{v}_1 - \mathbf{u}, \qquad \mathbf{b} = \mathbf{v}_2 - \mathbf{u}.$$

Equal magnitudes and perpendicularity mean

$$|\mathbf{v}_1 - \mathbf{u}|^2 = |\mathbf{v}_2 - \mathbf{u}|^2 \qquad \text{and} \qquad (\mathbf{v}_1 - \mathbf{u}) \cdot (\mathbf{v}_2 - \mathbf{u}) = 0.$$

The two equations

Write

$$\mathbf{w} = \mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2), \qquad \mathbf{d} = \tfrac{1}{2}(\mathbf{v}_1 - \mathbf{v}_2).$$

Then $\mathbf{v}_1 = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{d}$ and $\mathbf{v}_2 = \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) - \mathbf{d}$, so

$$\mathbf{a} = \mathbf{d} - \mathbf{w}, \qquad \mathbf{b} = -\mathbf{d} - \mathbf{w}.$$

Now expand each condition, using $\mathbf{w} \cdot \mathbf{d} = \mathbf{d} \cdot \mathbf{w}$:

$$\mathbf{a} \cdot \mathbf{b} = -(\mathbf{d} - \mathbf{w}) \cdot (\mathbf{d} + \mathbf{w}) = |\mathbf{w}|^2 - |\mathbf{d}|^2,$$
$$|\mathbf{a}|^2 - |\mathbf{b}|^2 = (|\mathbf{d}|^2 - 2\mathbf{d} \cdot \mathbf{w} + |\mathbf{w}|^2) - (|\mathbf{d}|^2 + 2\mathbf{d} \cdot \mathbf{w} + |\mathbf{w}|^2) = -4\,\mathbf{w} \cdot \mathbf{d}.$$

So perpendicularity, $\mathbf{a} \cdot \mathbf{b} = 0$, says exactly $|\mathbf{w}|^2 = |\mathbf{d}|^2$, which is

$$\left|\mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)\right|^2 = \left|\tfrac{1}{2}(\mathbf{v}_1 - \mathbf{v}_2)\right|^2,$$

and equal magnitudes, $|\mathbf{a}|^2 = |\mathbf{b}|^2$, says exactly $\mathbf{w} \cdot \mathbf{d} = 0$, which (multiplying by 2) is

$$\left(\mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)\right) \cdot (\mathbf{v}_1 - \mathbf{v}_2) = 0.$$

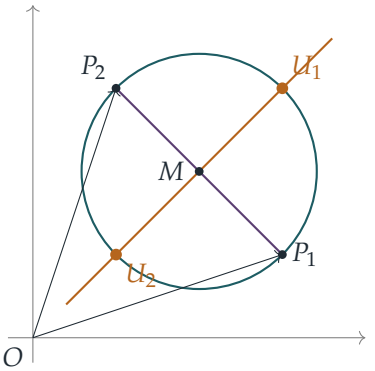
These are the two required equations. Each step above was an identity, so the argument also runs backwards: the two equations hold *if and only if* the two conditions hold.

Why there are two velocities for C

Draw velocities as position vectors from a common origin, and let P_1, P_2, U be the tips of $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{u}$, with M the midpoint of P_1P_2.

Ask yourself What does each equation say about where the point U can be?

The first equation says U is at distance $\frac{1}{2}|\mathbf{v}_1 - \mathbf{v}_2|$ from M: it lies on the circle with diameter P_1P_2. The second says MU is perpendicular to P_1P_2 (or $U = M$): U lies on the perpendicular bisector of P_1P_2.



In the plane, a line through the centre of a circle of positive radius meets it in exactly two points, at opposite ends of a diameter. The radius is $\frac{1}{2}|\mathbf{v}_1 - \mathbf{v}_2|$, which is positive precisely because $\mathbf{v}_1 \neq \mathbf{v}_2$. So there are exactly two possible velocities for C:

$$\mathbf{u} = \tfrac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2) \pm \tfrac{1}{2}\,\mathbf{d}^{\perp},$$

where $\mathbf{d}^{\perp}$ is $\mathbf{v}_1 - \mathbf{v}_2$ turned through a right angle. (If $\mathbf{v}_1 = \mathbf{v}_2$ the circle shrinks to the single point M, and then both relative velocities are zero, so the conditions say nothing useful.) In the diagram, the right angle P_1UP_2 is the angle in a semicircle, and $UP_1 = UP_2$ because U is on the perpendicular bisector.

The final part

Now $|\mathbf{v}_1| = |\mathbf{v}_2| = k$ and $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, where $k > 0$ (perpendicularity only makes sense for non-zero vectors). Then $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular and non-zero, so they span the plane, and we may write $\mathbf{u} = \alpha\mathbf{v}_1 + \beta\mathbf{v}_2$.

The second equation, using $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$ and $|\mathbf{v}_1|^2 = |\mathbf{v}_2|^2$, gives

$$\mathbf{u} \cdot (\mathbf{v}_1 - \mathbf{v}_2) = \tfrac{1}{2}\left(|\mathbf{v}_1|^2 - |\mathbf{v}_2|^2\right) = 0, \qquad \text{so} \qquad \alpha k^2 - \beta k^2 = 0,$$

and hence $\beta = \alpha$. For the first, expand:

$$|\mathbf{u}|^2 - \mathbf{u} \cdot (\mathbf{v}_1 + \mathbf{v}_2) + \tfrac{1}{4}|\mathbf{v}_1 + \mathbf{v}_2|^2 = \tfrac{1}{4}|\mathbf{v}_1 - \mathbf{v}_2|^2.$$

Since $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, both $|\mathbf{v}_1 + \mathbf{v}_2|^2$ and $|\mathbf{v}_1 - \mathbf{v}_2|^2$ equal $2k^2$, so these cancel and $|\mathbf{u}|^2 = \mathbf{u} \cdot (\mathbf{v}_1 + \mathbf{v}_2)$. With $\mathbf{u} = \alpha(\mathbf{v}_1 + \mathbf{v}_2)$ this is

$$2\alpha^2 k^2 = 2\alpha k^2 \quad \Longrightarrow \quad \alpha(\alpha - 1) = 0.$$

So $\alpha = 0$, giving $\mathbf{u} = \mathbf{0}$, or $\alpha = 1$, giving $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$. Hence if $\mathbf{u} \neq \mathbf{0}$ then $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$, as required.

This makes sense physically. If C is at rest, the relative velocities are $\mathbf{v}_1$ and $\mathbf{v}_2$ themselves, which already satisfy the conditions. If $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$, the relative velocities are $-\mathbf{v}_2$ and $-\mathbf{v}_1$, which again are equal in length and perpendicular.

Checks

Take $\mathbf{v}_1 = (3,1)$, $\mathbf{v}_2 = (1,3)$, as in the diagram. The formula gives $\mathbf{u} = (2,2) \pm (1,1)$, that is $(3,3)$ or $(1,1)$. For $\mathbf{u} = (3,3)$ the relative velocities are $(0,-2)$ and $(-2,0)$; for $(1,1)$ they are $(2,0)$ and $(0,2)$. Both pairs are equal in length and perpendicular. A computer testing 10^4 random pairs $\mathbf{v}_1, \mathbf{v}_2$ confirmed the two identities and the two solutions, and for 10^3 random equal perpendicular pairs it found the solutions to be exactly $\mathbf{0}$ and $\mathbf{v}_1 + \mathbf{v}_2$.

Common slips

- Writing the relative velocity of A to C as $\mathbf{u} - \mathbf{v}_1$. The sign doesn't affect these conditions, but it matters in most relative-velocity questions.
- Forgetting to say why there are *exactly* two velocities. The count depends on working in a plane (in three dimensions the solutions form a whole circle) and on the radius being non-zero.

Beyond STEP

Moving from $\mathbf{u}$ to $\mathbf{w} = \mathbf{u} - \frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$ is a change of reference frame, and the fact that relative velocities don't change when every velocity is shifted by the same vector is

Galilean invariance, the starting point of classical mechanics. The "velocity diagram" picture used here, with velocities drawn as points, also turns collision and pursuit problems into geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and reference frames carefully.

WORKED SOLUTION

STEP III 2001, Question 11

Mechanics • Rotational Energy Conservation

STEP III 2001, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q11

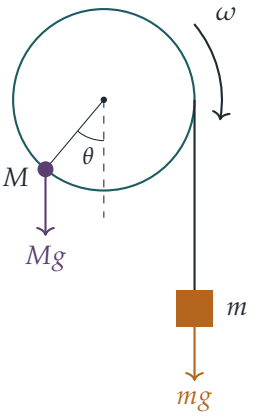
The key idea

Everything is smooth (the axle is free and the string is light), so mechanical energy is conserved, and one energy equation gives ω as a function of θ with no need for tension or forces. The cylinder halts exactly when ω^2 hits zero. So the second part becomes a question about a single function of θ: the hanging mass's loss of potential energy must always beat the particle's gain. Rearranged, that is a comparison between m/M and the largest value of $(1-\cos\theta)/\theta$, which is a straightforward maximisation.

Solution

The angular velocity

We model m and M as particles, the string as light and inextensible and not slipping on the cylinder, and the axle as smooth. Let the cylinder turn through angle θ in the sense that lets m descend, and let $\omega = \dot{\theta}$.



When the cylinder has turned through θ, a length $a\theta$ of string has unwound, so m has

fallen $a\theta$ and moves with speed $a\omega$. The particle M is on the rim, so it moves with speed $a\omega$; it started at depth a below the axis and is now at depth $a\cos\theta$, so it has risen $a(1-\cos\theta)$.

The kinetic energy is

$$\tfrac{1}{2}I\omega^2 + \tfrac{1}{2}M(a\omega)^2 + \tfrac{1}{2}m(a\omega)^2 = \tfrac{1}{2}\big(I + (M+m)a^2\big)\omega^2.$$

No work is done by the smooth axle, and the string, being light and inextensible, transfers energy without loss. So the gain in kinetic energy equals the net loss in potential energy:

$$\tfrac{1}{2}\big(I + (M+m)a^2\big)\omega^2 = mga\theta - Mga(1-\cos\theta),$$

giving

$$\boxed{\omega^2 = \frac{2ga\big(m\theta - M(1-\cos\theta)\big)}{I + (M+m)a^2}} \qquad \text{and} \qquad \omega = \sqrt{\frac{2ga\big(m\theta - M(1-\cos\theta)\big)}{I + (M+m)a^2}}.$$

One modelling point is worth checking: the string must stay taut. Differentiating the energy equation with respect to time gives $\big(I + (M+m)a^2\big)\ddot{\theta} = ga(m - M\sin\theta)$, and the tension is $T = m(g - a\ddot{\theta})$. Now $a\ddot{\theta} < g$ is equivalent to $a^2(m - M\sin\theta) < I + (M+m)a^2$, that is, $-Ma^2\sin\theta < I + Ma^2$, which is always true. So $T > 0$ throughout and the energy equation is valid for all θ.

Never coming to a halt

At the start, $\ddot{\theta} = gam/(I + (M+m)a^2) > 0$, so the cylinder does begin to turn in the direction we chose. Since ω is continuous, it can only change sign by passing through zero, so the cylinder keeps turning for ever provided $\omega^2 > 0$ for every $\theta > 0$. With $k = m/M$, the expression for ω^2 shows that this happens exactly when

$$k\theta > 1 - \cos\theta \quad \text{for all } \theta > 0, \qquad \text{i.e.} \qquad k > h(\theta) = \frac{1-\cos\theta}{\theta} \quad \text{for all } \theta > 0.$$

So we need the largest value of h on $\theta > 0$.

Stationary points. We have

$$h'(\theta) = \frac{\theta\sin\theta - (1-\cos\theta)}{\theta^2}.$$

Using $\sin\theta = 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta$ and $1 - \cos\theta = 2\sin^2\tfrac{1}{2}\theta$,

$$h'(\theta) = \frac{2\sin\tfrac{1}{2}\theta\,\big(\theta\cos\tfrac{1}{2}\theta - \sin\tfrac{1}{2}\theta\big)}{\theta^2}.$$

On $0 < \theta < \pi$ both $\sin\tfrac{1}{2}\theta$ and $\cos\tfrac{1}{2}\theta$ are positive, so $h'(\theta)$ has the sign of $\theta - \tan\tfrac{1}{2}\theta$. Let $u(\theta) = \tan\tfrac{1}{2}\theta - \theta$. Then $u(0) = 0$, $u'(\theta) = \tfrac{1}{2}\sec^2\tfrac{1}{2}\theta - 1$ is negative near 0 and $u''(\theta) = \tfrac{1}{2}\sec^2\tfrac{1}{2}\theta\tan\tfrac{1}{2}\theta > 0$, so u is convex on $[0,\pi)$. It dips below zero and then tends to $+\infty$ as $\theta \to \pi$. A convex function can cross zero at most twice, so u has exactly one zero in $(0,\pi)$, which is α. Hence h increases on $(0,\alpha)$ and decreases on (α,π).

Ask yourself What is the value of h at its maximum, and what about larger θ?

At $\theta = \alpha$ the numerator of h' vanishes, so $1 - \cos\alpha = \alpha\sin\alpha$ and therefore

$$h(\alpha) = \frac{1-\cos\alpha}{\alpha} = \sin\alpha.$$

For $\pi \le \theta \le 2\pi$ we have $\sin\frac{1}{2}\theta \ge 0$ and $\cos\frac{1}{2}\theta \le 0$, so $\theta\cos\frac{1}{2}\theta - \sin\frac{1}{2}\theta \le 0$ and $h' \le 0$: h keeps decreasing up to 2π, so $h < \sin\alpha$ on $(\alpha, 2\pi]$. For $\theta > 2\pi$, $h(\theta) \le 2/\theta < 1/\pi$. Finally, $u(\frac{3}{4}\pi) = \tan\frac{3}{8}\pi - \frac{3}{4}\pi \approx 2.414 - 2.356 > 0$ and $u(\frac{2}{3}\pi) = \sqrt{3} - \frac{2}{3}\pi < 0$, so $\frac{2}{3}\pi < \alpha < \frac{3}{4}\pi$, giving $\sin\alpha > \sin\frac{3}{4}\pi = \frac{1}{\sqrt{2}} > \frac{1}{\pi}$. So $\sin\alpha$ is the largest value of h on $\theta > 0$, attained only at $\theta = \alpha$.

Conclusion. If $m/M > \sin\alpha$, then for every $\theta > 0$

$$m\theta - M(1-\cos\theta) = M\theta\big(k - h(\theta)\big) \ge M\theta(k - \sin\alpha) > 0,$$

so $\omega^2 > 0$ for all $\theta > 0$ and the cylinder rotates without coming to a halt, as required.

(Conversely, if $m/M < \sin\alpha$ then $\omega^2 < 0$ at $\theta = \alpha$, so ω reaches zero before then and the cylinder stops and swings back. In the borderline case $m/M = \sin\alpha$ the system creeps towards the equilibrium position $\theta = \alpha$ without ever quite reaching it.)

Checks

- With $M = 0$ the formula gives $\omega^2 = 2gma\theta/(I + ma^2)$, the standard result for a mass unwinding from a pulley: ω^2 grows linearly, as for constant angular acceleration.
- Solving $\alpha = \tan\frac{1}{2}\alpha$ by bisection gives $\alpha \approx 2.3311$ (about 133.6°) and $\sin\alpha \approx 0.7246$; a numerical search of $(1-\cos\theta)/\theta$ over $0 < \theta < 100$ gives the same maximum 0.7246.
- Stepping the equation of motion numerically for random I, a, M, m reproduces the formula for ω^2, and with $m/M = 0.735$ the cylinder never stops while with $m/M = 0.715$ it does.

Common slips

- Forgetting the kinetic energy of M or of m: both move with speed $a\omega$, so the effective moment of inertia is $I + (M+m)a^2$.
- Getting the height of M wrong: it rises by $a(1-\cos\theta)$, not $a\sin\theta$.
- Showing only that h has a local maximum at α, without checking that h is smaller for all larger θ.

Beyond STEP

This is one-degree-of-freedom motion in a potential: $\frac{1}{2}J\omega^2 + V(\theta) = 0$ with $V(\theta) = Mga(1-\cos\theta) - mga\theta$, a tilted "washboard" potential. Starting from rest, the motion

runs away for ever exactly when V stays below its starting value for every $\theta > 0$, and the same picture describes a driven pendulum and the Josephson junction in physics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy diagrams of this kind.

WORKED SOLUTION

STEP III 2001, Question 12

Probability and Statistics • Expectation by Indicators

STEP III 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q12

The key idea

In part (i), don't try to find the distribution of the number of black balls: break the count into one indicator per black ball and use linearity of expectation. Drawing without replacement until the bag is empty is just a random ordering of all the balls, so each indicator comes down to a symmetry argument. In part (ii) the bag never changes between white draws, so the wait for each white ball is geometric, and the "hence" comes from noticing that after a white ball is removed we are back in the same situation with one fewer white ball.

Solution

Part (i)

Imagine continuing to draw until the bag is empty. Since every draw is at random and nothing is returned, the order in which the $b+w$ balls come out is a uniformly random ordering of them. The black balls drawn by the time the last white ball is removed are exactly the black balls that come before the last white ball in this ordering.

Label the black balls $1, 2, \ldots, b$ and let $I_j = 1$ if black ball j comes out before the last white ball, and $I_j = 0$ otherwise. If N is the number of black balls drawn when the last white ball is removed, then

$$N = I_1 + I_2 + \cdots + I_b, \qquad \text{so} \qquad \mathrm{E}(N) = \sum_{j=1}^{b} \mathrm{P}(I_j = 1)$$

by linearity of expectation (which needs no independence between the I_j).

? **Ask yourself** What is the chance that black ball j comes after every white ball?

Look only at the $w+1$ balls consisting of black ball j and the w white balls. In a uniformly random ordering of all the balls, the relative order of these $w+1$ is also

uniformly random, so black ball j is equally likely to be in any of the $w+1$ positions among them. It is last of the group with probability $\frac{1}{w+1}$, and that is precisely the event $I_j = 0$. Hence

$$\mathrm{P}(I_j = 1) = 1 - \frac{1}{w+1} = \frac{w}{w+1},$$

and

$$\boxed{\mathrm{E}(N) = \frac{bw}{w+1}.}$$

Part (ii)

Now black balls go back, so until a white ball is drawn the bag always holds b black and w white balls. Each draw is therefore black with probability $p = \frac{b}{b+w}$ and white with probability $q = \frac{w}{b+w}$, independently of earlier draws, since the contents are the same each time.

Let X be the number of black draws before the first white ball. Then $X = n$ means n blacks followed by a white, so

$$\mathrm{P}(X = n) = p^n q, \qquad n = 0, 1, 2, \ldots$$

If $b = 0$ then $X = 0$ and the result is trivial, so take $b \geq 1$; then $0 < p < 1$, and using $\sum_{n\geq 1} n x^{n-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$ (differentiate the geometric series),

$$\mathrm{E}(X) = \sum_{n=0}^{\infty} n p^n q = pq \sum_{n=1}^{\infty} n p^{n-1} = \frac{pq}{(1-p)^2} = \frac{p}{q} = \frac{b/(b+w)}{w/(b+w)} = \frac{b}{w},$$

since $1 - p = q$. This is the required result.

For the last white ball, write the total number of black draws as $X_w + X_{w-1} + \cdots + X_1$, where X_k is the number of black draws made while exactly k white balls are in the bag. While k white balls remain, the bag holds b black and k white balls, so this stage is exactly the first-white problem with w replaced by k, and the result just proved gives $\mathrm{E}(X_k) = \frac{b}{k}$. By linearity of expectation the expected number of black draws when the last white ball is removed is

$$\boxed{\sum_{k=1}^{w} \frac{b}{k} = b\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{w}\right).}$$

Checks

With $w = 1$ both parts are easy to see directly. In (i), the single black ball lands before the white one with probability $\frac{1}{2}$, giving $\frac{b}{2}$, which matches $\frac{bw}{w+1}$. In (ii), there is one stage only and the sum is just $\frac{b}{1} = b$. As $w \to \infty$ in (i), almost every black ball comes out before the last white one, and indeed $\frac{bw}{w+1} \to b$.

We checked (i) by listing every ordering for all $b \leq 4$, $w \leq 4$; for $b = 3$, $w = 4$ it gives exactly $\frac{12}{5}$. For (ii), a simulation of 2×10^5 runs with $b = 3$, $w = 4$ gave a mean of about 6.245, against $3\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = \frac{25}{4} = 6.25$.

Common slips

- In (i), assuming the black balls drawn are independent of one another. Linearity of expectation doesn't need independence, which is why indicators work so well.
- In (ii), using the geometric distribution that counts the white draw too, which gives $\frac{1}{q} = \frac{b+w}{w}$, one more than the answer.
- Using $b + w$ in the denominator at every stage of the last part: the number of white balls in the bag drops by one at each stage.

Beyond STEP

The second part is the coupon collector's problem in disguise: summing geometric waiting times over stages gives a harmonic sum, which grows like $\ln w$. Writing a count as a sum of indicators, as in part (i), is one of the most powerful techniques in probability, and you will meet it again when computing expected numbers of fixed points, records and cycles in random permutations. Blitzstein and Hwang, *Introduction to Probability*, has many examples of both ideas.

WORKED SOLUTION

STEP III 2001, Question 13

Probability and Statistics • Coin Sequence Games

STEP III 2001, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.8, *Penney's Game*: https://beyondstep.keithmcnulty.org/ch12/penney.html

The key idea

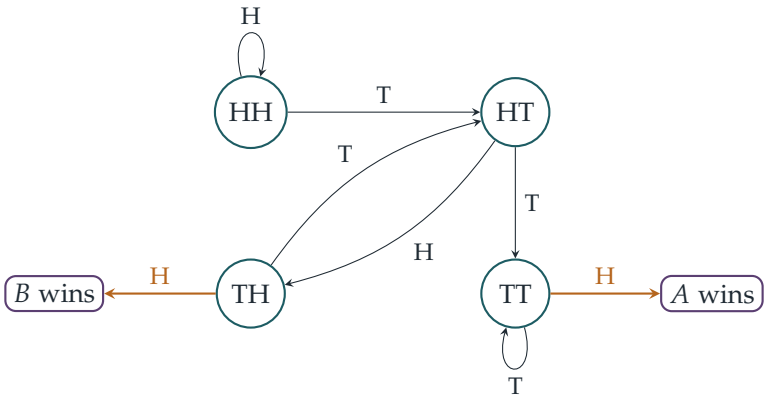
Both target sequences have length three, so nobody can win before the third toss, and from then on all that matters for the future is the *last two tosses*. That gives just four "states" (HH, HT, TH, TT), and conditioning on the next toss turns the game into a small system of linear equations in the winning probabilities from each state. Parts (ii) and (iii) can be done the same way, though in two of the four games a short direct argument is quicker.

Solution

Throughout, the tosses are independent and each shows H or T with probability $\frac{1}{2}$. Once at least two tosses have been made and nobody has won yet, the next toss together with the last two determines whether someone wins now, and if not, the new last two tosses. Earlier tosses are irrelevant, because each target sequence is only three long. So the probability of winning from a given point depends only on the last two tosses, and it is the same as if those two had been the first two tosses of a new game.

Part (i)

Let p_{XY} be the probability that A (TTH) beats B (THH) given that the first two tosses are XY. The diagram shows how the next toss moves us between states.



Why $p_{\mathrm{TT}} = 1$. Suppose the last two tosses are TT. With probability 1 a head eventually appears (the chance of n tails in a row is $2^{-n} \to 0$). Until it does, the tosses are all tails, so THH cannot appear; and the first head is preceded by at least two tails, so it completes TTH. Hence A wins for certain: $p_{\mathrm{TT}} = 1$.

Why $p_{\mathrm{HT}} = \frac{1}{2}p_{\mathrm{TH}} + \frac{1}{2}p_{\mathrm{TT}}$. After HT, neither HTH nor HTT is a target sequence, so nobody wins on the third toss. If it is H (probability $\frac{1}{2}$) the last two tosses are TH; if it is T they are TT. Conditioning on the third toss, and using the remark above that only the last two tosses matter,

$$p_{\mathrm{HT}} = \tfrac{1}{2}p_{\mathrm{TH}} + \tfrac{1}{2}p_{\mathrm{TT}}.$$

The other two states. After TH, a head gives THH and B wins, while a tail leaves HT. After HH, a head leaves HH and a tail leaves HT (and HHH, HHT are not targets here). So

$$p_{\mathrm{TH}} = \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}p_{\mathrm{HT}}, \qquad p_{\mathrm{HH}} = \tfrac{1}{2}p_{\mathrm{HH}} + \tfrac{1}{2}p_{\mathrm{HT}}.$$

The second equation gives $p_{\mathrm{HH}} = p_{\mathrm{HT}}$. Substituting $p_{\mathrm{TH}} = \frac{1}{2}p_{\mathrm{HT}}$ and $p_{\mathrm{TT}} = 1$ into the equation for p_{HT},

$$p_{\mathrm{HT}} = \tfrac{1}{4}p_{\mathrm{HT}} + \tfrac{1}{2} \quad\Longrightarrow\quad \tfrac{3}{4}p_{\mathrm{HT}} = \tfrac{1}{2} \quad\Longrightarrow\quad p_{\mathrm{HT}} = \tfrac{2}{3}.$$

Hence $p_{\mathrm{HH}} = p_{\mathrm{HT}} = \frac{2}{3}$ and $p_{\mathrm{TH}} = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}$, as required.

Ask yourself Why is it legitimate to "restart" the game from TH, as if TH were the first two tosses, when TH actually arose at tosses 5 and 6, say?

Because the future of the game depends only on future tosses and the last two tosses: any occurrence of TTH or THH that is not yet complete must use at most the last two tosses already made. The earlier history cannot help either sequence, and the future tosses are independent of it.

The overall probability. Nobody can win in the first two tosses, and the four possible pairs each have probability $\frac{1}{4}$. By the law of total probability,

$$\mathrm{P}(A \text{ wins}) = \tfrac{1}{4}\left(\tfrac{2}{3} + \tfrac{2}{3} + \tfrac{1}{3} + 1\right) = \tfrac{1}{4}\cdot\tfrac{8}{3} = \tfrac{2}{3}.$$

Part (ii)

Now B has THH and C has HHT. Here a direct argument is neatest.

If the first two tosses are HH, then C wins: the first tail after them (which comes with probability 1) is preceded by HH and completes HHT, and before it every toss is a head, so THH cannot have appeared.

Otherwise there is a tail among the first two tosses. With probability 1 the pattern HH eventually occurs. Consider its first occurrence. It does not start at toss 1, so the toss just before it exists, and it is not H (else there would be an earlier HH), so it is T. Thus the first HH completes a THH. Meanwhile HHT cannot have occurred before, since it would need an earlier HH. So B wins.

Therefore B wins unless the first two tosses are HH:

$$\mathrm{P}(B \text{ wins}) = 1 - \tfrac{1}{4} = \tfrac{3}{4}.$$

Part (iii)

C (HHT) against D (HTT). Let q_{XY} be the probability that C wins from last two tosses XY. From HH, C wins for certain, by the same reasoning as in (ii): the first tail completes HHT, and HTT needs a tail after HT, which cannot come first. So $q_{\mathrm{HH}} = 1$. Conditioning on the next toss in each other state:

HT:	T gives HTT (D wins), H gives TH:	$q_{\mathrm{HT}} = \frac{1}{2} q_{\mathrm{TH}}$,
TH:	H gives HH, T gives HT:	$q_{\mathrm{TH}} = \frac{1}{2} + \frac{1}{2} q_{\mathrm{HT}}$,
TT:	H gives TH, T gives TT:	$q_{\mathrm{TT}} = \frac{1}{2} q_{\mathrm{TH}} + \frac{1}{2} q_{\mathrm{TT}}$.

The last gives $q_{\mathrm{TT}} = q_{\mathrm{TH}}$. Substituting the first into the second, $q_{\mathrm{TH}} = \frac{1}{2} + \frac{1}{4} q_{\mathrm{TH}}$, so $q_{\mathrm{TH}} = \frac{2}{3}$, $q_{\mathrm{TT}} = \frac{2}{3}$ and $q_{\mathrm{HT}} = \frac{1}{3}$. Hence

$$\mathrm{P}(C \text{ wins}) = \tfrac{1}{4}\left(1 + \tfrac{1}{3} + \tfrac{2}{3} + \tfrac{2}{3}\right) = \tfrac{1}{4} \cdot \tfrac{8}{3} = \tfrac{2}{3} > \tfrac{1}{2},$$

so C is more likely to win than D.

D (HTT) against A (TTH). This mirrors part (ii) with H and T swapped. If the first two tosses are TT, A wins: the first head completes TTH, and HTT would need a head followed by two tails, which cannot happen before that first head. Otherwise there is a head among the first two tosses. With probability 1 the pattern TT eventually occurs; at its first occurrence it does not start at toss 1, and the toss before it cannot be T (there would be an earlier TT), so it is H and the first TT completes HTT. TTH cannot have appeared earlier, as it needs an earlier TT. So D wins, and

$$\mathrm{P}(D \text{ wins}) = 1 - \tfrac{1}{4} = \tfrac{3}{4} > \tfrac{1}{2}.$$

So D is more likely to win than A.

? **Ask yourself** A beats B, B beats C, C beats D and D beats A. How can that be?

"Is more likely to beat" is not a transitive relation, so there is no contradiction: the four players form a cycle, with A, B, C, D winning with probabilities $\frac{2}{3}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{3}{4}$ in their respective games. Whoever chooses second can always pick a sequence that beats the first player's.

Checks

- In part (i), substitute back: $\frac{1}{2}\cdot\frac{1}{3}+\frac{1}{2}\cdot 1=\frac{2}{3}=p_{\mathrm{HT}}$, and $\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{3}=p_{\mathrm{TH}}$.
- The state method also works for part (ii) and the D–A game: for B against C the winning probabilities for B from HH, HT, TH, TT are $0, 1, 1, 1$, averaging to $\frac{3}{4}$.
- A simulation of 200 000 games for each pairing gave winning frequencies 0.664, 0.752, 0.666 and 0.749, against the exact $\frac{2}{3}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{3}{4}$.

Common slips

- Forgetting that the game can end on the third toss, for example writing $p_{\mathrm{TH}}=\frac{1}{2}p_{\mathrm{HH}}+\frac{1}{2}p_{\mathrm{HT}}$ instead of recognising that TH followed by H is an immediate win for B.
- Averaging the four state probabilities without saying that each pair of first tosses has probability $\frac{1}{4}$ and that nobody can win before the third toss.

Beyond STEP

This is Penney's game, a famous example of non-transitive probabilities. The state method is the theory of absorbing Markov chains, where the probabilities of absorption satisfy exactly the kind of linear equations solved here. There is even a closed formula for the odds between any two sequences, Conway's "leading numbers" algorithm, which can be proved using martingales. For Markov chains and first-step analysis, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP III 2001, Question 14

Probability and Statistics • Unbiased Variance Estimator

STEP III 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q14

The key idea

Everything here turns on one fact: the "variance of the sample" uses divisor $n = 2$, so it underestimates the population variance, and doubling it puts that right. For two values the sample mean is the midpoint, so each value sits half the gap away from it. That makes V a function of the single difference $X_1 - X_2$, and the final probability becomes an area in the square $[0, a]^2$.

Solution

The variance of X

The density of X is $f(x) = \frac{1}{a}$ on $[0, a]$. By symmetry $\mathrm{E}(X) = \frac{1}{2}a$, and

$$\mathrm{E}(X^2) = \int_0^a \frac{x^2}{a}\,\mathrm{d}x = \frac{a^2}{3}.$$

So

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = \frac{a^2}{3} - \frac{a^2}{4} = \frac{a^2}{12},$$

as required.

The sample variance

For the sample X_1, X_2 the sample mean is $\bar{X} = \frac{1}{2}(X_1 + X_2)$, and the variance of the sample is the mean squared deviation from it:

$$V = \tfrac{1}{2}\big[(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2\big].$$

Now $X_1 - \bar{X} = \frac{1}{2}(X_1 - X_2)$ and $X_2 - \bar{X} = -\frac{1}{2}(X_1 - X_2)$, so both squared deviations equal $\frac{1}{4}(X_1 - X_2)^2$. Hence

$$V = \tfrac{1}{2} \cdot 2 \cdot \tfrac{1}{4}(X_1 - X_2)^2 = \tfrac{1}{4}(X_1 - X_2)^2,$$

which is the given form.

Ask yourself How do we find $\mathrm{E}\big((X_1 - X_2)^2\big)$ without a double integral?

Since X_1 and X_2 have the same distribution, $\mathrm{E}(X_1 - X_2) = 0$, so $\mathrm{E}\big((X_1 - X_2)^2\big) = \mathrm{Var}(X_1 - X_2)$. Because X_1 and X_2 are independent, their variances add (the minus sign doesn't matter, as $\mathrm{Var}(-X_2) = \mathrm{Var}(X_2)$):

$$\mathrm{Var}(X_1 - X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) = \frac{a^2}{12} + \frac{a^2}{12} = \frac{a^2}{6}.$$

Therefore

$$\mathrm{E}(2V) = \tfrac{1}{2}\,\mathrm{E}\big((X_1 - X_2)^2\big) = \tfrac{1}{2} \cdot \frac{a^2}{6} = \frac{a^2}{12} = \mathrm{Var}(X).$$

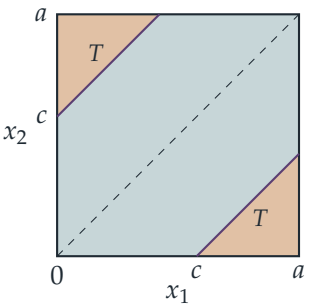
The expected value of $2V$ equals the quantity it estimates, whatever the value of a, so $2V$ is an unbiased estimator of the variance of X. (By contrast $\mathrm{E}(V) = \frac{a^2}{24}$, only half the true variance.)

The probability

Since $V \geq 0$ and $\frac{1}{12}a^2 > 0$, we may take square roots:

$$V < \frac{a^2}{12} \iff (X_1 - X_2)^2 < \frac{a^2}{3} \iff |X_1 - X_2| < \frac{a}{\sqrt{3}}.$$

The pair (X_1, X_2) is uniformly distributed over the square $[0, a]^2$, because X_1 and X_2 are independent and each is uniform. So the probability is the fraction of the square's area lying in the band $|x_1 - x_2| < c$, where $c = a/\sqrt{3} < a$.



The complement consists of the two shaded corner triangles T, each right-angled and isosceles with legs $a - c$. Their total area is $2 \cdot \frac{1}{2}(a - c)^2 = (a - c)^2$, so

$$\mathrm{P}\Big(V < \frac{a^2}{12}\Big) = 1 - \frac{(a-c)^2}{a^2} = 1 - \Big(1 - \frac{1}{\sqrt{3}}\Big)^2 = 1 - \Big(\frac{4}{3} - \frac{2}{\sqrt{3}}\Big) = \frac{2}{\sqrt{3}} - \frac{1}{3}.$$

That is, the exact probability is

$$\mathrm{P}\Big(V < \frac{a^2}{12}\Big) = \frac{2\sqrt{3} - 1}{3}.$$

With $\sqrt{3} \approx 1.732$ this is $\frac{2.464}{3} \approx 0.821$, so to one significant figure the probability is 0.8.

Checks

- The answer doesn't depend on a, as it shouldn't: rescaling X by a factor scales both V and $\frac{1}{12}a^2$ by its square.
- A simulation of 4×10^5 pairs with $a = 3$ gave a mean of $2V$ of 0.751 (target 0.75) and a proportion 0.821 with $V < \frac{1}{12}a^2$, against $\frac{1}{3}(2\sqrt{3} - 1) = 0.8214$.
- Sanity: the probability lies in $(0, 1)$, and replacing $\frac{1}{12}a^2$ by $\frac{1}{4}a^2$ (the largest possible V) makes $c = a$ and the formula $1 - (1 - c/a)^2$ give 1.

Common slips

- Using divisor $n - 1$ for the "variance of the sample": that gives $\frac{1}{2}(X_1 - X_2)^2$, not the stated $\frac{1}{4}(X_1 - X_2)^2$. The question wants the divisor n.
- Forgetting the absolute value, and so counting only one of the two triangles (which gives a probability below $\frac{1}{2}$).
- Writing $\mathrm{Var}(X_1 - X_2) = \mathrm{Var}(X_1) - \mathrm{Var}(X_2) = 0$.

Beyond STEP

The result here is the case $n = 2$ of a general theorem: for any sample of size n from a distribution with variance σ^2, the mean of $\sum(X_i - \bar{X})^2$ is $(n - 1)\sigma^2$. That is why the divisor $n - 1$ (Bessel's correction) gives an unbiased estimator. First-year statistics goes on to ask which unbiased estimator has the smallest variance. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both cover expectation and variance of sums in the style used here.

Beyond STEP

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